



SIG.TRAINING DAY CAIRO SIG ENG

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Cairo – Mai 3, 2020

Agenda

1. Introduction
2. Railway Signalling System
3. Basic Component of signalling system
4. Automatic Block & Headway concept
5. Solutions & Products
6. Rules and Regulations

01

Introduction



Introduction

- Signaling is one of the most important components of a railway system.
- Train movement safety depends on it and the control and management of trains depends on them.
- In the very early days of railways there was no fixed signaling - no system for informing the driver of the state of the line ahead. Trains were driven "on sight". Drivers had to keep their eyes open for any sign of a train in front so they could stop before hitting it.

Very soon though, practical experience proved that this didn't work and there needed to be a way of preventing trains running into each other.

Signaling is essentially a sophisticated traffic light system for the railway.

The signaling system is very complicated and comprises a great many parts.

Signaling systems are designed to fail to a safe state, so faulty components can often result in signals remaining at red, causing train delays

Here is an attempt to explain, in simple terms, how railway signaling developed and how it really works, based on the railways standards.

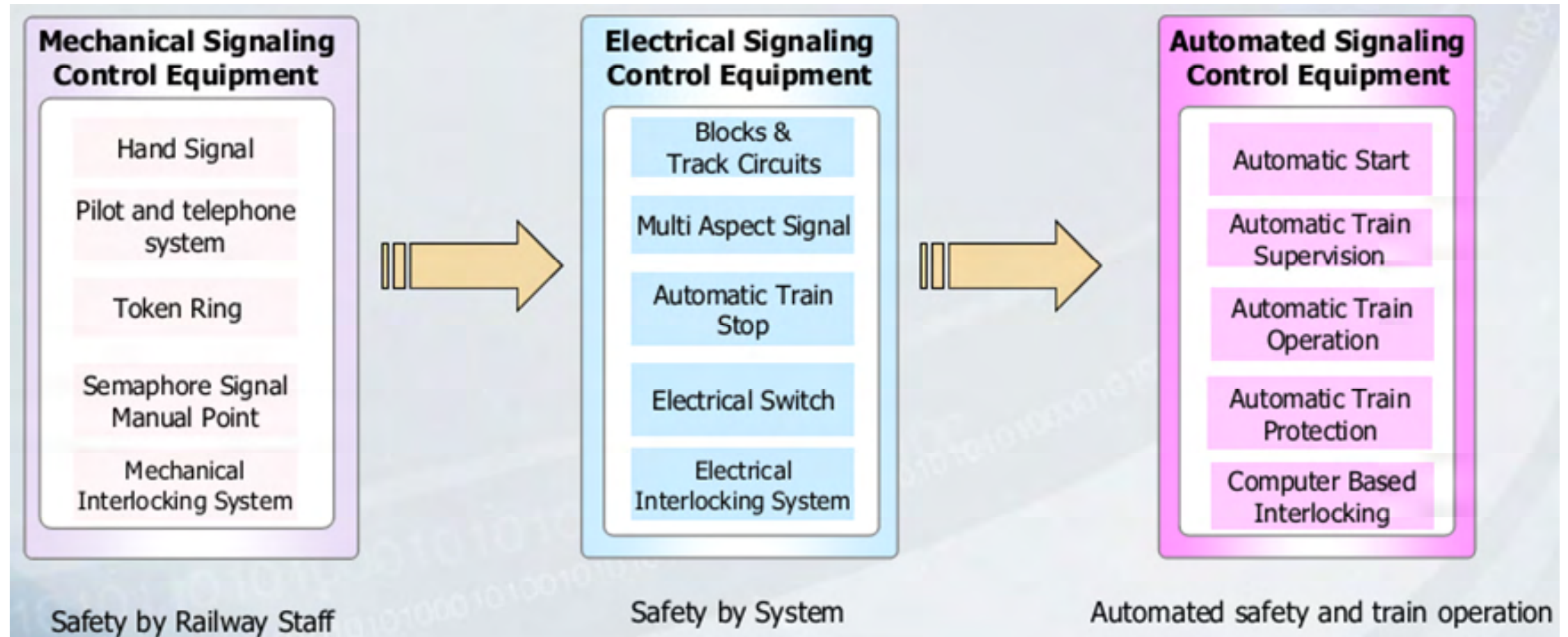
Introduction – Signaling : Why? What for?

- For years and years safety of railways was based on respect of operation procedures



But man is not safe...

Signaling Types / Mode



The development of signaling must not diminish the importance of the correct application of operational rules

Aims of railway signaling

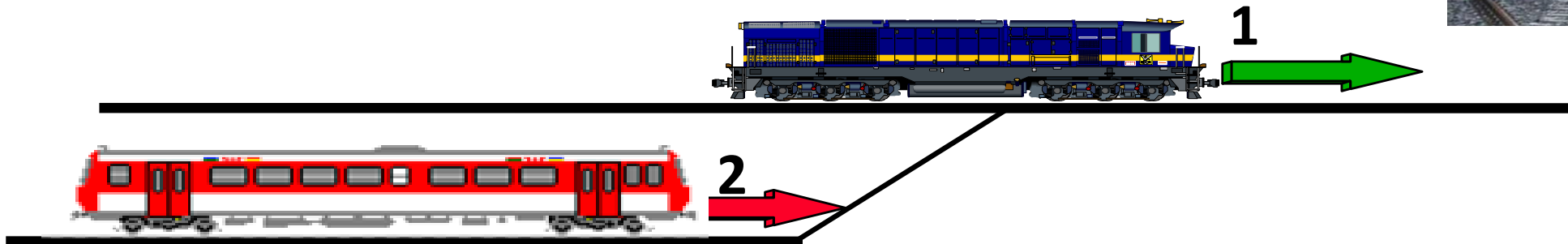
- 
- Secure a safety for train running
 - Increase the capacity of line
 - ✓ Improve the train speed
 - ✓ Reduce the headways
 - Increasing the efficiency for train operation
- Faster
 - To more operational
 - More efficient
 - Safer

The 5 reasons for railway signalling (1/5)

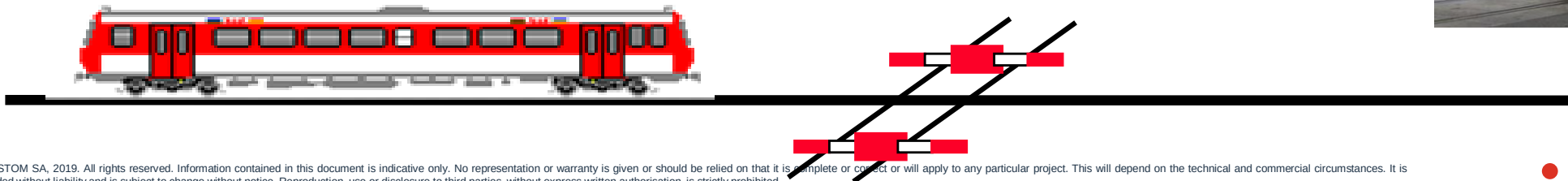
Priority

Conflicts between two vehicles

Conflicts between two railway routes that merge (flank protection)



Level crossings (and other crossings between different transport modes, such as with mobile bridges, ...)



The 5 reasons for railway signalling (2/5)

- Derailments and over-speeds

Speed



2013

Speed profile

100 km/h

50 km/h

160 km/h



2013

2013



The 5 reasons for railway signalling (3/5)

Catching

- Rear-end collisions

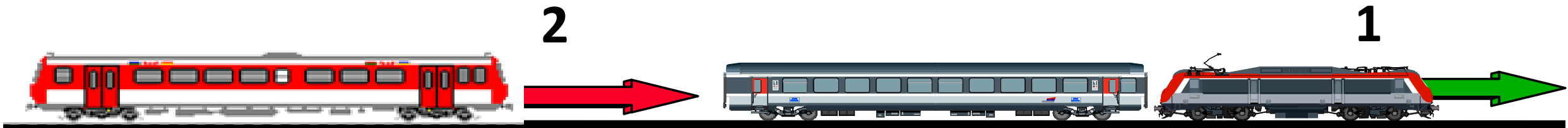


2009



2010

Two trains circulating on the same track, with the same direction



The 5 reasons for railway signalling (4/5)

Nose to Nose

- Head-on collisions

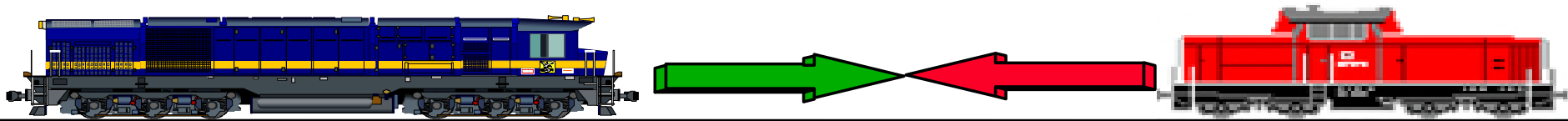


2013



2013

Two trains in opposite directions on the same track

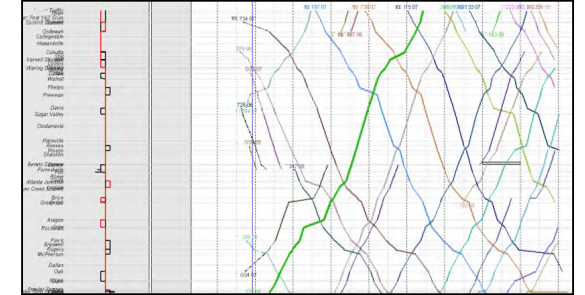


The 5 reasons for railway signalling (5/5)

- Traffic management

Time schedules, intervals, line throughput, priorities given to certain trains, etc ...

Traffic Mgt



Signaling importance

- When trains run on railway tracks they follow rules of operations in which safety plays a very important role.
- The most important rule in respect of safety is ensuring that two trains do not occupy the same position on the track at the same time.
- To make this rule work operation of trains uses signaling to control movement of trains on tracks and divides tracks into several sections which are protected by the signals.
- Transportation efficiency vs safety: **In case both cannot be achieved simultaneously we prefer to lost efficiency, but guarantee safety.**
- The need to ensure safety is always the topmost consideration and under no circumstances the signaling arrangement can compromise with this primary requirement.
- Fail-safe principle is critical for railway signaling systems.
- Railway accidents teach us the importance of signaling safety.

Controlling Train



Controlling Train.mp4

02

Railway Signalling system

Signaling is about operations



- Transporting efficiently passengers / freight from point A to point B enforcing
 - The best safety level (Safety management)
 - Train schedules (Traffic management)
- Whatever the conditions
 - Traffic density
 - Perturbations and failures

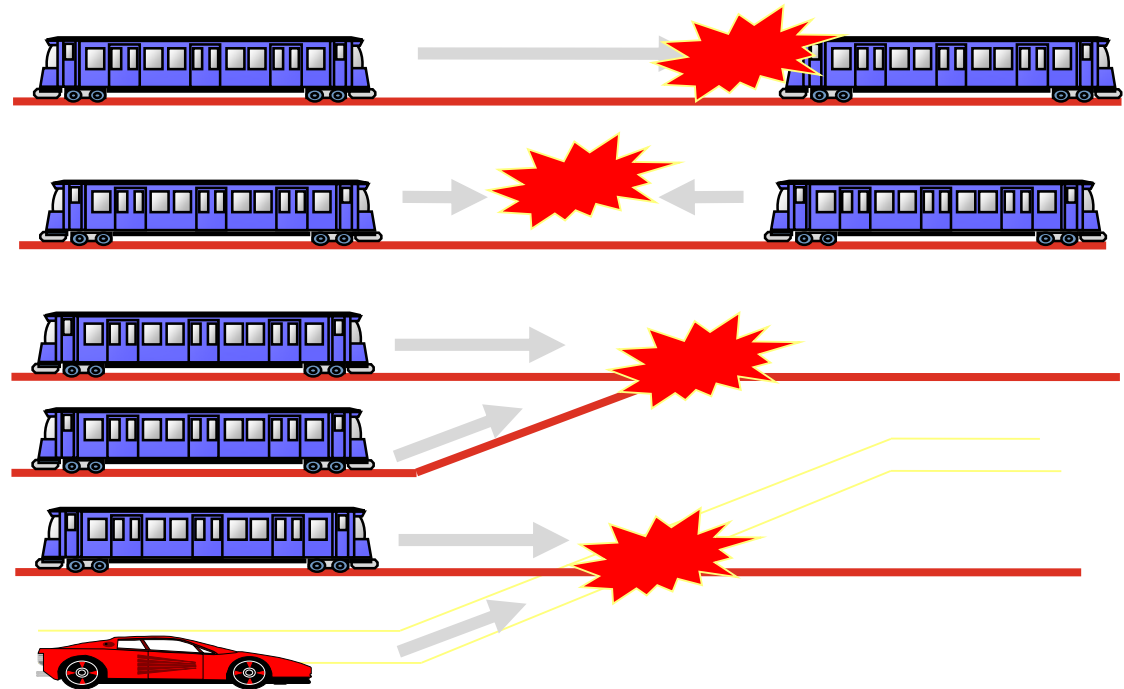
Prochains départs de Paris Nord				
	Num	Heure	Destination	Vole
ter	848535	18h10	SAINT-JUST	12
TOV	9051	18h13	LONDON	--
ter	847475	18h18	BEAUVAIS	21
ter	847853	18h19	COMPIEGNE	
TOV	7145	18h22	VALENCIENNES	14
TER	9355	18h25	AMSTERDAM	09
TER	9455	18h25	OOSTENDE	09
TOV	7281	18h28	DUNKERQUE-V	
TOV	7481	18h28	LILLE EUROPE	
	12039	18h34	AMIENS	

Signaling – Mission

Signaling basics

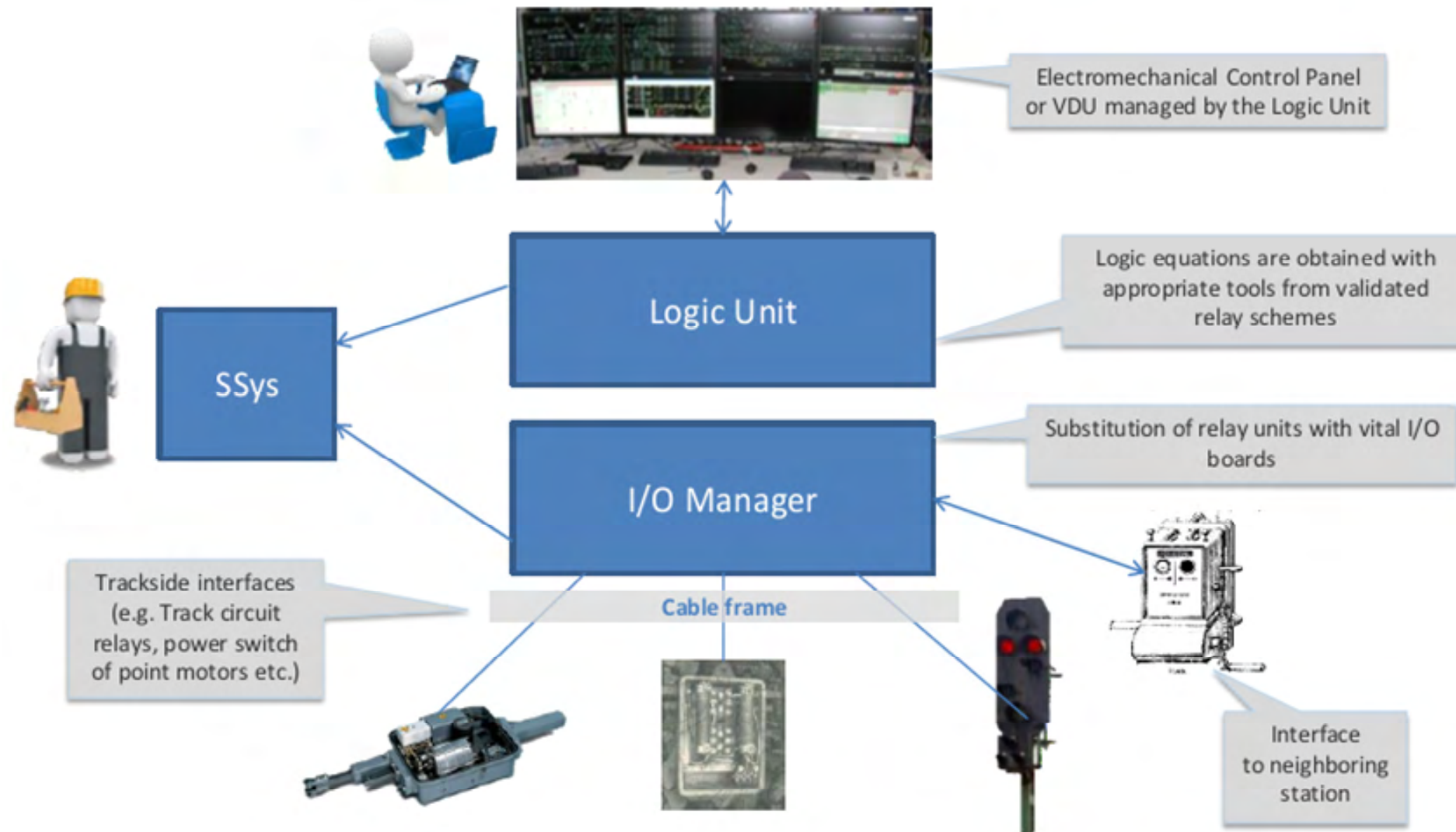
- A train movement is authorized only if there is zero risk of:

- Overtaking
- Front collision
- Merging or crossings conflicts
- etc...



- By design, no unsafe train movement can be controlled

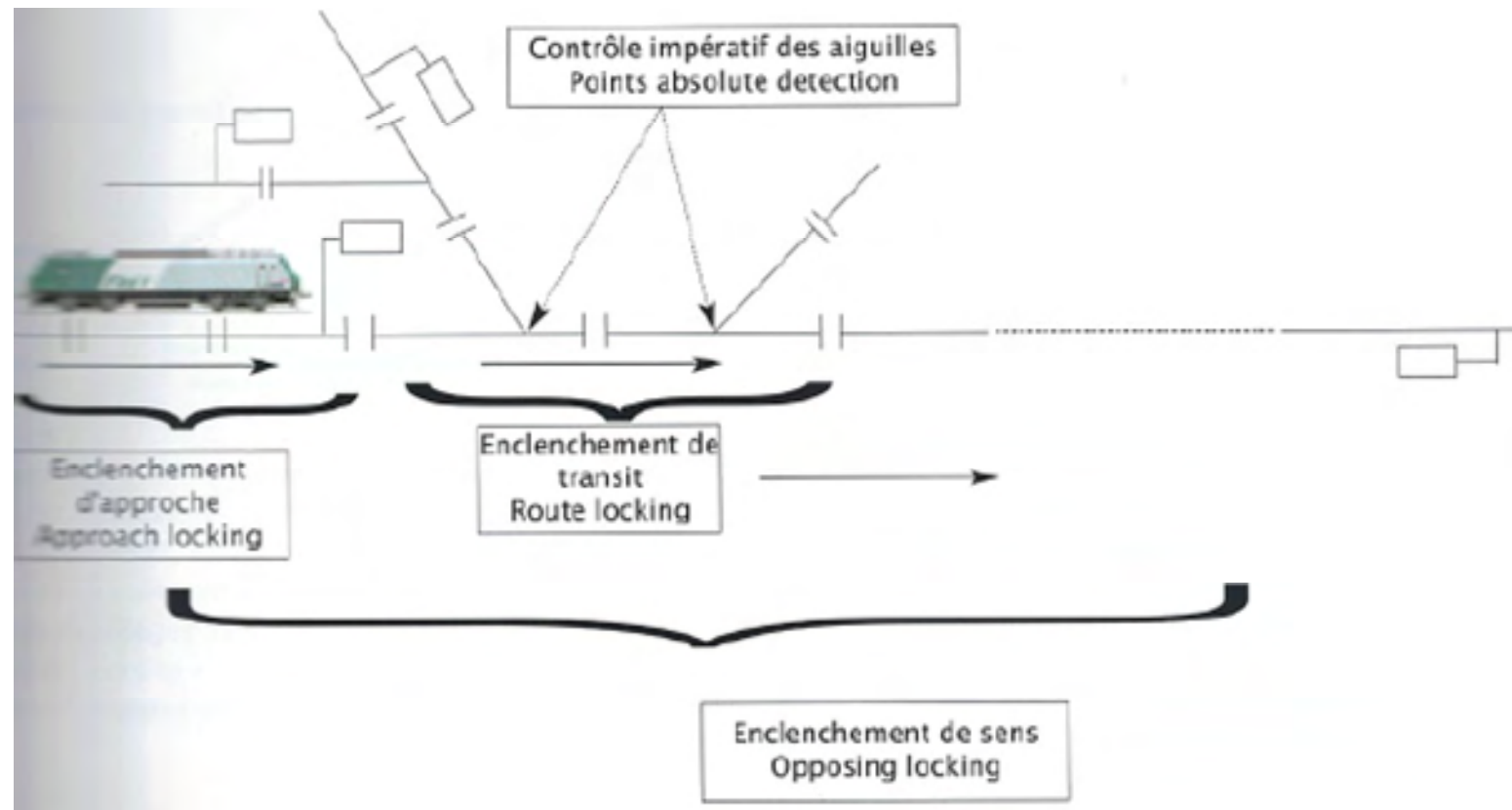
Run-time architecture of the Computer Based IXL (CBI)



Control of routes for trains

To allow train movement to take it is
Necessary to ensure that:

1. The route is free
2. The point are set, locked and detected
3. The route is protected against all conflicting movement (converging, crossing, opposing..)
4. And either that the conditions cannot change until the movement reach the end of route (German principle) or that each set of point within the route becomes free to be moved after the passage of train (French and British principles)



Point locking by train detection

Route locking: prohibits the movement of points until the 1st train clear the point of conflict or clear whole route with rigid route locking

Approach locking: prevent any modification of route set once the driver approaching could have seen clear aspect

Absolute detection of points: immobilized and locked

Opposite direction: locking opposite route “nose to nose”

Signaling – Mission

- Ensure **Safety** of people and trains, thanks to route control & Train protection management

- Reduced **operational costs** through traffic and asset management



- Ensure **Comfort** through traffic supervision, passenger information,...

- **Speed control** thanks to train management (automatic regulation)

Signaling is the **heart** of the transport system

Signaling Documentation Plan

signalling plans, (track plans, working sketches)

control tables

circuit drawings and books

track insulation plans (track bonding plans)

level crossing layout plans

installation drawings

equipment housing layout plans.

Signaling layout plan : Schematic view of signaling objects arrangement in railway station. Each separated object in this figure is provided with a unique identification code. The layout plan of the stations is considered as the **first stage of the interlocking design**, based on the operation requirements provided by the railway operator.

A “control table” is a functional specification of the signaling system of a railway section. It specifies the routes on which trains are allowed to pass. It contains the various safety aspects such as

- interlocking of conflicting routes,
- requirements of points for each route,
- the track circuit controls for the points,
- the route holding requirements such as approach locking and back or route locking and other controls such as crank handle controls, gate controls, block control and overlap release etc.

All possible and required routes in the stations, which are derived from the signaling layout plans, are represented in the route table of the stations. Generation and verification of control table is the design stage after the route table generation and before the wiring diagram designing in relay based interlocking systems (or software flowchart development in CBI). The format and contents of tables is not standardized, and may vary even within the same railway administration. Nevertheless, general principles of control table design are evident.

Control Table Format

Control Table for Points		Control/Location	
Point			
Set and Locked Normal By			
Set Only Normal By			
Set and Locked Reverse By			
Set Only Reverse By			
Route Locked Normal By	Track Circuit	After Using Route	Released By
Route Locked Reverse By			
Locked Normal for Overlap Maintenance By			
Locked Reverse for Overlap Maintenance By			
Remarks			

Control Table for Signals		Control/Location	
Signal		Route	
Commence		Finish	
First Proceed Aspect		Train Stop	
Auto-Reclearing		Auto-Normalising	
Requires Routes Normal			
Requires Routes Reverse		Requires Routes Clear	
Also Sets		Set By	
Sets, Locks and Detects Points Normal			
Sets, Locks and Detects Points Reverse			
Sets and Indirectly Locks Normal			
Sets and Indirectly Locks Reverse			
Sets Points in Sequence			
Requires Tracks Clear			
Clearing Delayed Until			
Requires Lever Stick			
Higher Aspects Require			
Route Locked By	Track Circuit	After Using Route	Released By
Approach Locked By	Track Circuit		Released By
Remarks			

Fail-safe principle

- A number of decades ago when relays were widely used in railway signaling systems. The fail-safe principle said that railway signaling had to guarantee the safety when any element in the system was faulty, including signals degraded running.



- All signaling must be designed in accordance with accepted railway fail safe principles for both mechanical and electrical equipment, e.g., the failure of any component is not to present an unsafe condition.
- be reliable but fail-safe such that any predictable type of failure of an item of signaling equipment will lead to a more rather than less restrictive operating condition.
- The fail-safe principle is to identify a critical fault and mask its effect until recovery is taken.
- A system is fail-safe if any critical state will transit to a safe state when any single critical fault occurs, which will be detected in time.
- The fail-safe requirement is that the probability of keeping train operation safe larger than (1-10⁻¹⁰), MTBF is typically 100 years.
- The shortest signal path principle for fail-safe is to guarantee the safety with highest reliability in critical situations when fault occurs in the system.

Safety Integrity Level (SIL)

Safety Integrity Level (SIL) are measures of the safety risk of a given process. It is incorrect to call a particular device “SIL 1” or “SIL 2.” This is **inaccurate** because the entire control loop must be taken into account. Technically, it is accurate to say a device is “**suitable for use within a given SIL environment.**” For example, suitable for use in a SIL 2 environment.”

SIL has three main purposes:

- to define the safety integrity requirements for a safety related system or function,
- to provide a guide to appropriate development processes,
- to provide a basis for claiming achievement of safety requirements

DEMAND MODE OF OPERATION		
Safety Integrity Level (SIL)	Average Probability of Failure on Demand	Risk Reduction
4	$\geq 10^{-5}$ to $<10^{-4}$	$>10,000$ to $\leq 100,000$
3	$\geq 10^{-4}$ to $<10^{-3}$	>1000 to $\leq 10,000$
2	$\geq 10^{-3}$ to $<10^{-2}$	>100 to ≤ 1000
1	$\geq 10^{-2}$ to $<10^{-1}$	>10 to ≤ 100

Safety integrity levels: probability of failure on demand

CONTINUOUS MODE OF OPERATION	
Safety Integrity Level (SIL)	Frequency of Dangerous Failures Per Hour
4	$\geq 10^{-9}$ to $<10^{-8}$
3	$\geq 10^{-8}$ to $<10^{-7}$
2	$\geq 10^{-7}$ to $<10^{-6}$
1	$\geq 10^{-6}$ to $<10^{-5}$

Safety integrity levels: frequency of dangerous failures per hour

Why use a Certified product?

A product certified for use within a given **SIL environment** offers several benefits to the customer.

- The ability to purchase a “Black Box” with respect to SIL requirements.
- Reliability calculations for such products are already performed and available to the end user. This can significantly cut lead times in the implementation of a SIL rated process.
- Additionally, the customer can rest assured that associated reliability statistics have been reviewed by a neutral third party.
- The most important benefit to using a certified product is that of the associated certification report. Each certified product carries with it a report from the certifying body. This report contains important information ranging from restrictions of use to diagnostics coverage within the certified device to reliability statistics.
- Additionally, ongoing testing requirements of the device are clearly outlined. A copy of the certification report should accompany any product certified for functional safety.

Most common causes of signaling failure

- Point failure
- Power failure
- Train/Track detection failure
- Extreme heat : Very high temperatures can affect signaling systems, causing them to shut down. They can also affect train detection systems.
- Faulty components: Faults in all types of electronic, electrical and mechanical components can result in a system failure. A complete system is made up of many thousands of individual components and duplication is not always possible.

Maintenance action levels				
Action levels	Associated Actions briefs	Associated actions features	Associated intervention features	Associated locations
1st level	Maintain or re establish system availability (interlocking, power supplies, trains)	Check, diagnose, adjust, change sub units, cleaning	Short durations (less than 30 min), light tools, individual tooling, test cases	- On site (track, interlocking) - Terminus station, - Inspection ditches, maintenance workshops (for on board equipment)
2nd level	Maintain or re establish sub unit availability	Check, diagnose, adjust, change component(s) cleaning	Durations and means may become quite important	- Equipment maintenance workshops - Centralised workshops - suppliers
3rd level	Maintain or re establish component availability	Check, diagnose, adjust, change component(s), cleaning	Specialised industrial means and tools	Devoted teams from : - Equipment maintenance workshops

03

Basic Components of Signalling system

AGENDA

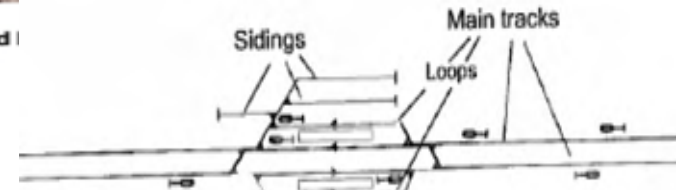
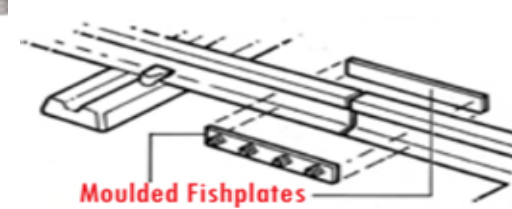
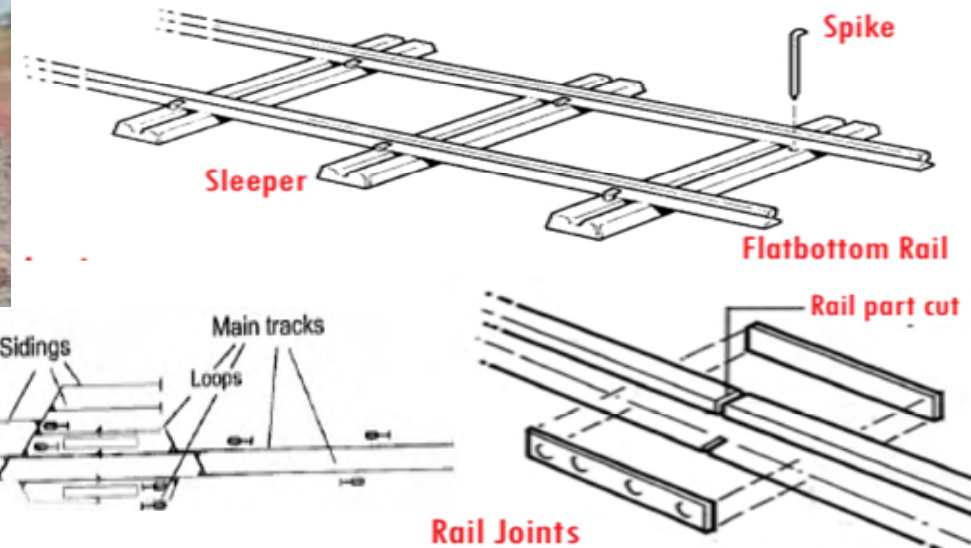
1. Train & Track
2. Signals
3. Level Crossings
4. Point Machine
5. Train detection
6. Interlocking
7. Automatic Train Control
8. Automatic Train Supervision

Train & Track



Figure 1: Typical railway track showing good width

Figure 6: Well laid track



Rail Joints

Moulded Fishplates

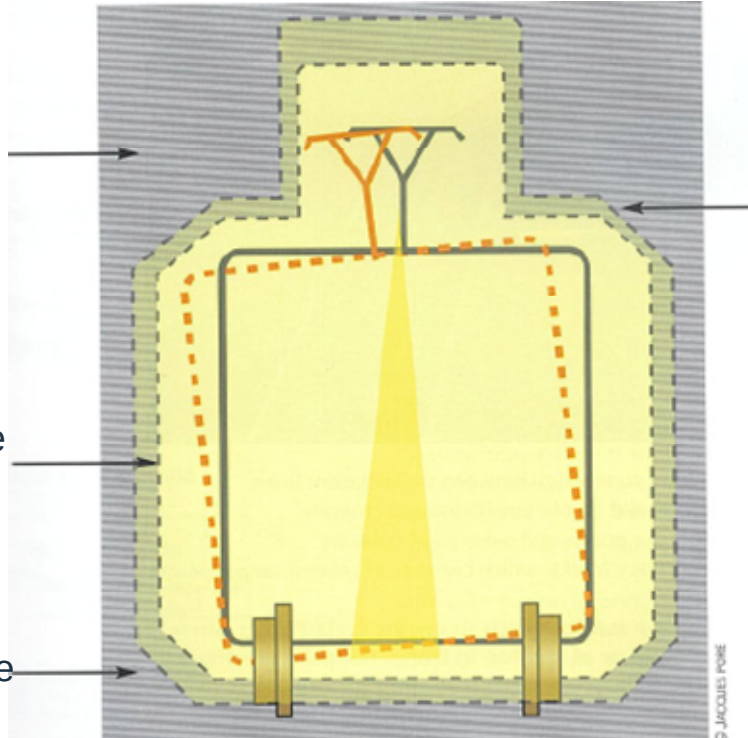
The Track

For Electrical vehicles

Infrastructure Gauge

Clearance Gauge reference

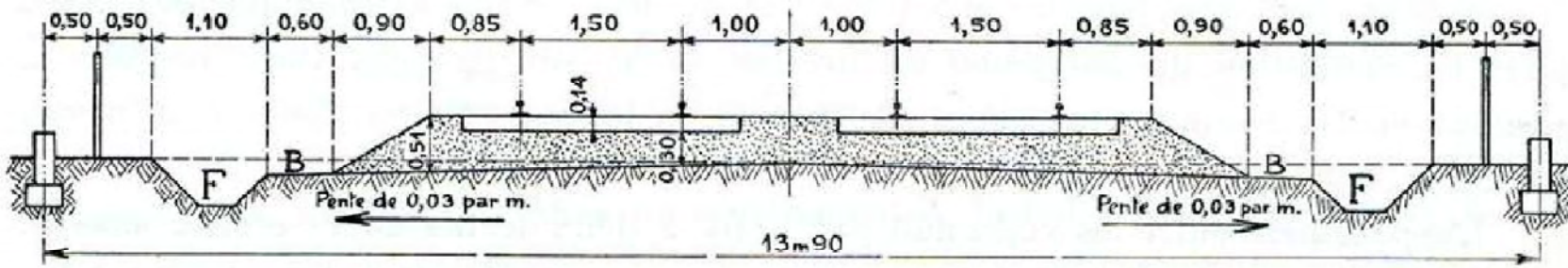
Low level Gauge



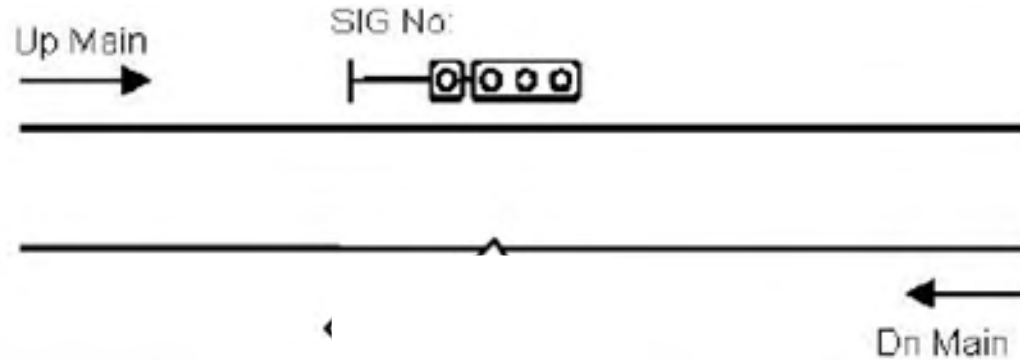
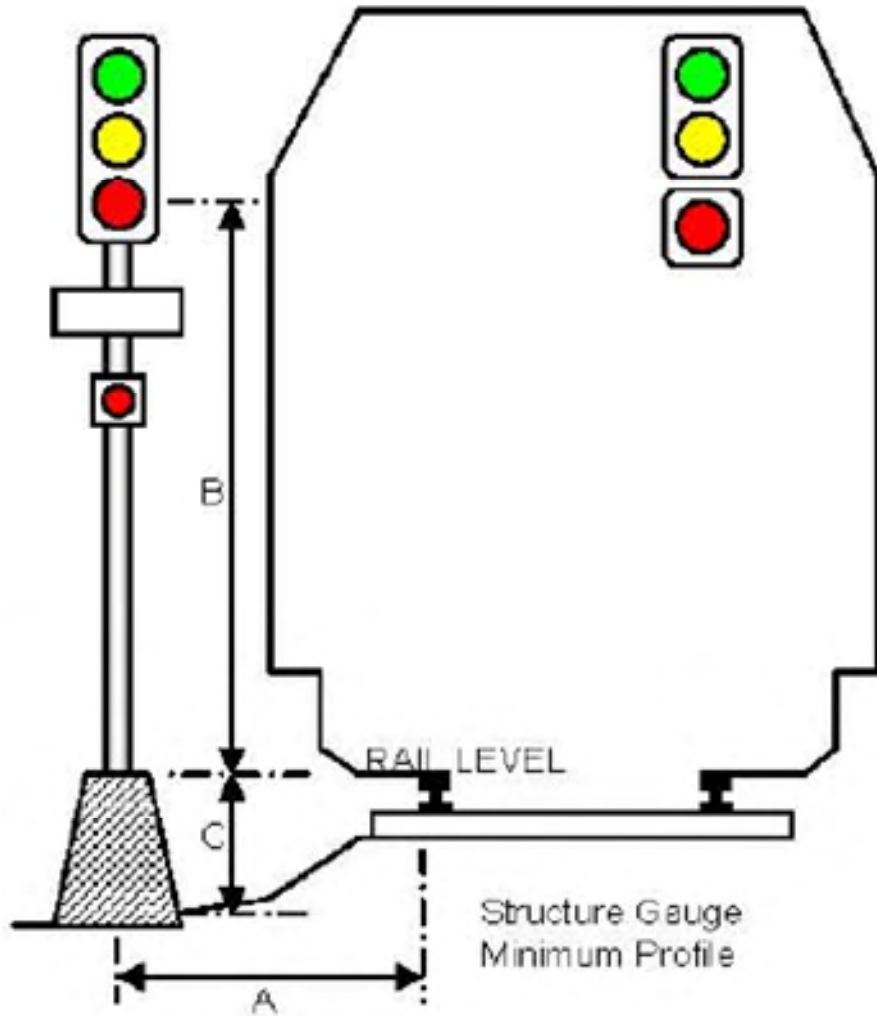
Insulated rail joint (IRJ)



Glued insulated rail joint (IRJ) São Paulo metro line 4.



Signals positions - Clearance

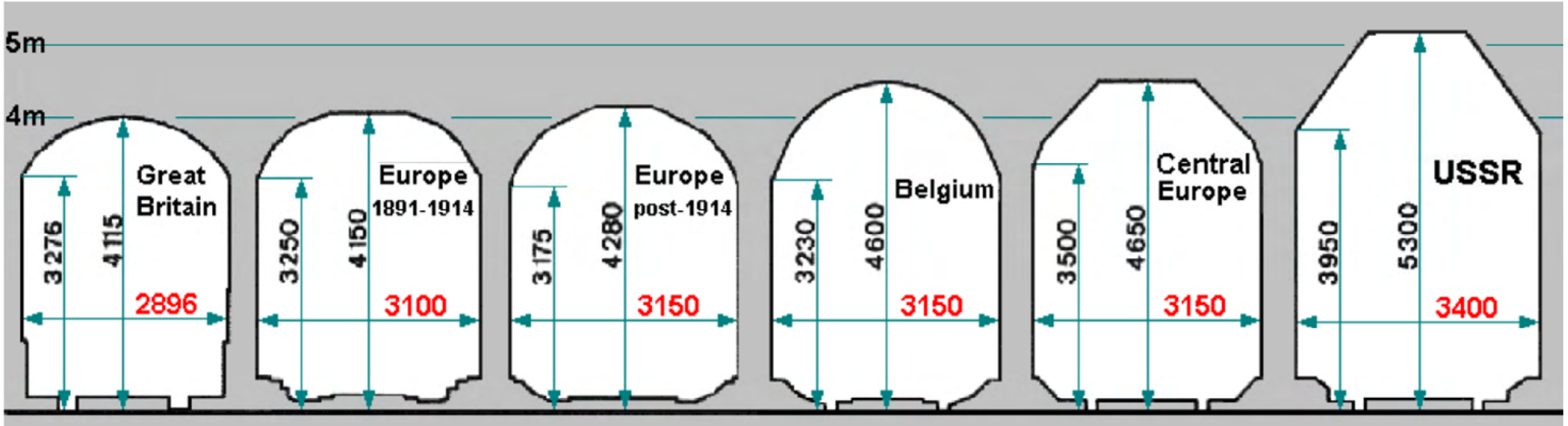


up means the **train** is moving in north direction and **down means** moving towards south direction.

The position of all signals, point indicators, Train Authority indicators and nominated signs (notice boards) will be considered during a initial site survey during the Preliminary Design Phase of the project

Signals must be positioned to afford structural clearance between all parts of the signal, its associated structure and screens, and adjacent lines using the kinematic envelope.

The Loading Gauges



Gauging frequently refers to gauges, a term whose original definition related to an envelope in which a vehicle is to remain, or within which a structure is not to intrude (vehicle and structure gauge, respectively, separated by a clearance).

The gauging principles are based upon a requirement for maintaining adequate clearance between the vehicle and adjacent structures, and maintaining adequate passing clearance between the vehicle and other vehicles operating on adjacent tracks.

Railway Ballast



Figure 1: Typical railway track showing good ballast height and width

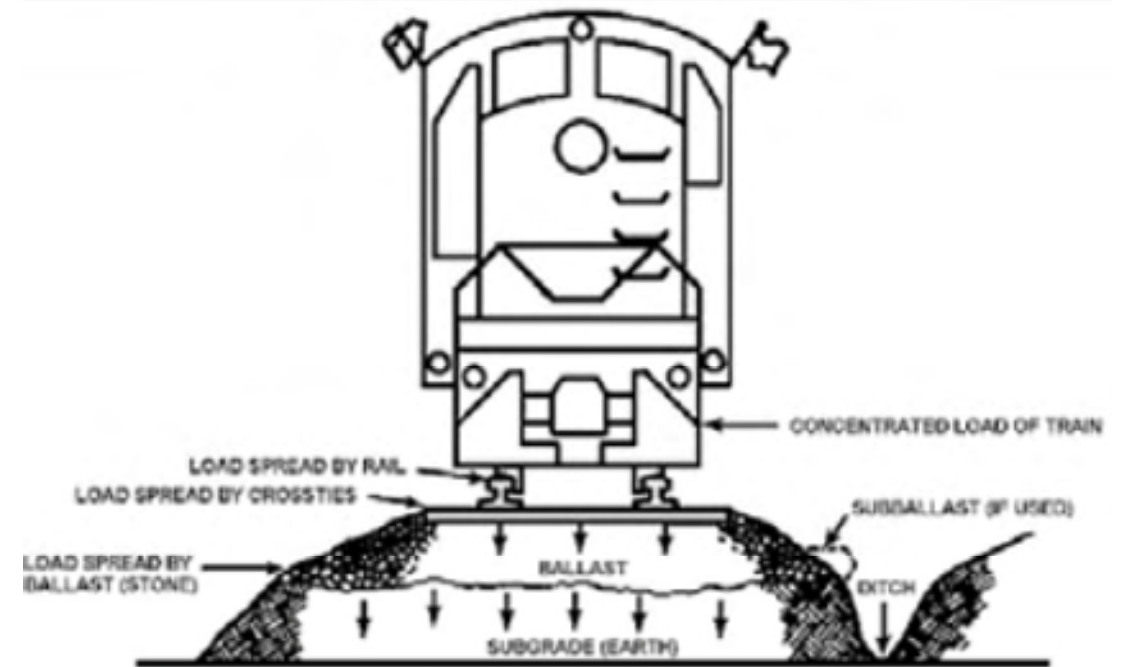


Figure 2: Typical construction layout

If ballast becomes contaminated with other materials, it can lose its ability to do some or all of his jobs. A common problem is fouling of rail ballast by coal spillages or other where the drainage ability of the ballast is affected. Track circuits with ballast contamination will be **shorter** than those with good ballast, thus needing more cut tracks. Conditions in the track bed (roadbed) which create stray electrical signals, such as muddy ballast (which can generate a "battery effect") or parasitic electrical currents from nearby power transmission lines.

Ballast Degradation



Water ponds
in wet weather

Ingress of vegetable



spillage from trains

Wet beds



Ballast completely
choked with sand



Measurement of ballast

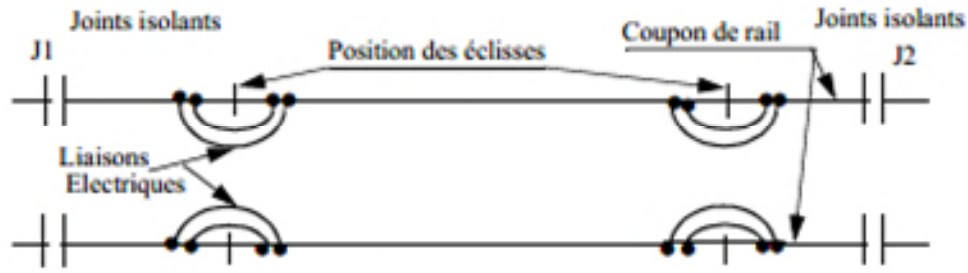


Figure 1

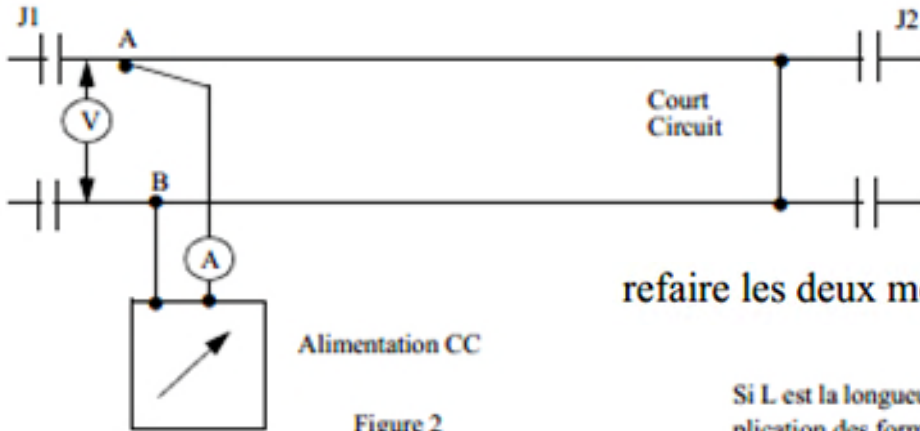


Figure 2

$$R_{1cc} = \frac{V_{1cc}}{I_{1cc}}$$

$$R_{2cc} = \frac{V_{2cc}}{I_{2cc}}$$

$$R_{cc} = \frac{R_{1cc} + R_{2cc}}{2}$$

refaire les deux mesures précédentes en inversant les polarités à la voie (I2cc, V2cc)

Si L est la longueur de voie en kilomètre entre J1 et J2 on obtient $R_{linéique}$ et $R_{ballast}$ par l'application des formules :

$$R_{linéique}(\Omega/km) = 1,15 \frac{\sqrt{R_{co} \times R_{cc}}}{L} \times \log \frac{\sqrt{\frac{R_{co}}{R_{cc}}} + 1}{\sqrt{\frac{R_{co}}{R_{cc}}} - 1}$$

$$R_{ballast}(\Omega \times km) = 0,87L \frac{\sqrt{R_{co} \times R_{cc}}}{\log \frac{\sqrt{\frac{R_{co}}{R_{cc}}} + 1}{\sqrt{\frac{R_{co}}{R_{cc}}} - 1}}$$

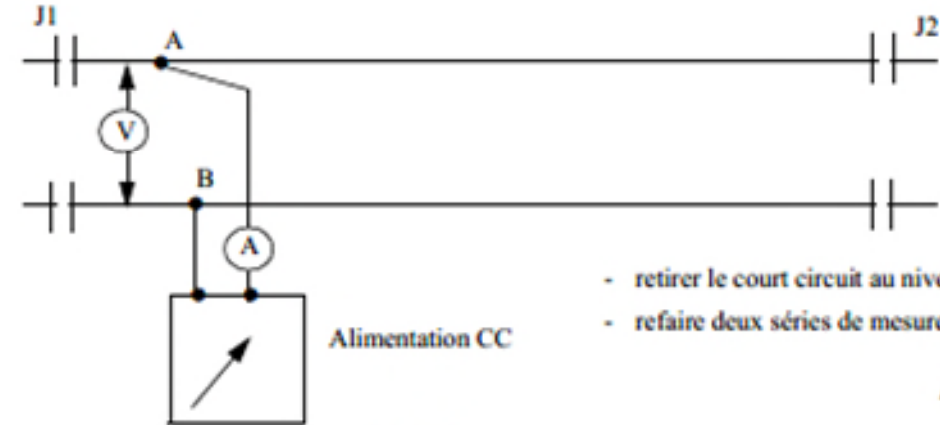


Figure 3

- retirer le court circuit au niveau des joints J2,
- refaire deux séries de mesures comme en 3.1

$$R_{1co} = \frac{V_{1co}}{I_{1co}}$$

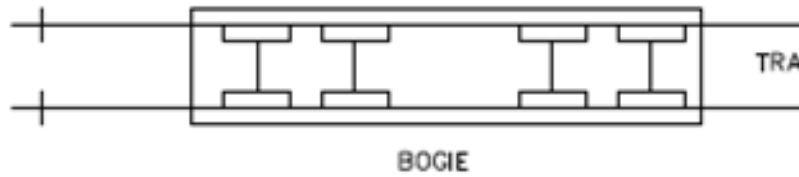
$$R_{2co} = \frac{V_{2co}}{I_{2co}}$$

$$R_{co} = \frac{R_{1co} + R_{2co}}{2}$$

Measuring the rail-to-rail, or ballast, resistance is important for proper setup and operation of track circuits and predicting their performance.

Basic Track Structure

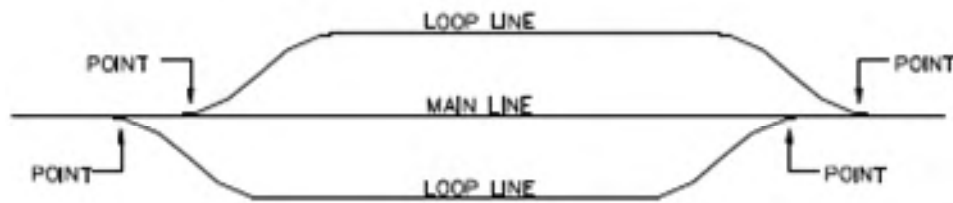
Trains run on dedicated line. A line consists of two rails running parallel to each other.



The track is represented in Signalling plans by a single line as shown below.

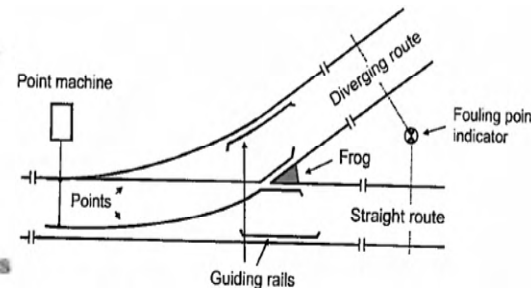
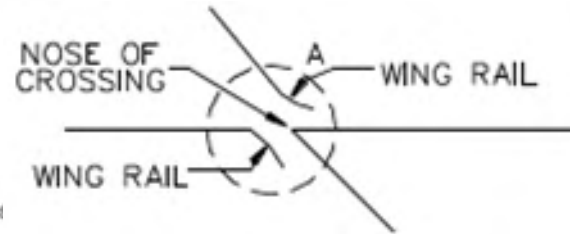
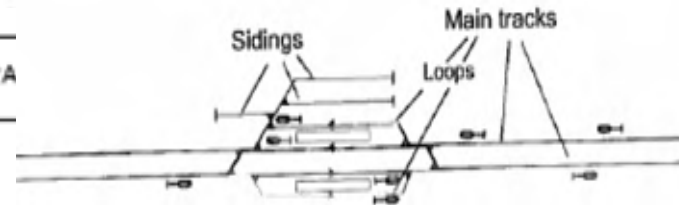


In station yards there will be more than one track for receiving and dispatching train.

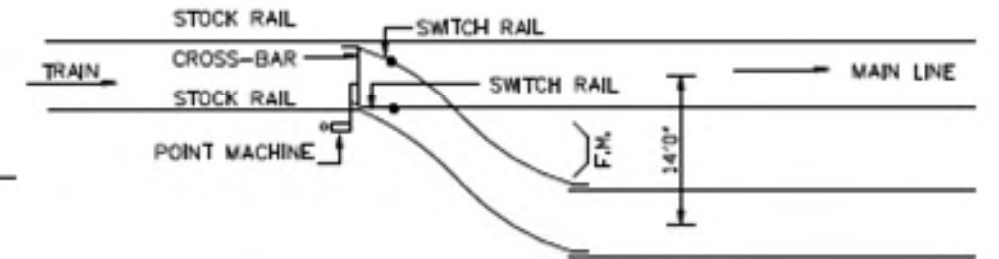
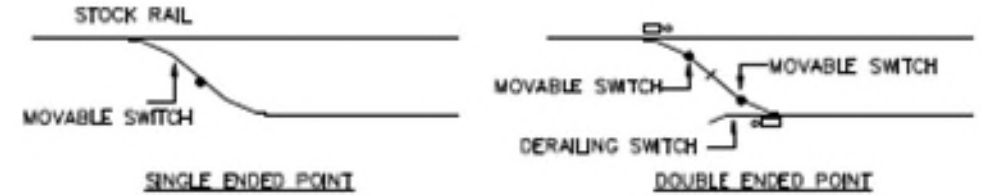


Points are provided to divert the running trains from one track to another. The points have movable switches which can be operated electrically by a point machine.

When a train drawn on a track, it should stand clear of the **FOULING MARK** as otherwise, the train may foul the movement of train on the adjacent track, thereby causing a side-collision.

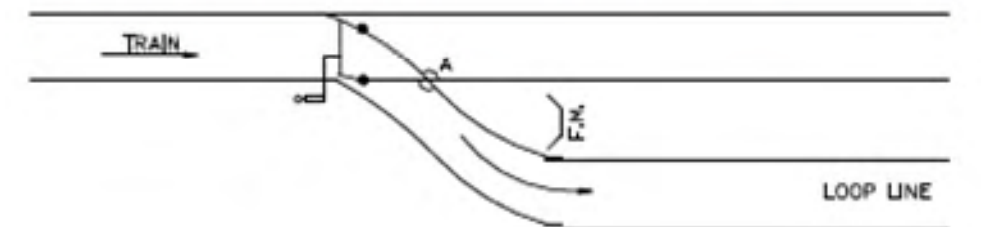


A point can be single-ended point or double-ended point depending on whether the movable switches are provided at one end or two ends of the point.



The two switch rails of the point are rigidly connected together by a cross bar so that they can be moved from one position to the other position together by the point machine. If the position of the switches is such that the train is moved on to the main line as shown above, then the point is said to be in its normal (N) position.

If the switch rails are moved to the other position as shown below so that the train is diverted to the loop line, then the point is said to be in its REVERSE (R) position.

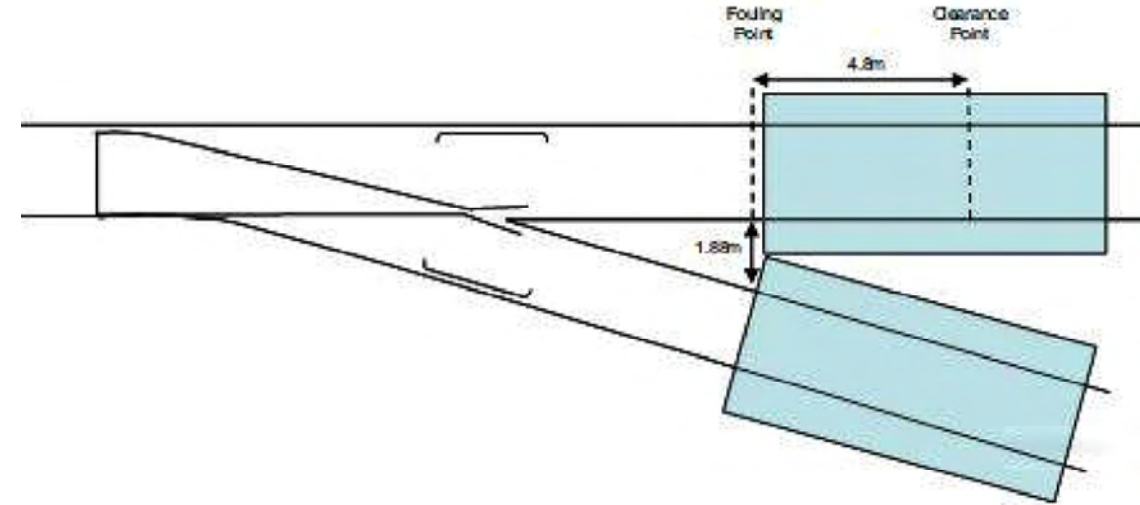


Fouling Mark



MEANING OF FOULING MARK

- Fouling mark means the mark at which the infringements of fixed dimensions occurs where two railway lines cross or join one another.
- Fouling Mark is a point to the rear of a converging junction, such train must be to the rear of that point in order to ensure that any train moving on the other converging line can proceed without being obstructed.
- Purpose is to avoid collision between two trains on adjacent track. (in between diverging / converging tracking rear t/in curve).
- It should be fixed at a point where the spacing between the tracks begins to reduce to less than the minimum as specified in schedule of dimensions (i.e. 4.72 Metre track centre normally).
- They are made up of stone or concrete blocks and painted in black and white.



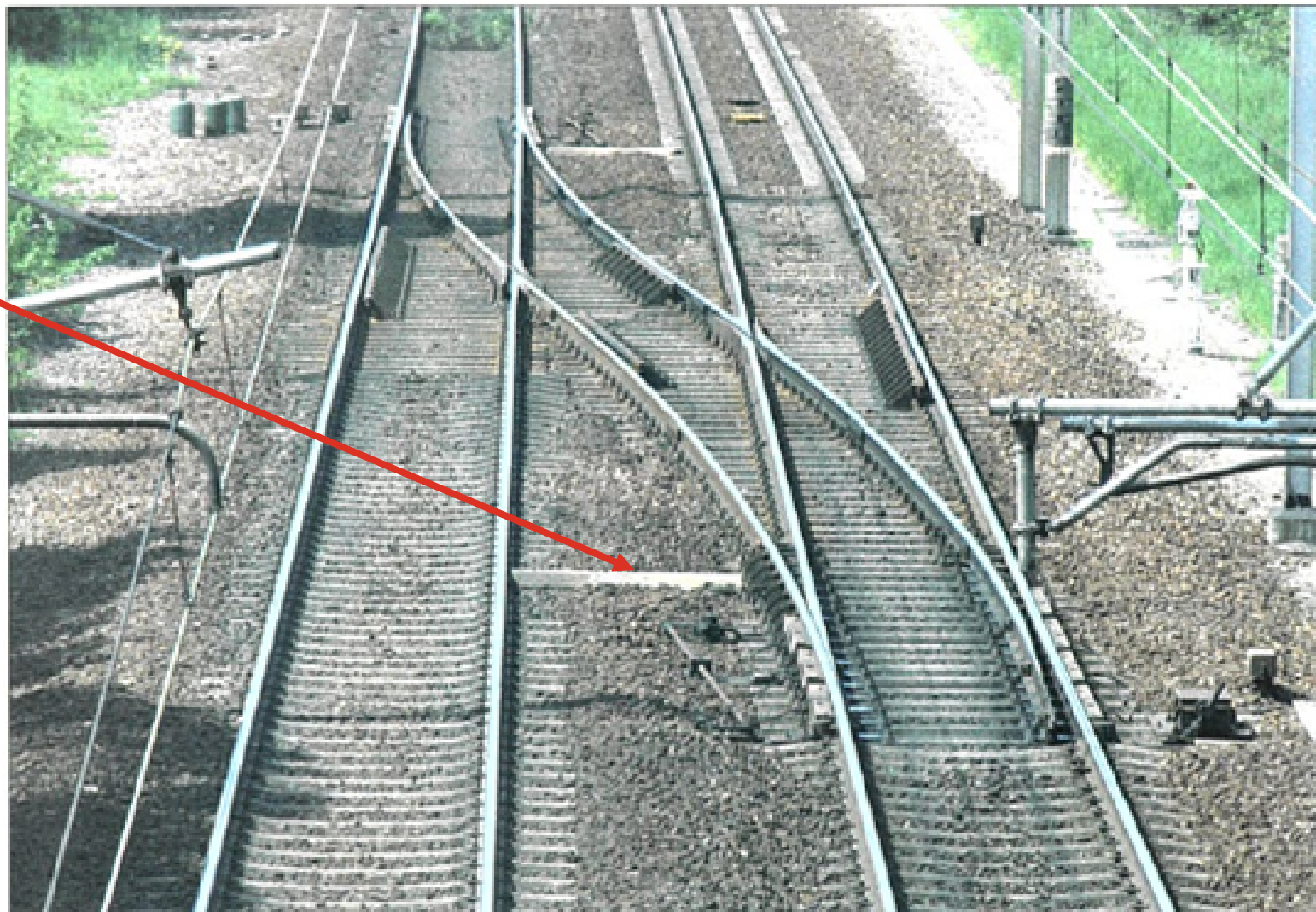
DETERMINATION OF FOULING MARK

The easiest way to determine this point, is to literally place a wagon on two adjacent lines and then gently move them along to the nearest point (turnout), until they literally physically touch. Make a note of that 'point of impact' (POI).

Having found the POI, move the wagons BACK along the tracks until they are a sufficient distance from each other so that they will not hit each other. Don't forget to include the couplers in this measurement, as they extend out beyond the ends of the wagons and can sometimes hit each other even though the wagon 'body' is further back. This 'safe' distance is usually 2-3 sleepers (ties) back from the POI.

Communication cross-over – Fouling Mark

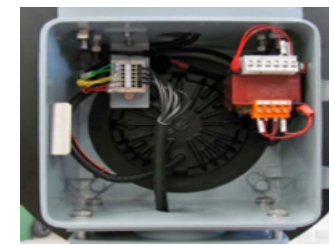
Sleeper paint in white used as fouling Mark



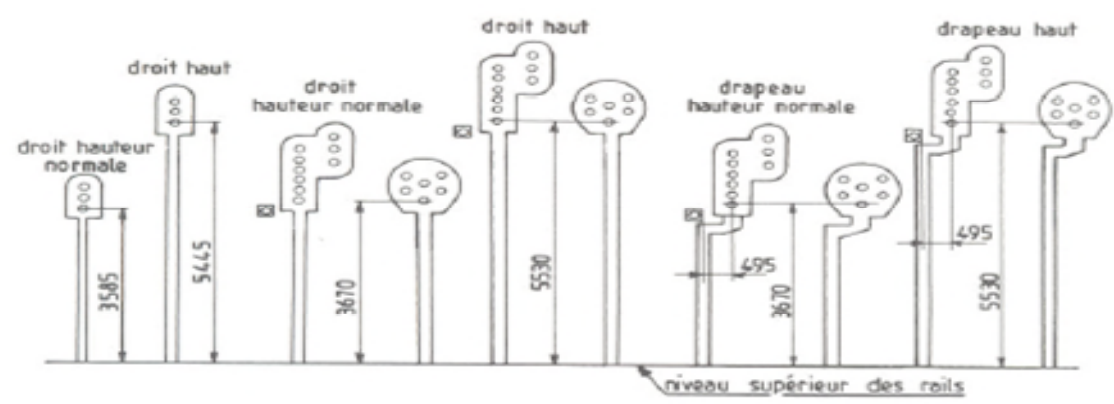
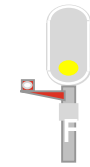
Signals

There is a great variety of different signalling systems in the world. Concerning the classification of signal aspects, these systems can be divided in two basic principles of signalling:

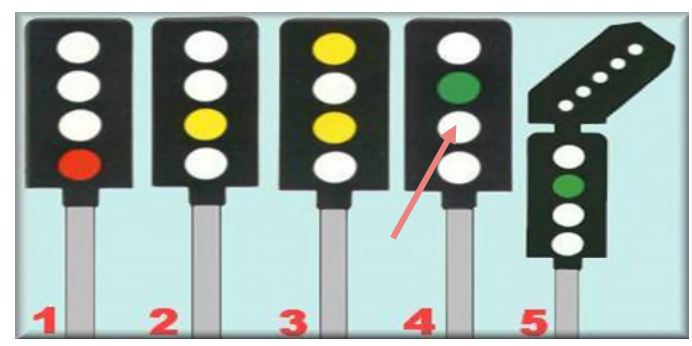
- speed signalling,
- route signalling.



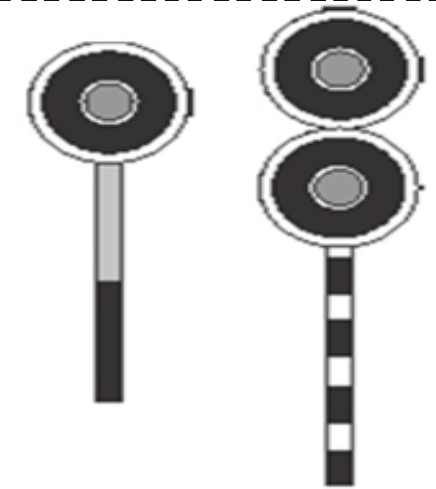
SIGNAL



French railway SNCF

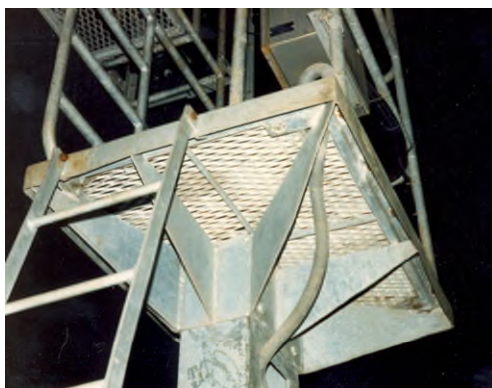


British Railway



Italian railway FS

Signals



Signal sub segments :

1. Mainline

- Main LED / Bulb
- Dwarf LED / Bulb
- Indicator LED / Bulb (residual)

2. Metro

- Main LED / Bulb
- Indicator LED / Bulb

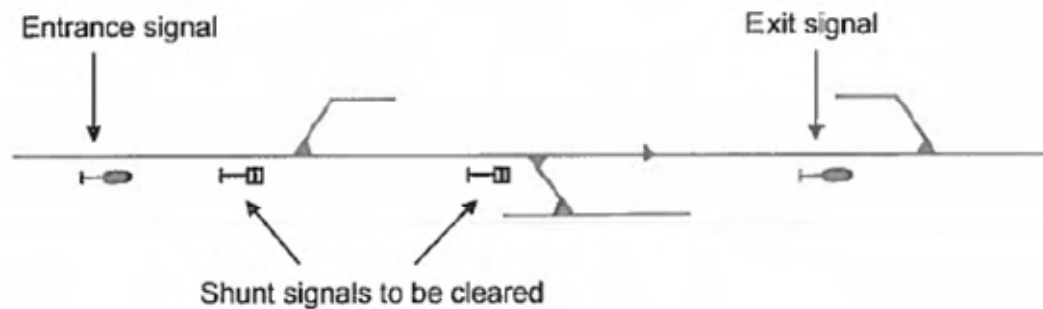
3. Level Xing LED



Shunt Signals

Shunt signals can only be found on railways where the operating procedures require a different signalling for trains and shunting movement. There, shunt signals are used to

- authorise shunting movements,
- provide flank protection.

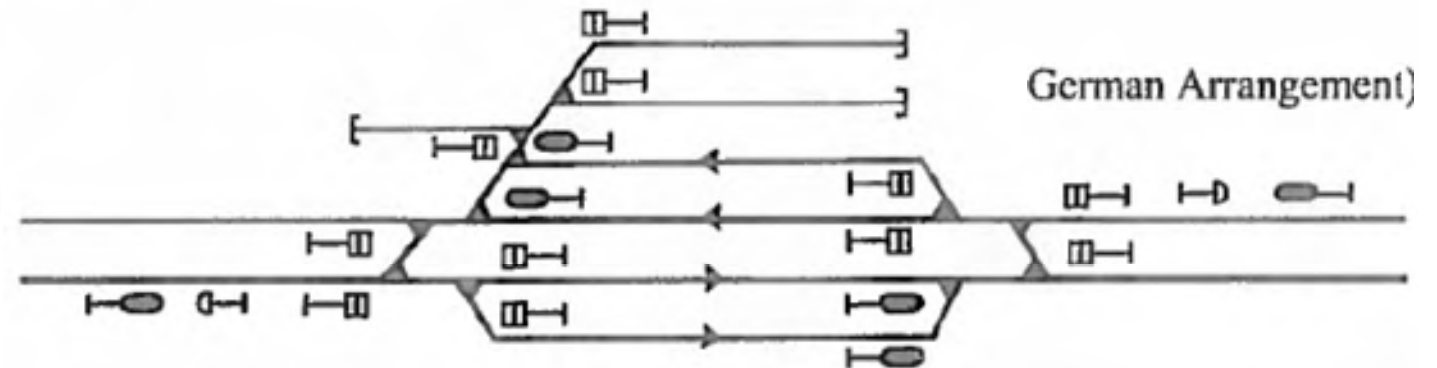


Except from authorising a shunting movement, a shunt signal must also be cleared when it is passed by a train route. The reason is to remove the stop aspect of the shunt signal when a train is passing

Shunt signals are placed at all points

- where flank protection is required at a track without an interlocking signal,
- at all locations where it is useful for the control of shunting operations.

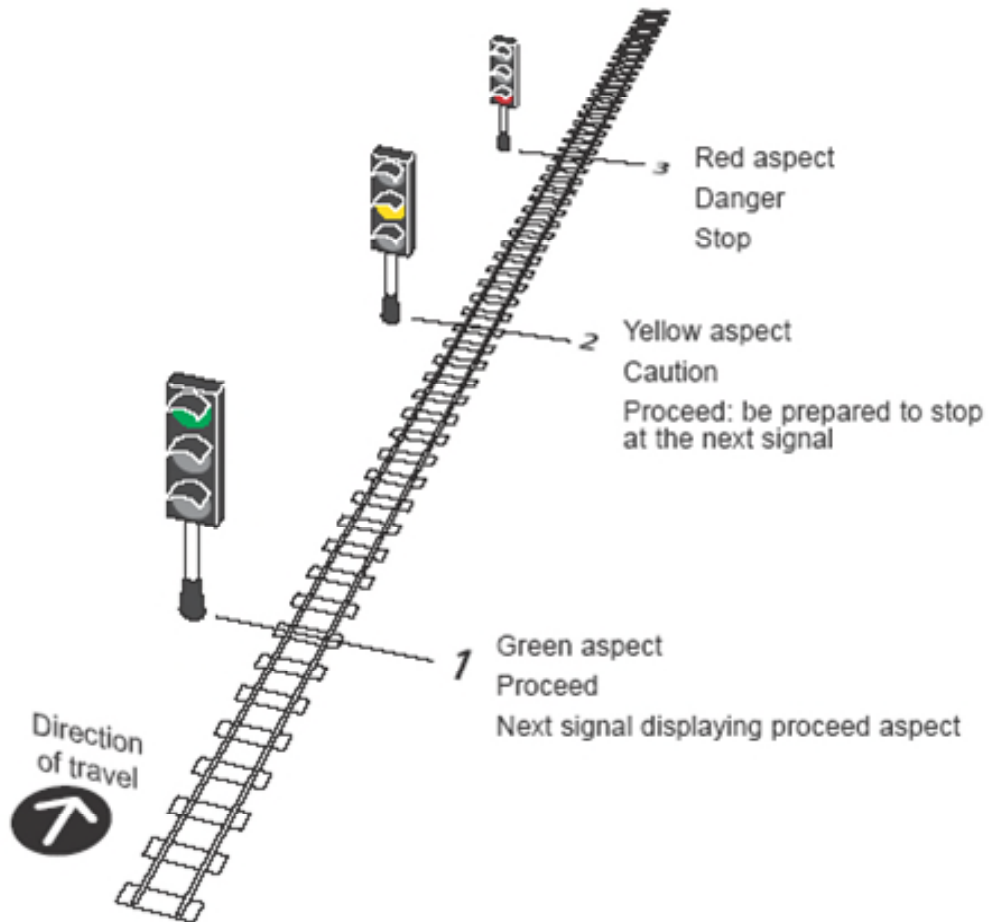
Shunt signals are not only used where shunting movements are governed on main tracks but also in sidings. In modern interlockings, there are a lot of shunt signals in the track layout because all shunting movements are carried out on interlocked shunt routes. Figure 4.20 shows a typical example of a track layout with shunt signals.



UK Signalling

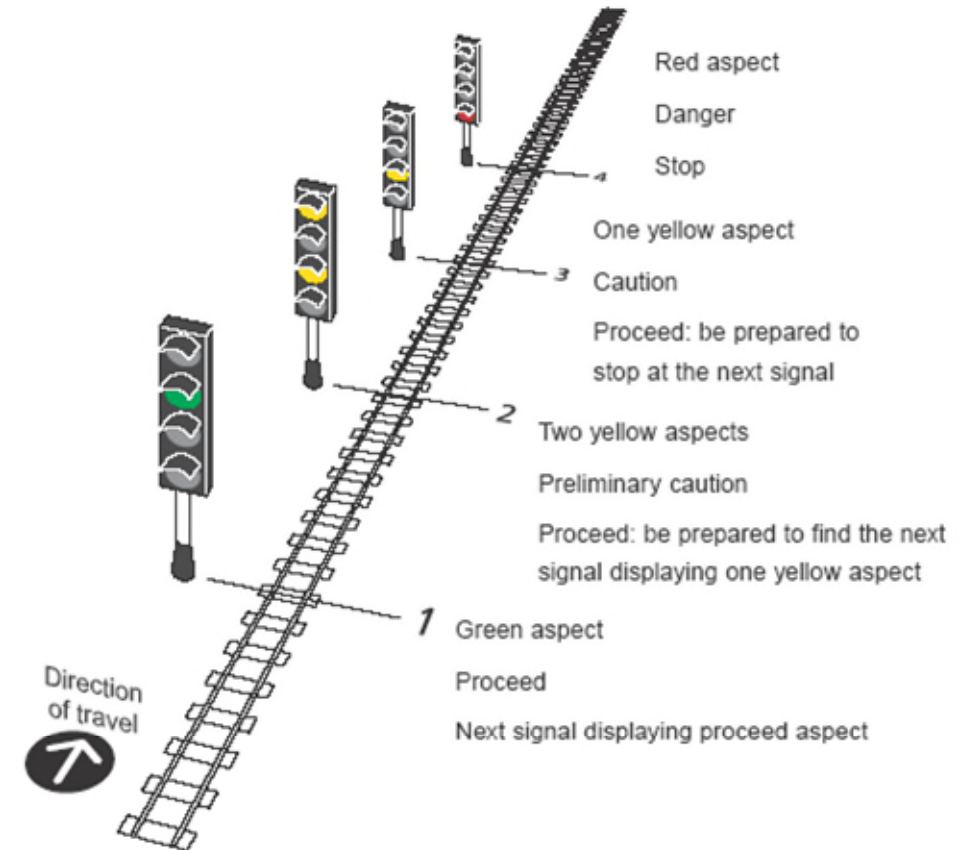
2.1 Three-aspect signalling - normal sequence

The normal sequence of three-aspect signalling is:



2.2 Four-aspect signalling - normal sequence

The normal sequence of four-aspect signalling is:

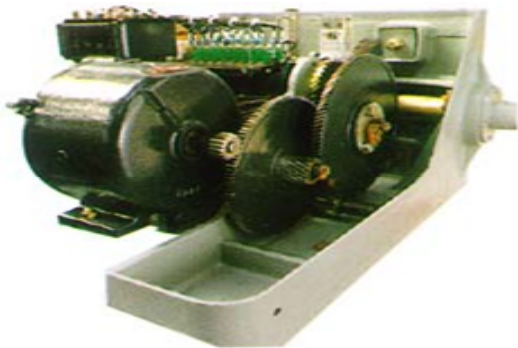


Level Crossings

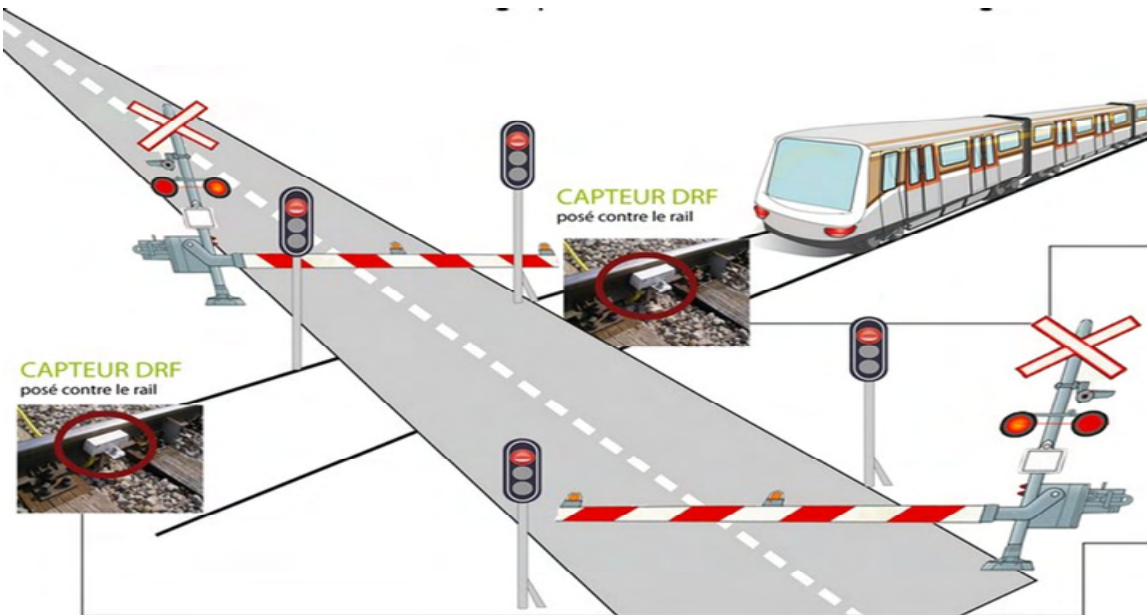
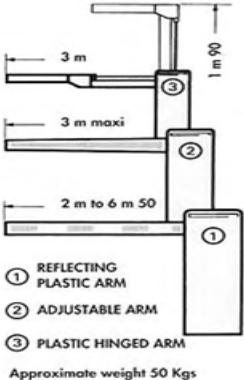
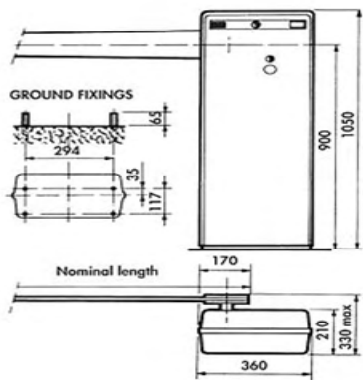


Introduction to Level Crossings.mp4

LEVEL CROSSING



Technical characteristics



Types of Level Crossings

The potential risk posed by a level crossing depends on the intensity of traffic on the railway and on the road. This risk is assessed by calculating the "**running time**" which is the product of the number of trains and the number of road vehicles using the crossing on average per day.

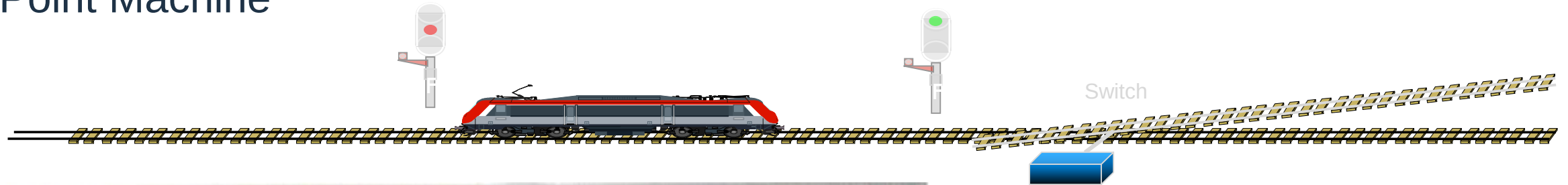
There are several types of level crossing:

- Guarded (always fitted with barriers)
- not guarded;
- lights and barriers (2 half-barriers)
- 4 half-barriers to close the entire road
- 2 complete barriers to close the entire road

The most common level crossings are LX Sound Signaling Automatic Light (LX SAL):

- **SAL0** : does not have barriers. low traffic. two or four flashing red road lights and an audible signal.
- **SAL2: most widespread**. flashing red road lights (two for each direction of traffic) , an audible signal and **two half-barriers** each blocking the right half of the road.
- **SAL2B** : equipped with **two barriers**, completely blocking the road.
- **SAL4: four half-barriers** (SAL 4) - flashing red road lights (two per direction of road traffic) an audible signal;
four half-barriers:
two half-barriers, called entry barriers,
two half-barriers, called exit barriers,

Point Machine



- POINT MACHINE**
1. Tip of the switches with locking mechanism.
 2. Multiple drive.
 3. Point operating mechanism.

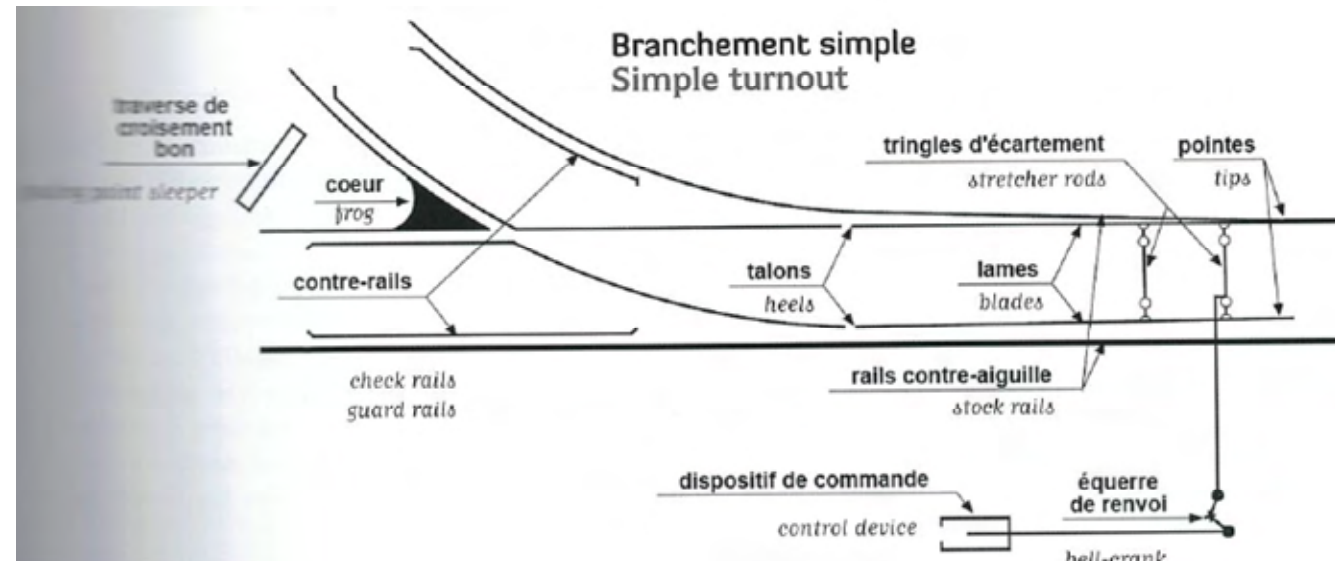
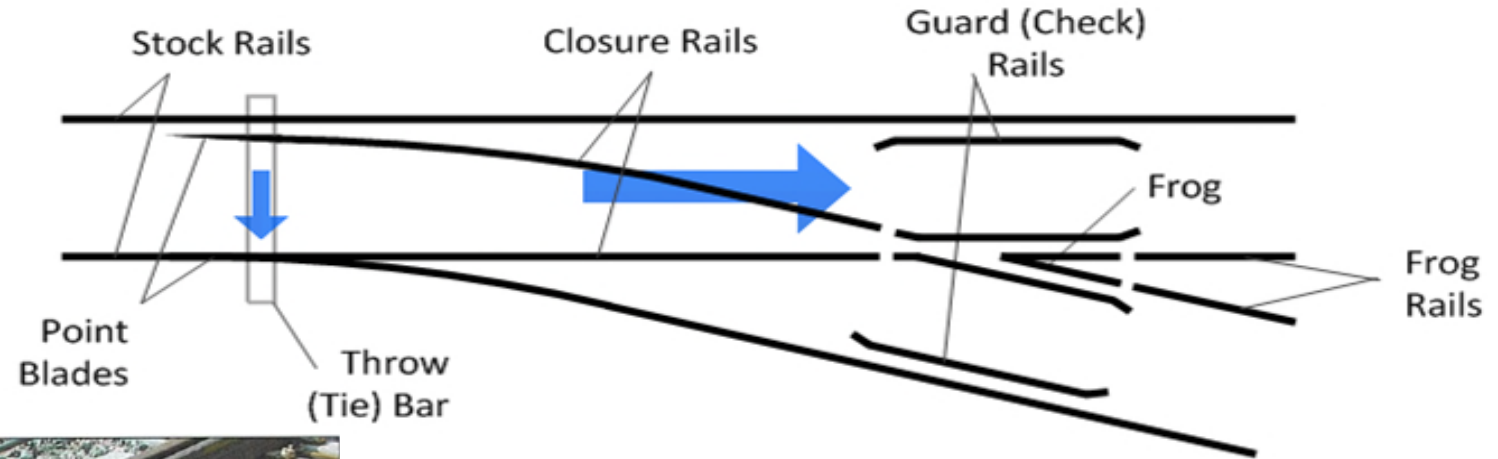
Point position detector.



Turnout is the device that allow the train changing direction

Turnouts are required to be:

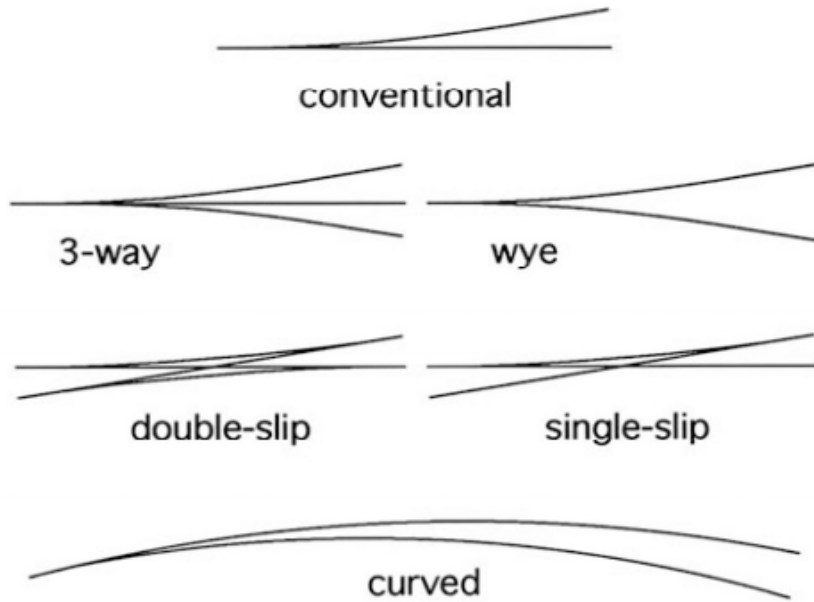
- Safe
- Functional
- Reliable
- Maintainable



Point machine is the actuation moving, locking and controlling

Example of Turnout Geometry

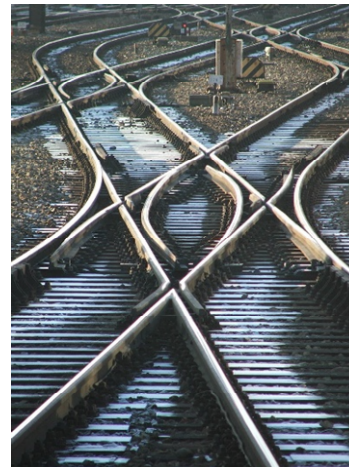
Some type of turnouts :



SINGLE SWITCH



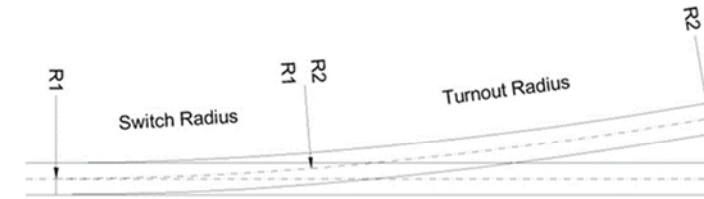
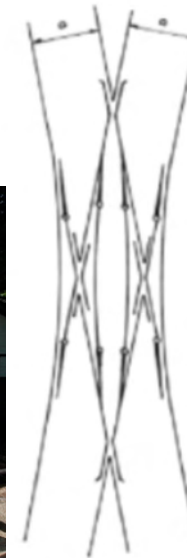
DOUBLE SLIP SWITCH



A three-way stub switch



'Double English'



wye switch (Y points)



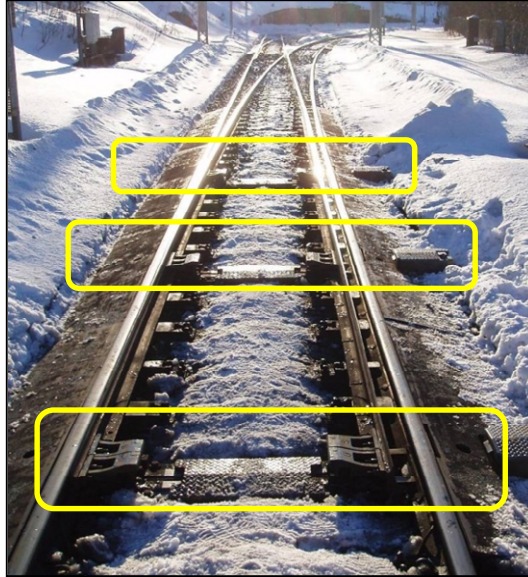
Among the 5 railways risks , 2 have a direct link with points : the side collision and the derailment

Number of manoeuvre points is based on Turnout TANGENT & turnout RADIUS

SINGLE POINT TURNOUT



MULTI-POINT TURNOUT

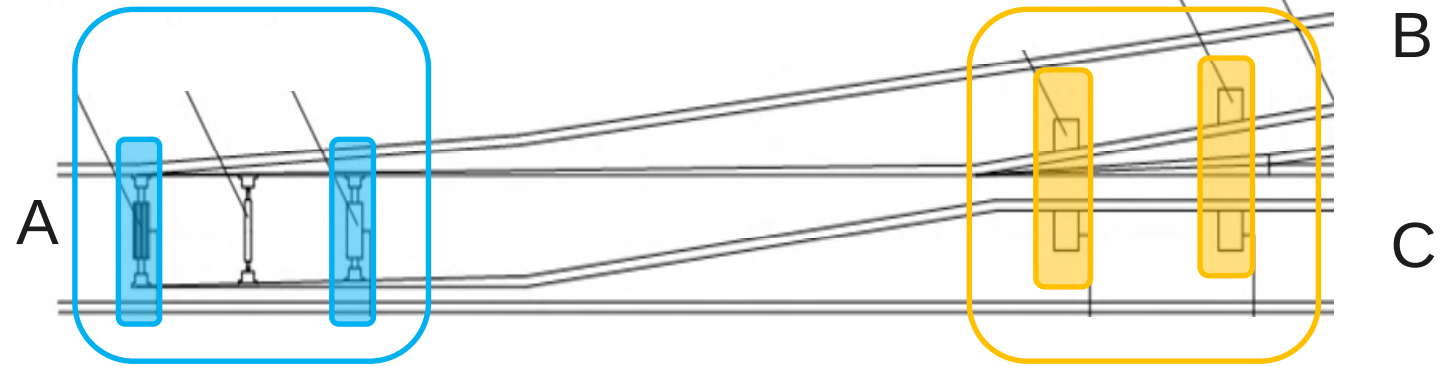
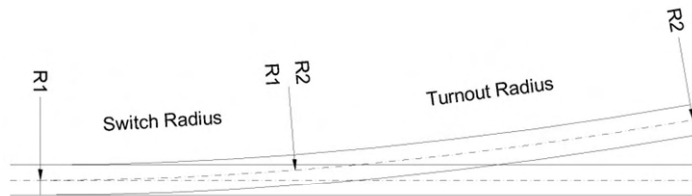


Switch Blade Drives

1 drive minimum

Frog Drives

fixed Frog → zero
mobile Frog → 1 or more

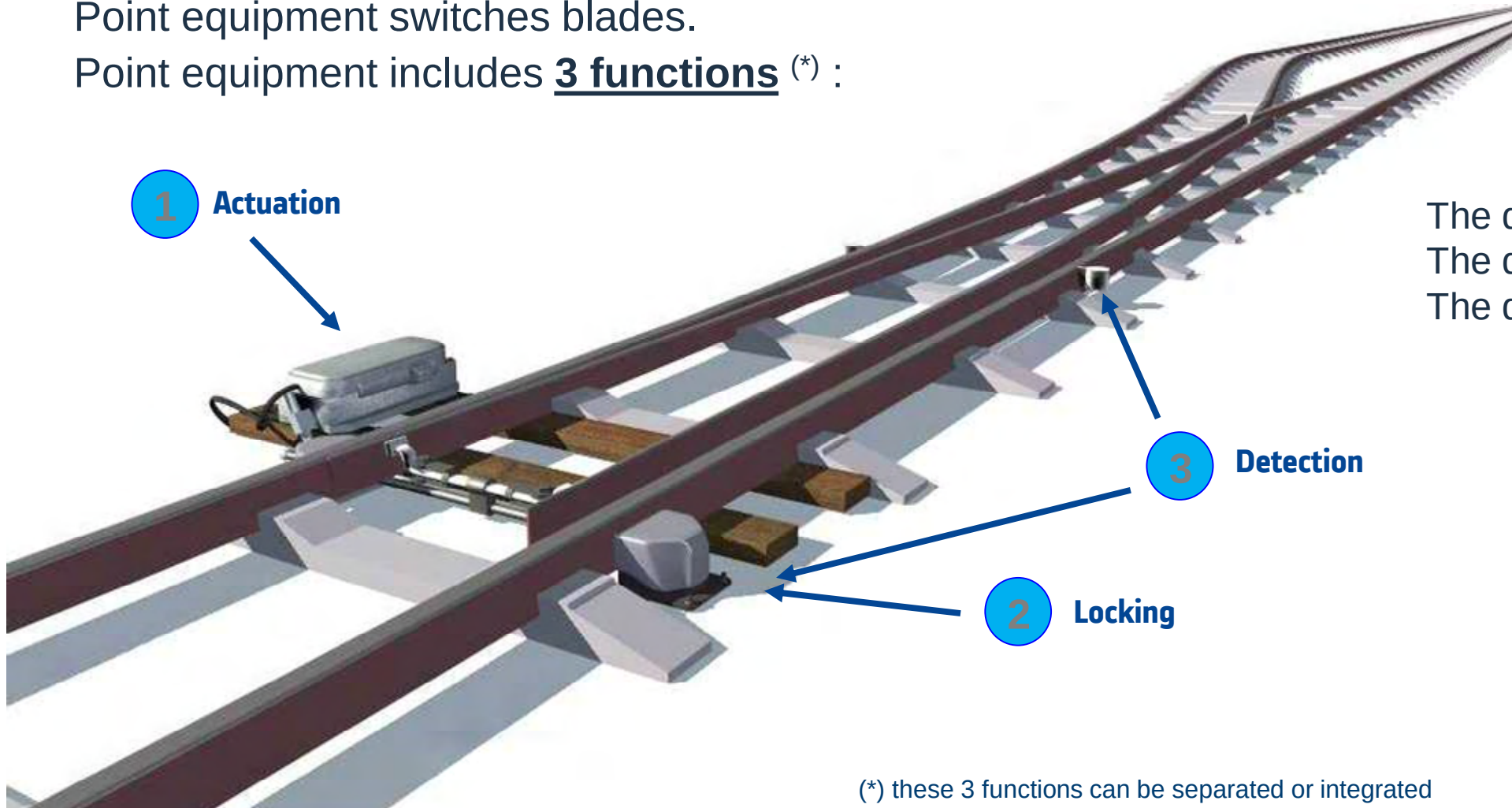


Example : Italy Turnout - tg 0,074 - based on SO → 2 drives Switch Blade + 2 drives Frog

Building the path with Point Machine

Point equipment switches blades.

Point equipment includes **3 functions** ^(*) :

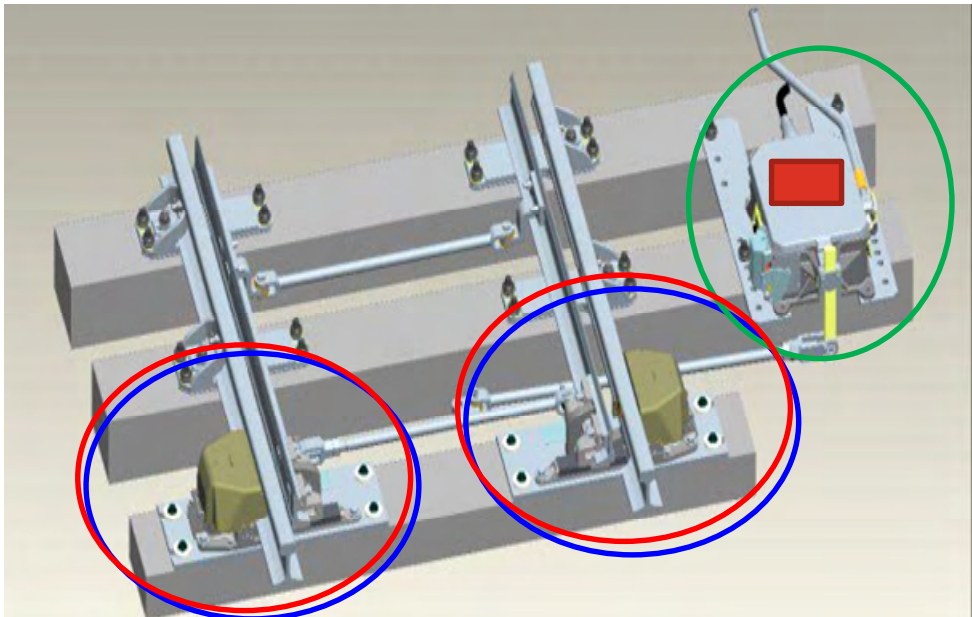


The drive: setting movement
The drive lock or blade lock
The detection of both blade

(*) these 3 functions can be separated or integrated

Internal Lock vs External Lock

External lock (example)

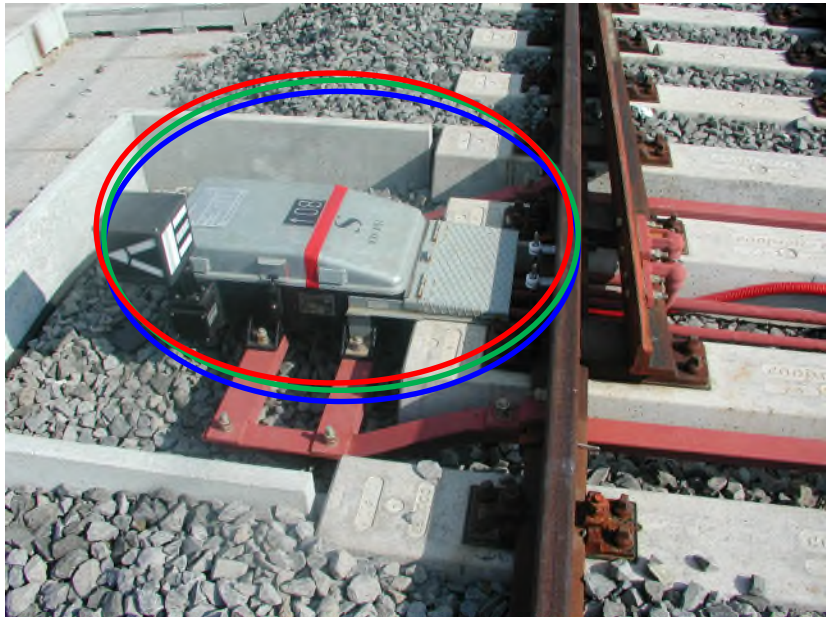


Separated functions

Point Machine
External Detection Function
External Locking Device

— Motor
— Detection
— Locking

Internal lock (example)

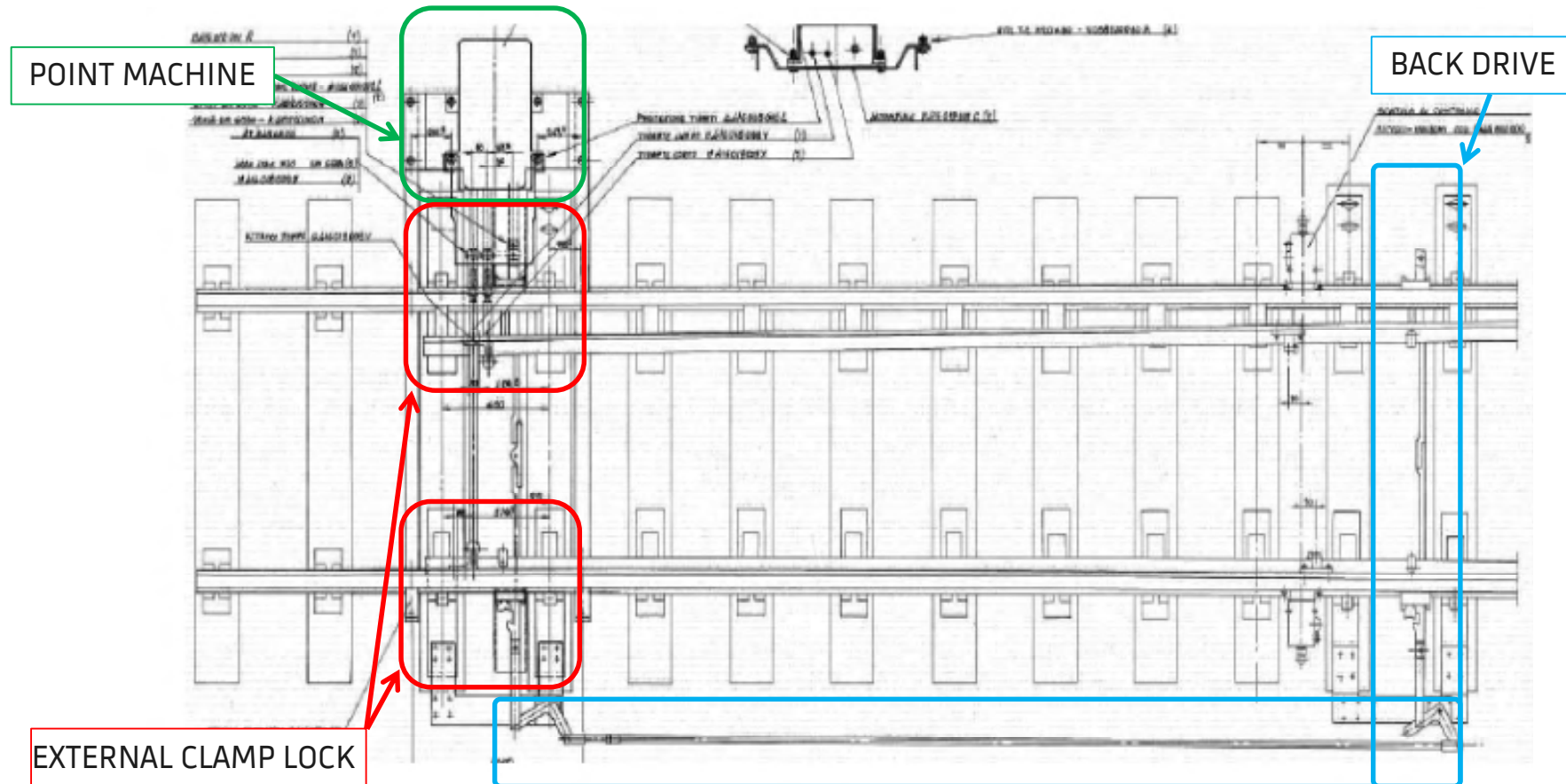


All-in-one functions

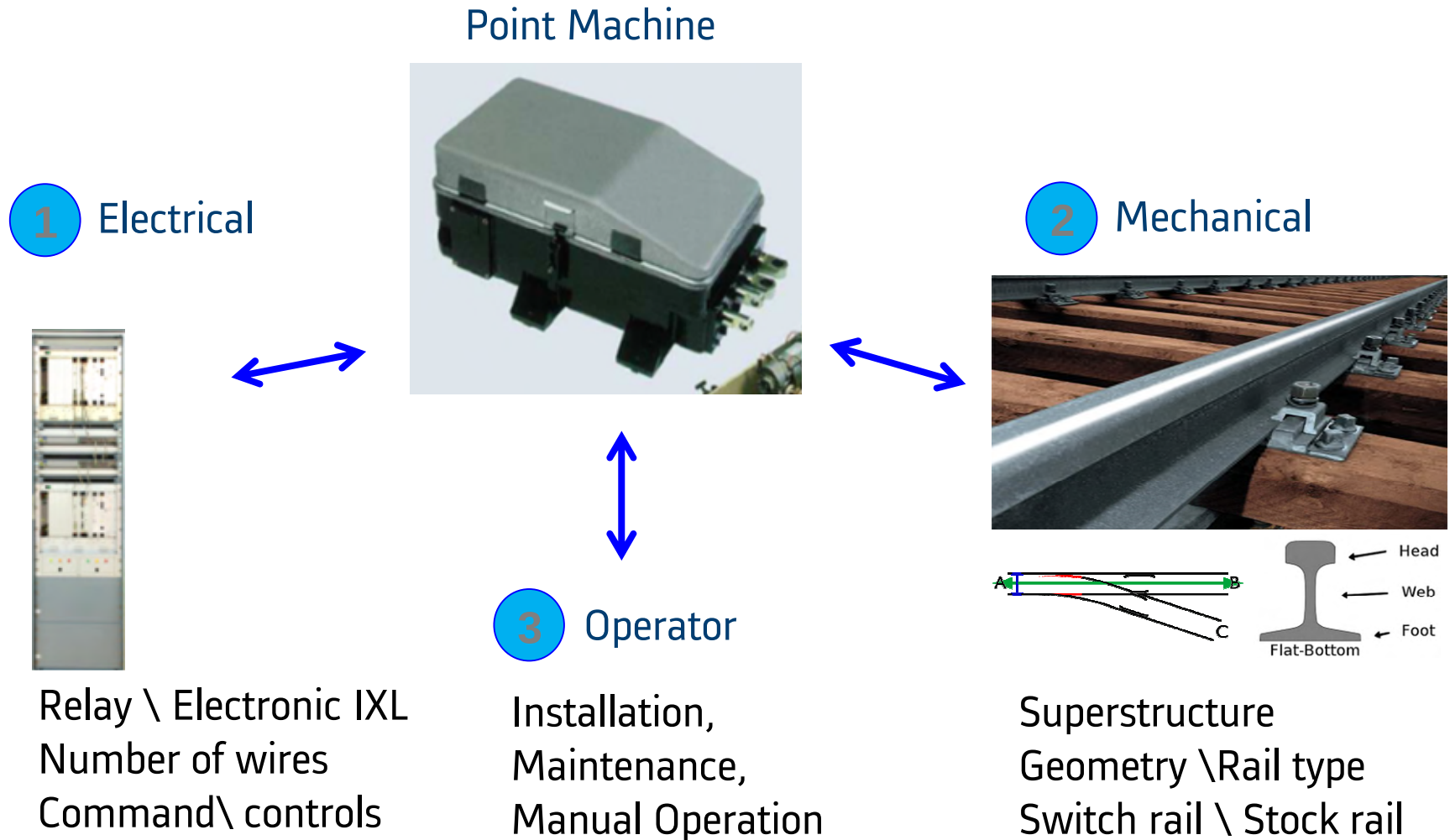
Internal Motor
Internal Detection
Internal Locking Device

Back Drive

Example of multi-drive turnout in back drive mode and with external clamp lock.



Key Interfaces



Speed limits on turnouts depending on the tangent

Désignation des appareils Type of points			VITESSE de franchissement km/h
Branchement à déviation Turnout to the left or right	tg	0,0154 (1/65)	220
	tg	0,0218 (1/46)	160
	tg	0,034 (1/29)	120
	tg	0,05 (1/20)	100
	tg	0,085 (1/11)	70
	tg	0,11 (1/9)	40
	tg	0,13 (1/7)	30
Branchement symétrique Symmetrical point	tg	0,034 (1/29)	150
	tg	0,05 (1/20)	120
	tg	0,085 (1/11)	90
	tg	0,11 (1/9)	60
	tg	0,13 (1/7)	50
	tg	0,167 (1/6)	40

The smaller the tangent or deviation angle, the longer the blade length and the higher the train speed can be towards the deviated route

Point Machine comparative catalogue

ProductName	Alstom internal Reference	Clone(Father)	Reference countries/Projects	ManufactorCompany	Tangent	Type of locking
MATR	8511010--1	Not applicable	France-Hong Kong- Singapour	Alstom France	0.117647059	external
MJ80MJ81	93907799390764	Not applicable	France-Hong Kong- Singapour	Alstom France	up to tg = 0,022	external
MJ2004-4MJ2004-6	26329(L11 SNCF)	MJ80	France(SNCF)	Alstom France	up to tg = 0,022	external
MET modular	P424008A00C (first point)P424008A10P (second point)	MET FS	SwedenNorway	Alstom Italy	up to tg = 0,04	external-double way
MET modular US	P424008A07L	MET FS	US- CANADA	Alstom Italy	up to tg = 0,074	external-double way
MET modular US	P424008A08M	MET FS	US- CANADA	Alstom Italy	up to tg = 0,074	external-double way
MET FS	P424004A00E	Not applicable	Italy	Alstom Italy	up to tg = 0,074	external-double way
P80	P41800100WP41800200H	Not applicable	Italy-Brazil-Chile	Alstom Italy	up to tg = 0,074	internal-double way
P80	P417004000A	Not applicable	Tunisia	Alstom Italy	up to tg = 0,074	internal-double way
P80	P417002000BP417003000P	Not applicable	Greece	Alstom Italy	up to tg = 0,074	internal-double way
L90	P42000100K	P80	Italy-Greece-Brazil-Chile	Alstom Italy	up to tg = 0,074	internal-double way
HW 1000	1121 - UK	Not applicable	UK-Australia-South Africa-Brazil-Canada	Alstom UK	up to tg 0,040	Internal + clamp lock
HW 2000	2121 - UK	Not applicable	UK-Australia-South Africa-Brazil-Canada	Alstom UK	up to tg 0,040	Internal + clamp lock
IRS S24		Not applicable	India	Alstom India		External
GM4000A	WMALWPEBBLBKNB	Derived from Model 5 Series	US, CANADA, ASIA, Mexico	Alstom Rochester	up to tg = 0,074	Internal, Single & Dual Locking
GRS Model 5 Series (E,F,G,H, & T)	54529-011-22	-	US, CANADA, ASIA, Mexico	Alstom Rochester	up to tg = 0,074	Internal, Single & Dual Locking
MD2000		S 700	Spain	DIMETRONIC	up to tg 0,074	External
S 700/S700K		MD2000	German railways	SIEMENS	up to tg 0,022	External
S 700 V	Not applicable	S700K	German railways	SIEMENS	up to tg 0,022	External
NSE	PM-TNG-120-120-2-2	Not applicable	Holland-Indonesia-South East	VIALIS NMA	up to 0,083	Internal
EP 600	EP633.1,2	Not applicable	CzechBulgaria	AZD Praha		External
SOA1M	Not applicable	S700	Czech,Bulgaria	BALKAN-SAST	radius 140-600 m	External
SOA3M	Not applicable	S700	Czech,Bulgaria	BALKAN-SAST	radius 140-600 m	Internal
JEA50/74	Not applicable	Not applicable	Sweden-Brazil-Poland	BOMBARDIEREricsson	up to tg = 0,074	Internal
Ebiswitch/JEA 95	Not applicable	Not applicable	Holland-Sweden	BOMBARDIEREricsson	up to tg = 0,074	Internal
ZD6	Not applicable	Not Applicable	South-East Asia	Tianjin Railway Signal Factory		
ZDJ9	Not applicable	Not Applicable	South-East Asia	Tianjin Railway Signal Factory		
CTS2	Not applicable	Not applicable	ItalyUSA China	General Electric	up to tg = 0,074	Internal
WSSB	Not applicable	MD2000		GANZ		
C1H		Not applicable	South- Africa-Vietnam	Alstom South Africa		external
SO	P425019A00R (0,022), P425018A00D (0,040), P425017A00	Not applicable	Italy- UK	Alstom Italie	up to tg = 0,022	external
L826H	Not applicable	L700H	German railways -Spain	ALCATEL	up to tg = 0,022	external
A900H	Not applicable	Not applicable	Spain	BOMBARDIER	up to 0,040	external
AH950	Not applicable	A700H	Austria-Turkey	EKA		external
Hydrostar	HB & ZB	Not applicable	Austria	VAEE	up to 0,040	external

Train Detection

There are two types of technology generally used for train detection:

- **Track circuit**

The track circuit continuously proves the absence of a train from a given section of track in a fail-safe manner. It cannot absolutely prove the presence of a train, since any failure mode will give the same indication as if a train is present, but, by proving the absence of a train, a clear track circuit can be used to confirm that it is safe to set a route and permit a train to proceed.

DC, AC and coded track circuits

- **Axle counter**

The axle counter system track mounted equipment counts axles entering and leaving a track section at each of its extremities. This information is evaluated to determine whether the track section is occupied or clear.

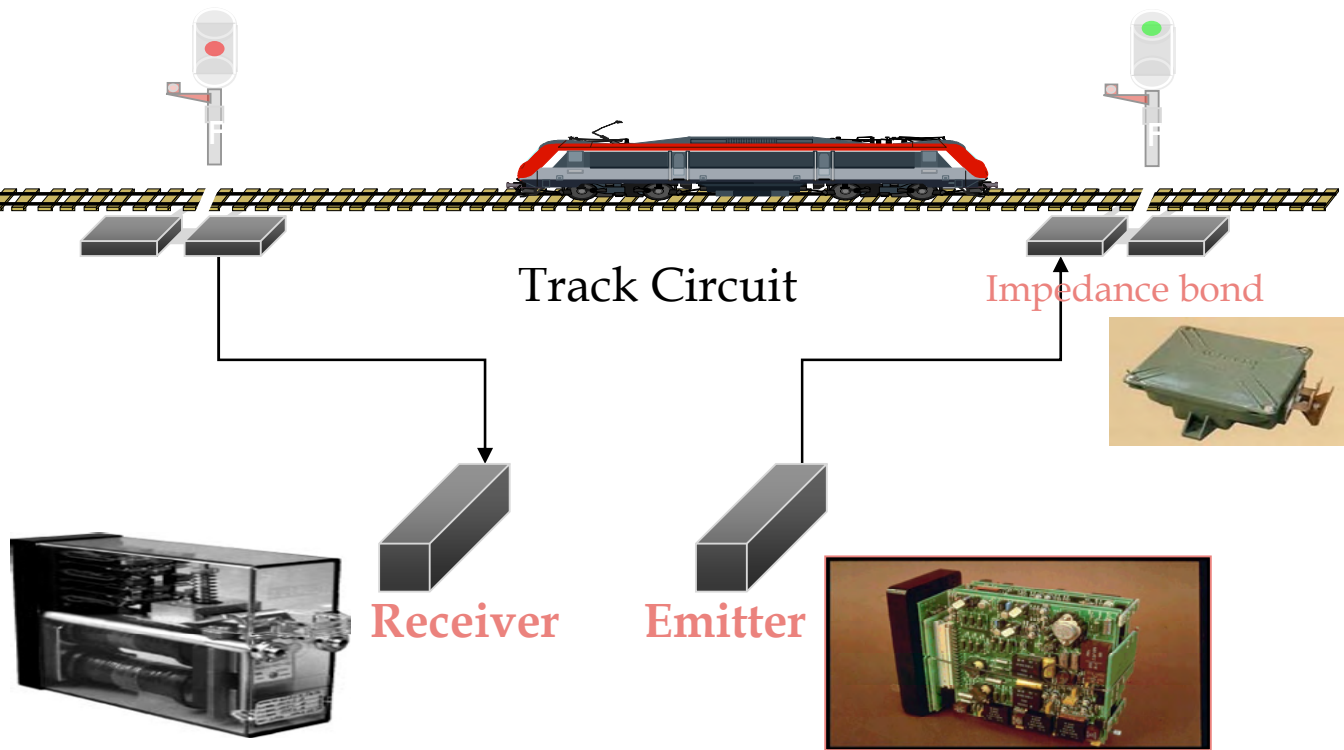
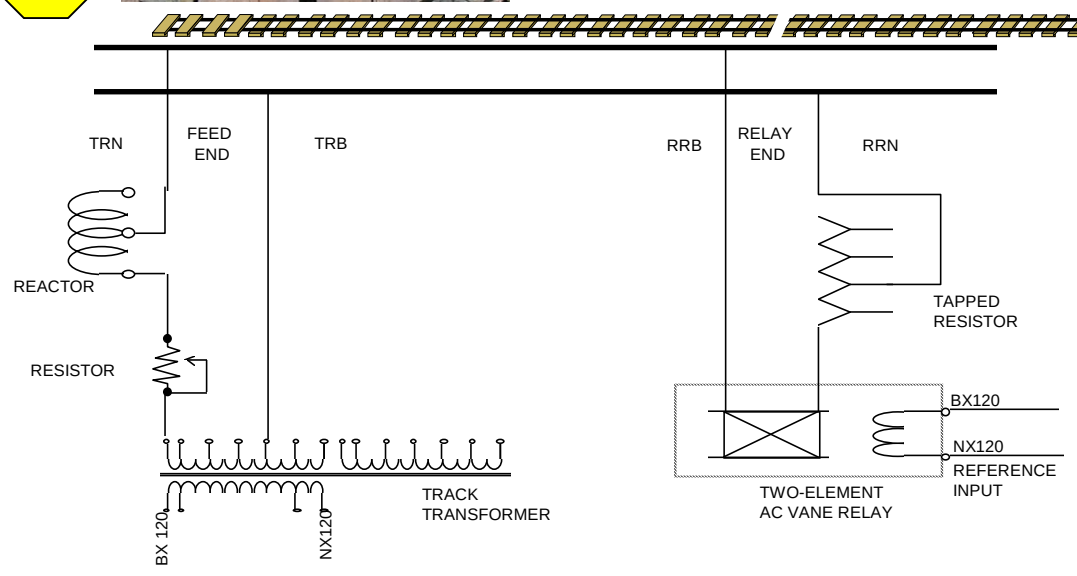
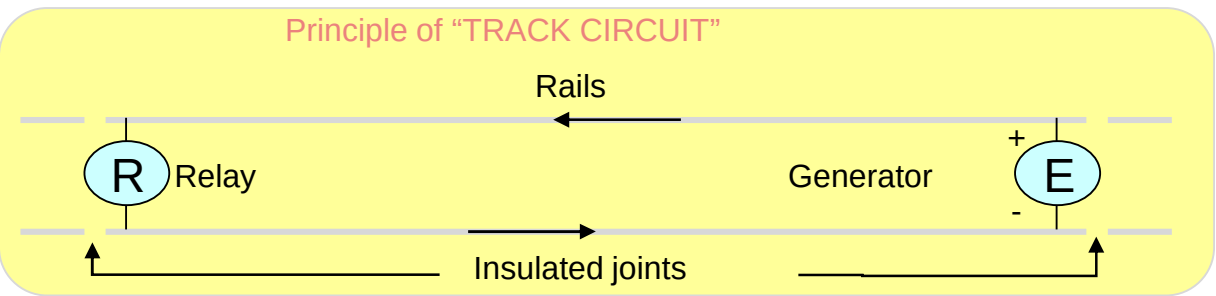
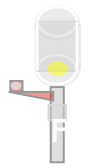
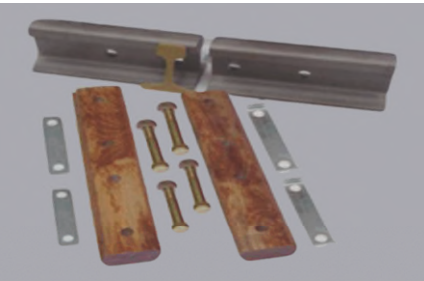
The main problem of Axle Counter : non detection of broken rails. Another problem with axle counters is that a right-side failure can occur when a wheel stops directly above the axle counter inductive sensor, known as 'wheel rock'

Both methods of train detection, track circuits and axle counters, have their supporters and detractors. However, for now, the world seems to be moving in the direction of axle counters, that is, until something else comes along.



Track Circuit

TRACK
PROD



Axle Counter

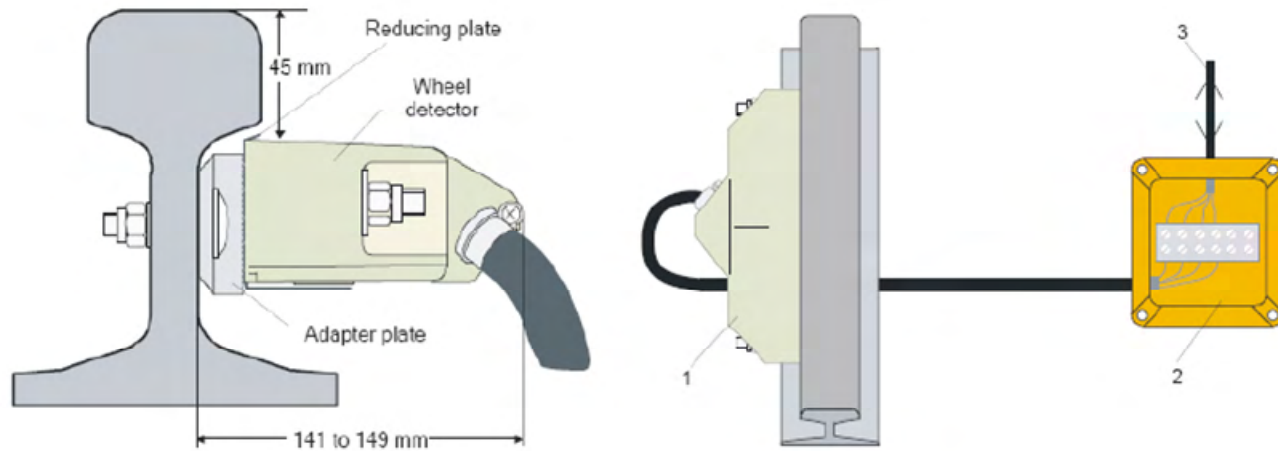


Figure 1, Axle Counter field equipment Layout.

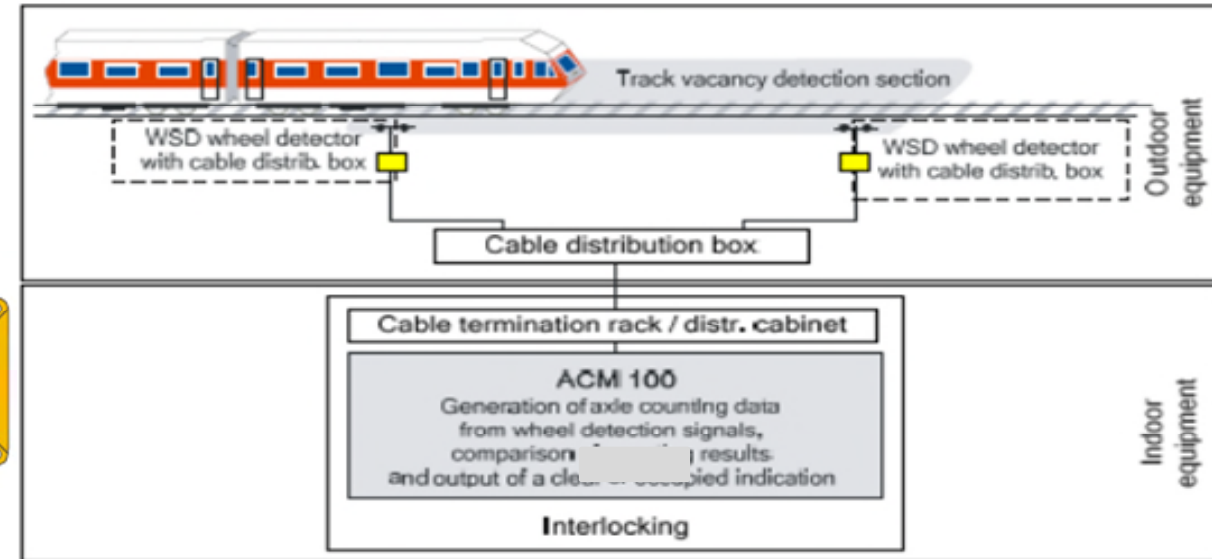


Figure 2, Typical layout interfacing to interlocking control

Axle counters primarily consist of two types of components, **the axle counting head**, and **the axle counting computer** (evaluation computer).

The axle counting heads are mounted along side the rail profile and actually detect the presence of wheels. They typically use LC oscillators to detect when the flange of a railroad wheel has modified the magnetic field created around the axle counter.

The evaluation computer typically resides in a signal control enclosure alongside the wayside. The evaluation computer generates data based on the axle counting heads and generates either a vacant or occupied declaration of one or more blocks (track sections).

The axle counting heads and the computer are physically connected, typically by copper wire or fiber optic cable.

Benefits of axle counters over track circuits

DISADVANTAGES OF THE TRACK CIRCUIT

- The quality of the electrical signal transmitted by the rail is dependent on the insulation resistance of the ties and the ballast. Causes leakage currents (if insufficient).
- This resistance is a limiting factor for the maximum length of the track circuit.
- This resistance may depend on hydrometeorological circumstances and on condition of the top layer (ties and mud). A simple rotten tie soaking in water or mud can reduce resistance and to leak too much current so that the receiver is no longer excited and declares a false occupation.
- The track circuit function also depends on the quality of the conductivity of the axle. If rusty, the function is not guaranteed.
- The traction return current can influence the track circuit if, for example, the upper harmonics are similar to the frequency of the main signal, which complicates signal detection. This is amplified if there is a tension with thyristor.
- The track circuits require insulated rail joints. (Even if jointless track circuits are available, in the punctual areas, is less reliable).
- Track circuits require more wiring and connections which increase the cost of installation and maintenance as well as the cases of failure.
- Track circuits are facing problems when the rail head is contaminated, for example with rust or accumulated leaves.
- Track circuits are unreliable under wet conditions, so they cannot be used to detect a train in a tunnel.
- Track circuits may be unreliable on the steel structures (e.g. steel sleepers).

AXLE COUNTERS ADVANTAGES

- Virtually unlimited length of the section.
- No insulating joints between sections which means better reliability and reduced maintenance.
- Independence of isolated conditions with the track which means better reliability.
- Independent from the quality of axles of the train shunt which means improved safety.
- Works perfectly where track circuits cannot work because of the ballast poor condition, steel sleepers, bridges, metal, wet tunnels.
- Applications on long blocks lines where track circuits are not a cost effective solution.
- Can operate in addition to track circuits to make temporary areas or increase the reliability of a section.
- No connection constraint and less wiring. The axle counters are generally less expensive to install and maintain.
- Very useful on steel structures that prevent normal operation of track circuits.
- Very useful on long sections on which several intermediate track circuits can be recorded.
- Some analyzes and experience with axle counters in continental Europe show that they regularly reach up to five times the reliability of track circuits with the same function.
- Immediate improvement in service reliability as a track circuit failure is often the most important cause of late trains.
- May be readjusted in the event of functional modification of a section or addition of a disconnection for the extension of a track while the cost of a track circuit embedded in the concrete is lost.

Axle Counting vs. Track Circuits - Advantages

- Track circuits are sensitive to ballast resistance, power fluctuations, and other track conditions while Axle Counters are not. In wet conditions, when the ballast resistance generally drops, track circuit operation is less reliable. Similarly with rusted rails, rails covered with leaves or flying dust/flakes, or a large amount of salt or conductive minerals impairing its surface conduction properties, track circuit operation is often unreliable. Axle counter applications are independent of track bedding and ballast, which means they can be used with steel structures (bridges), concrete or wood ties, and embedded rails.
- Track circuits are, by design, reliant on a very specific shunting characteristic of train axles (often 0.06 or 0.25 Ohms). If a train axle actually differs from this resistance (perhaps due to environmental influences such as those mentioned above), track circuits can “overlook” trains and incorrectly determine track vacancy.
- No insulated joints are required with axle counters.
- Track circuits consume a significant amount of electrical power and involve elaborate power supply equipment and cabling which can lead to frequent maintenance. Axle counters consume very low electrical power, simple cabling and are virtually maintenance free.
- High traction currents do not influence the detection method used in axle counters, and their installation and maintenance is faster and simpler than the installation of track circuits and related hardware.
- Less traction power bonding is required with axle counters than with track circuits. This has the additional benefit of minimizing the risk of DC stray current leakage into the ground which causes corrosion of nearby utility services.
- No impedance bonds are required with axle counters.
- Axle counters are essentially unlimited in block length. Track circuits vary greatly depending on type (AF, AC, DC), but are all very limited in track circuit length.
- Experience with axle counters in Europe demonstrates an improvement of up to five times the reliability of track circuits carrying out the same function. This has an immediate improvement in service reliability as track circuit failure is often the most significant cause of train delay. It also has safety benefits as it reduces the use of degraded modes of operation outside of the control of the signaling system due to failure.

Axle Counting vs. Track Circuits - Disadvantages:

- 1. Reset Requirements – If for any reason there is a failure (possibly a loss of power during a train movement), axle counters must be manually reset (the default start up mode is to create an occupied block). This can be done remotely, but requires a pro-active, manual manipulation.
- 2. No “shunt” – track circuits can be easily physically shunted by maintenance personnel (hard wired shunt applied to the rails), axle counters do not provide this function. A manual occupancy can also be accomplished with axle counters by hand manipulation of a metal object in front of the sensor. However, there is no visualization of the state (as is the case with a physical shunt), and the system must be reset.
- 3. Axle counters, unlike closed loop track circuits, do not provide broken rail protection. Although there is a great deal of controversy in the industry as to the percentage of broken rails actually detected by track circuits before they are traversed by a train, axle counters will not detect any break whatsoever.
- 4. By very small wheel profiles (less than 12 inches in diameter), axle counters can suffer from degraded reliability.

One of the disadvantages of axle counting systems is the lack of broken rail detection that is inherently provided with track circuits. Track circuits are not specifically required by customer, they do require broken rail detection.

Several types of breaks that even track circuits will not detect

Type of Rail Break	Cause	Location	Example
Transverse Cracks in Rail Head	Develops from an original internal defect inside the rail head, an internal horizontal crack, or very occasionally deep shelling of the gauge corner	At the Joint	
		Beyond the Joint	
Transverse cracks in rail head	Develops from an internal defect in the head of the weld	Flash butt Welding Zone	
	Develops along a plane near a normal cross-section of the profile	Thermit Welding Zone	
	Cracks at the periphery of the welding	Electric arc welding position	
Transverse cracks in rail head	Transverse fatigue crack across the resurfaced part of a rail head	Welding or resurfacing position	
	Transverse cracking under	At right angles with an electrical connection for return current	
Horizontal Rail head delaminations	Manufacturing defect	Above the rail web, at rail end	
		Above the rail web, away from rail end	

Vertical rail head end delaminations	Manufacturing defect	In rail head, $\pm 5\text{mm}$ of vertical rail axis, at rail end	
		In rail head, $\pm 5\text{mm}$ of vertical rail axis, away from rail end	
Vertical Rail Web end delaminations	Manufacturing defect	In rail web, $\pm 5\text{mm}$ of vertical rail axis, at rail end	
		In rail web, $\pm 5\text{mm}$ of vertical rail axis, away from rail end	
Horizontal cracking at head-to-web transitions	Manufacturing defect	Head/web interface at rail end	
		Head/web interface away from rail end	
Horizontal cracking at web-to-foot transitions	Manufacturing defect	Web/foot interface at rail end	
		Web/foot interface from rail end	

Rail breaks that can avoid detection by Track Circuits.

TC causing faulty signaling

When the track is frozen or completely dry, the dissipation is practically zero. In warmer autumn rain, the ballast resistance will fall to its minimum value and the diversion is therefore at its maximum.

The train detection system may warn that a train is occupying the track when there is no train. The opposite may occur as well, by the possibility that a train is not detected - **dangerous failure that cannot be allowed.** Some factors influence the diversion and track voltage, causing low resistance between rails and triggering a faulty signal. These influences are:

- Ballast resistance.
- Grounding of the rail,
- Rail structure.
- Switches and crossings.
- Obstacles on the track.

Ballast condition and drainage	Poor ballast condition, ballast height above foot of rail, and poor drainage can cause failures in wet conditions, as the ballast resistance change between wet and dry is too large for the normal track adjustment to handle.
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The size of the diversion and track voltage are influenced by several factors:

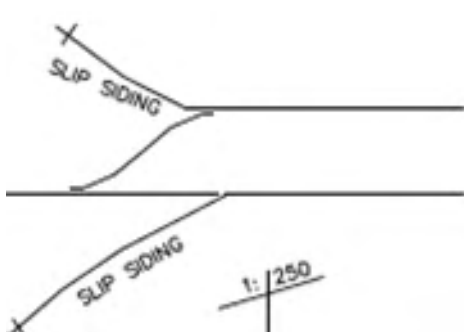
- Material of the ballast and its consistency
- Ballast humidity.
- Temperature.
- Type of sleeper and rail fastenings.
- Switches & crossings over the distance of the track circuit.
- Pollution in the ballast.
- Ballast going up against the rail foot or alternatively not going up

Concept of Isolation

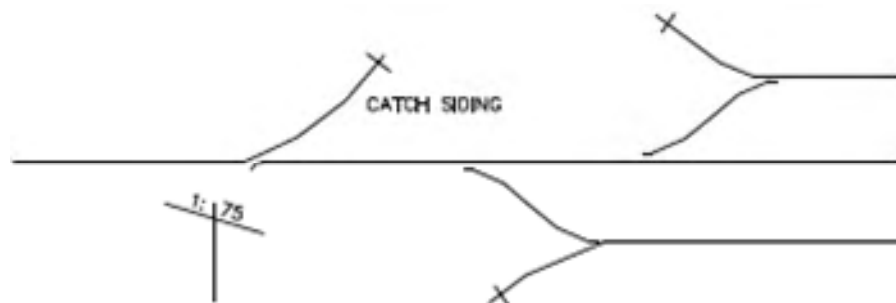
The term isolation denotes the condition in which line for a particular movement of a train is separated from all adjoining lines connected to it in such a manner that it cannot be fouled or interfered with by any movement taking place on the adjoining lines .

Rules regarding isolation:

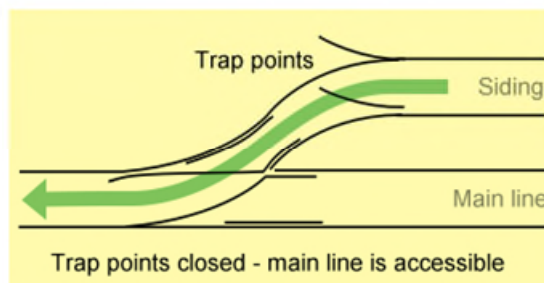
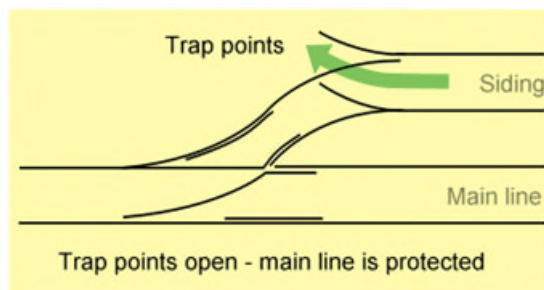
- A line, on which train movements at speeds higher than 50 KMPH are permitted, should be isolated from all connected lines.
- Passenger lines should be isolated from all connected goods lines and sidings, whatever the speed may be.
- Isolation of goods reception lines from sidings is considered desirable.
- To maintain safety in through running, points and trap sidings should be not be inserted in the main or through line except with the permission of CRS.
- Where other means cannot be adopted to permit simultaneous reception on a single line.
- To trap vehicles running away from a station.
- To avoid trains entering from block section due to heavy falling gradient



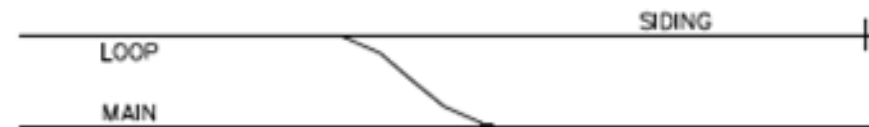
Slip siding will prevent vehicles entering block section.



Catch siding will prevent vehicles entry into block section.



1) Connecting to another line or a siding.



2) Provision of short dead end siding
(Vehicles not to be stabled on siding)



3) Provision of traps derailing switches



4) Provision of double derailing switch with lead rails without crossing.

5) Long dead end siding with trap (vehicles can be stabled on this siding).



6) Provision of sand humps.



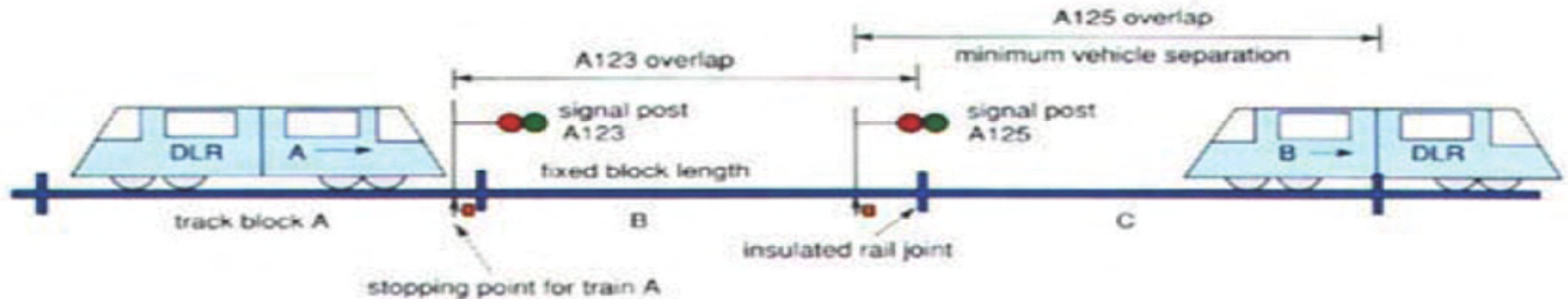
04

Automatic Block & Headway concept

Fixed Block System

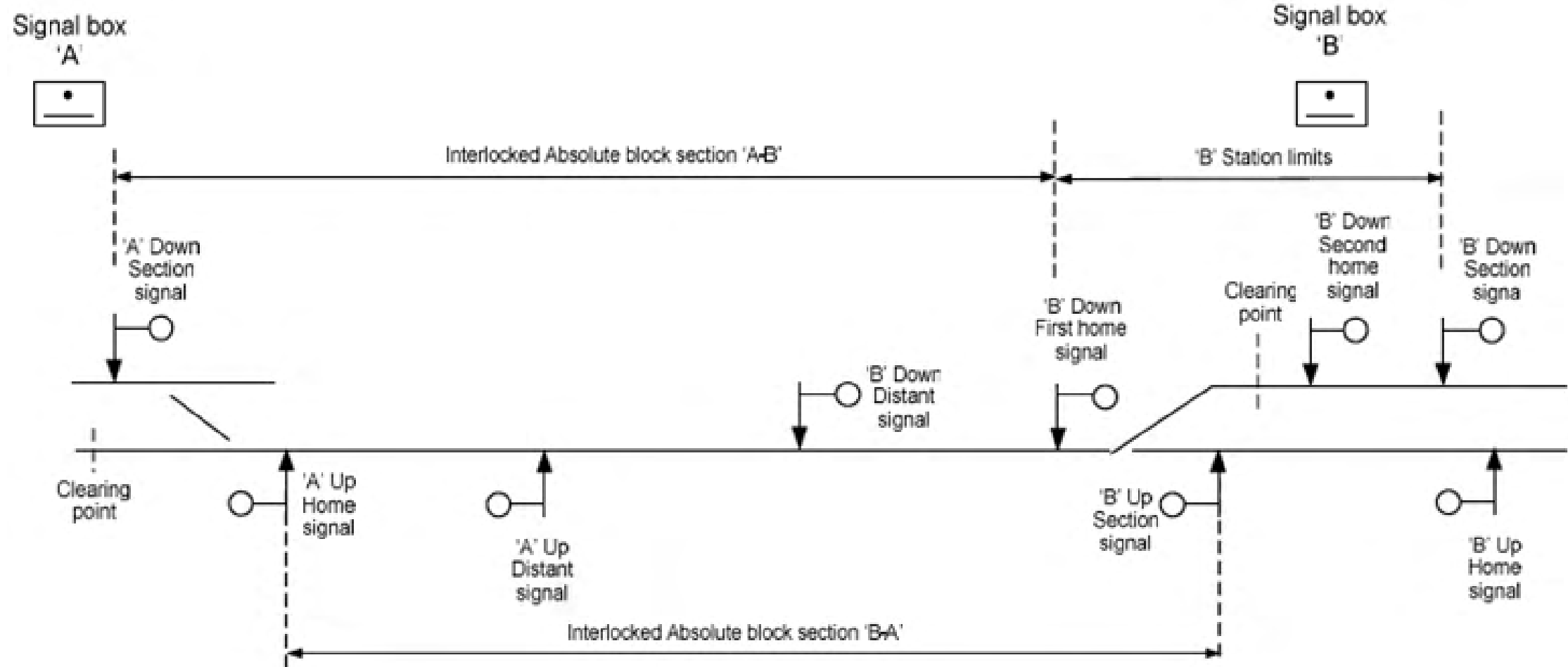
The fixed-block system

In an automated train system it is important to know exactly where all trains are and to keep them a safe distance apart from each other. Most train systems use a so called 'fixed block' system to do this. This system divides the track into a number of blocks and will only allow a train into a block if the next block is cleared.

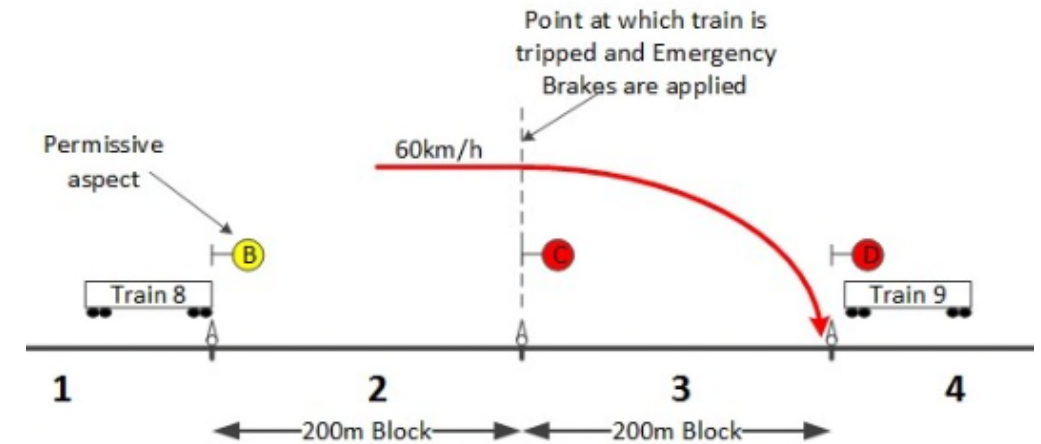
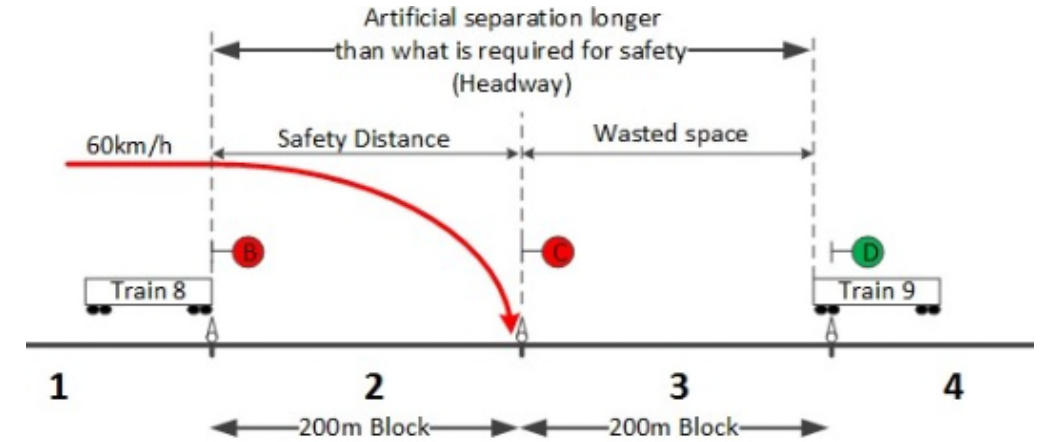
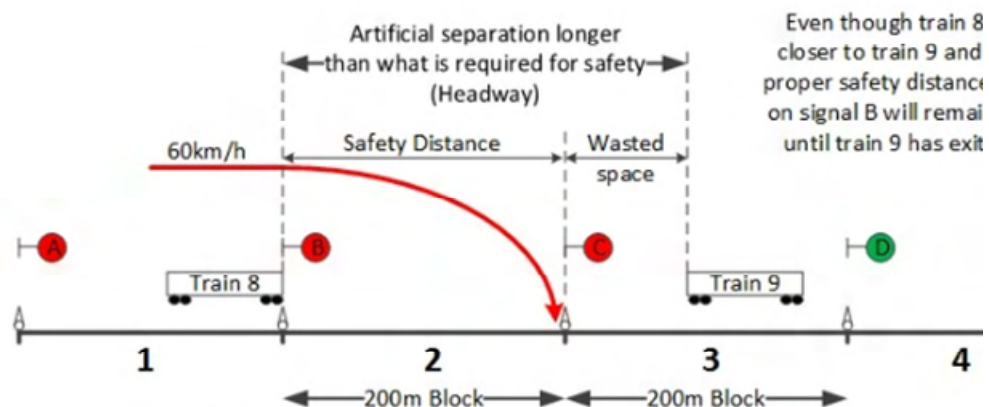
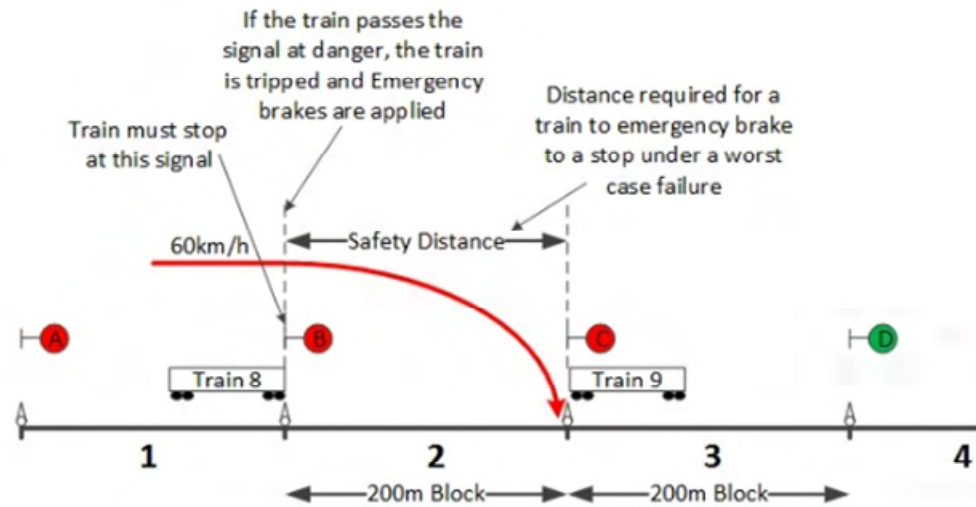


In this example train A can only enter block B (A123) when train B has cleared block C (A125).

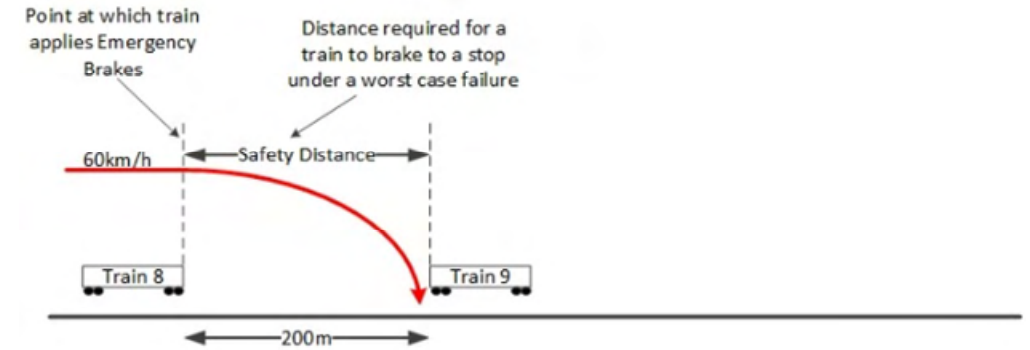
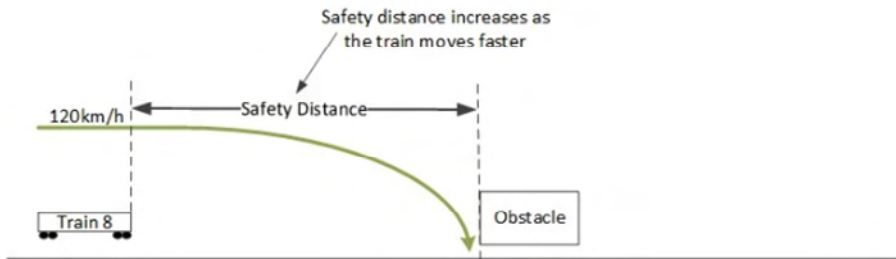
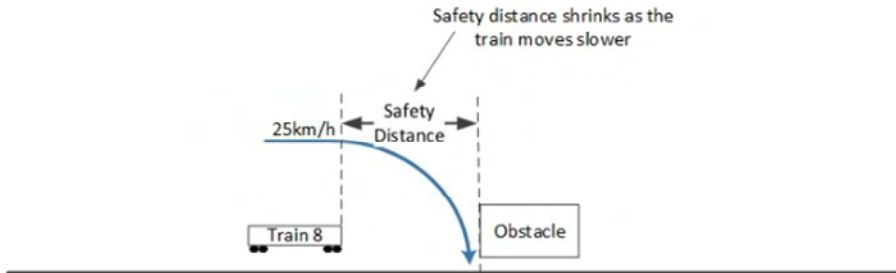
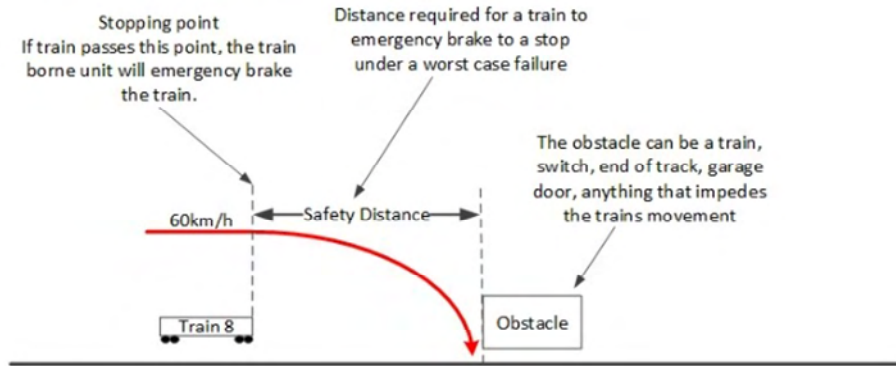
Single line interlocked absolute block system and station limits



Fixed block signaling



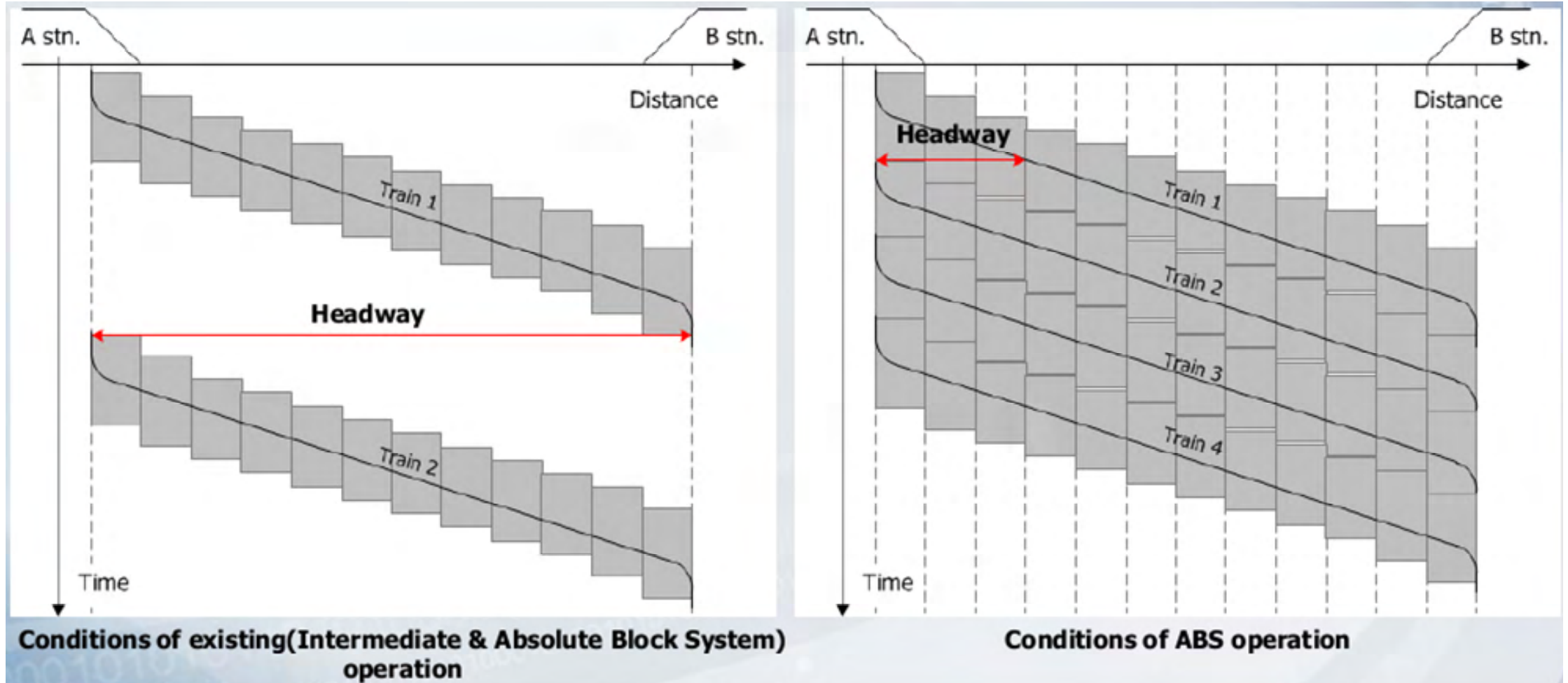
Moving Block



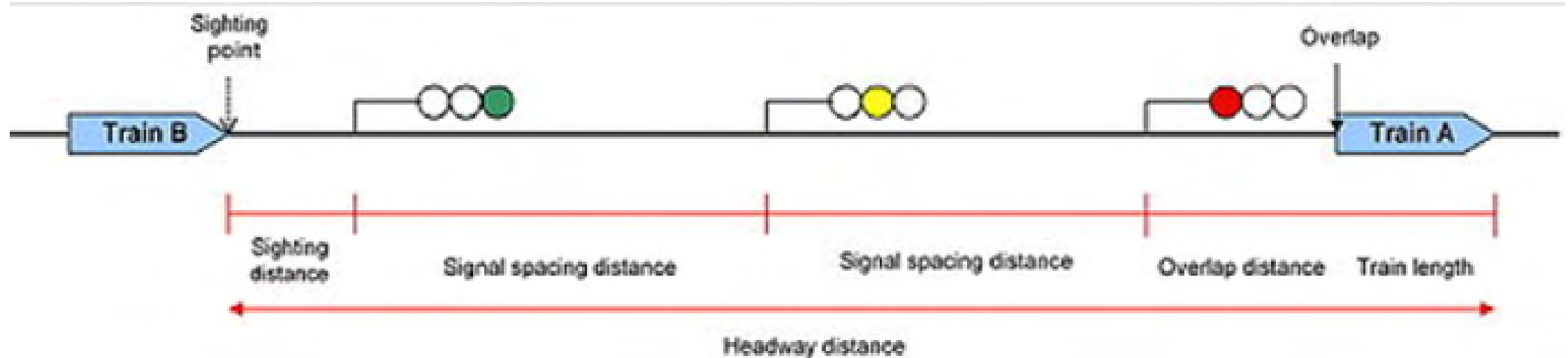
If train 8 reduces speed, the safety distance is also reduced and train 8 can move closer to train 9 (only if train 9 reduces it's speed at the same time)



Line capacity : Intermediate & Absolute << Automatic Block System



An example of headway on a railway system with multiple blocks



An example of headway on a railway system with multiple blocks. Train B can only enter a block with a green or yellow "aspect" (light), and must reduce speed in the yellow blocks to the point where they can stop within the sighting distance.

Concept of overlap

Every stop signal by its indication to the driver controls the movement of train upto the next stop signal as the next stop signal will control the movement beyond it. Hence the track between the stop signal and the next has to be clear and the points have to be correctly set and locked before a movement is permitted by it. However due to any unforeseen reasons like with sudden brake inadequacy the driver may not be able to stop at the next stop signal. So an extra safety margin of the track beyond the next stop signal is also to be kept free so that if the train overshoots the next signal, he will be able to bring the train to stop within that margin. This safety margin is called "overlap".

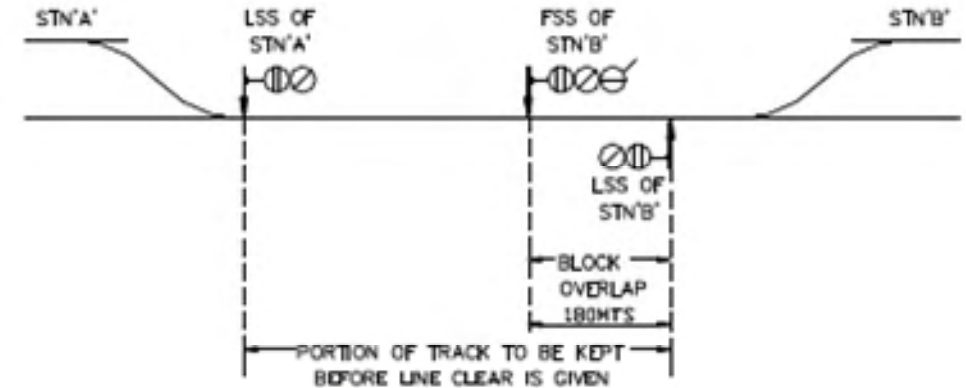
Similarly we have to ensure that when a train moves on the track the other rail vehicles from the adjoining track should not roll down and infringe with the movement. To prevent this "isolation" between adjoining lines is required.

Overlaps

Overlaps are referred to as ADEQUATE distance. Overlaps are of two types:

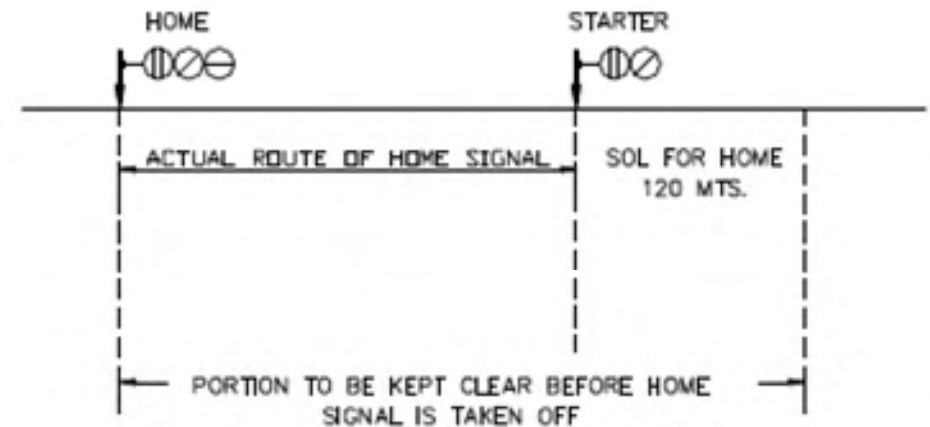
- 1) Block Over Lap (BOL)
- 2) Signal Over Lap (SOL)

Block over lap : It is the extra length of track in advance of the FSS (First Stop Signal) of a station, which must be kept clear, before Line clear can be given to the station in rear.



Signal over lap : The length of track in advance of a stop signal of station, which must be kept clear, before the signal next in rear could be taken 'OFF'.

SOL in MACLS sections shall not be less than 120 Mts.



05

Solution and Products

Main Functions of a signalling system

Traffic Control Non Safety Systems



Manage Traffic in
Control Centers

ICONIS: Mainlines and Urban
RailEdge: Freight



Inform & Secure
Passengers

Trackside Products Safety Systems



Signals

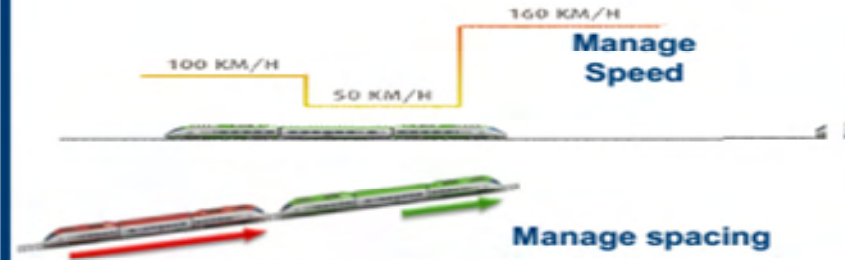


Point
Machines

Train
Detection

Train Control Safety Systems

ATLAS : mainlines
URBALIS : urban
ITCS : freight

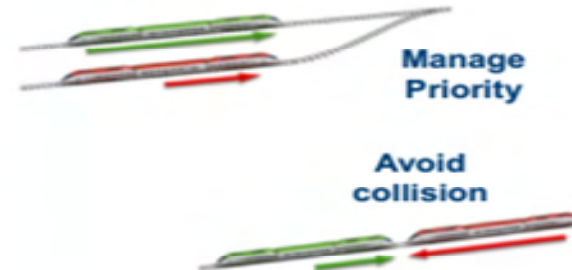


Assist
driver

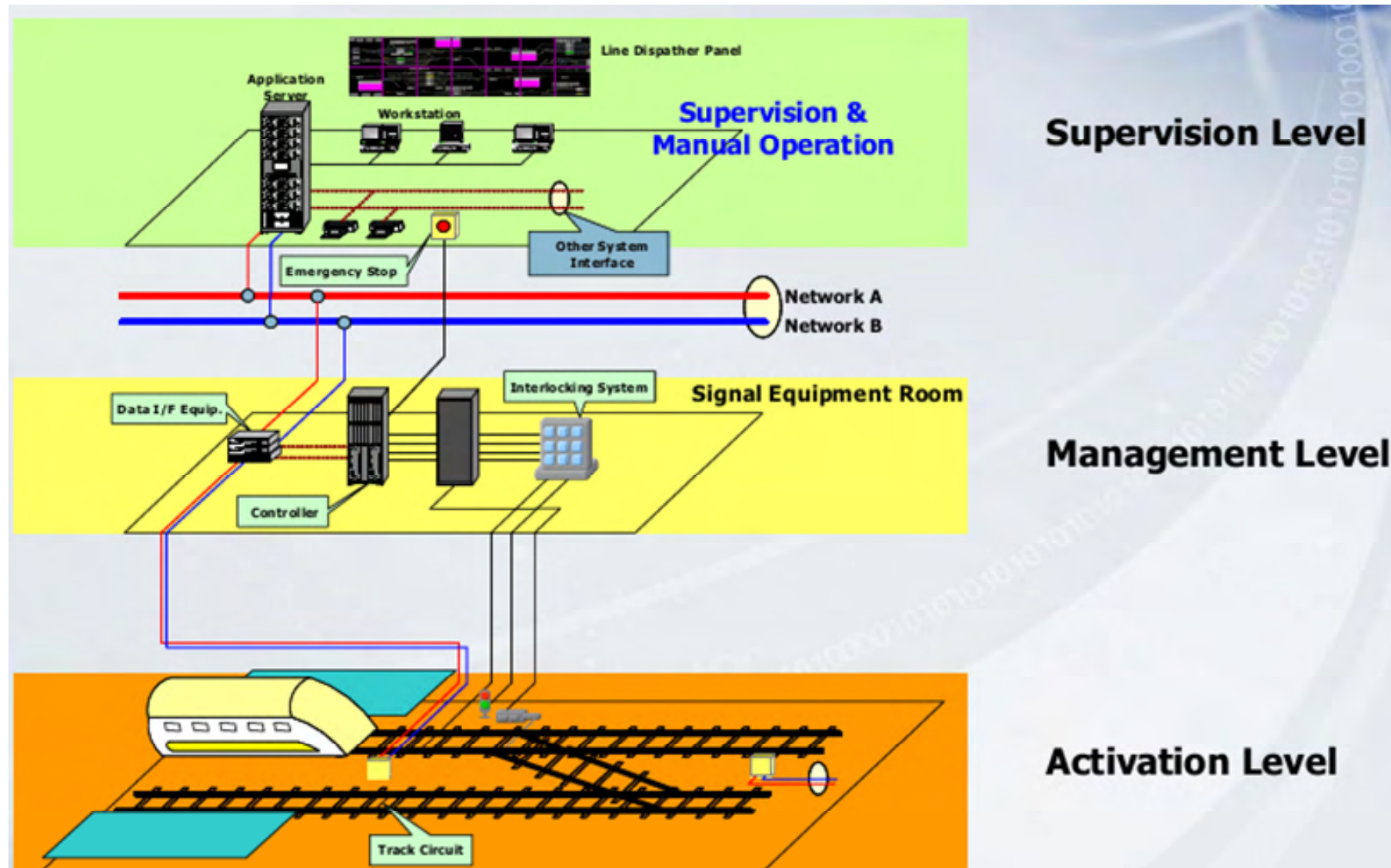
Eco
Driving

Interlocking Safety Systems

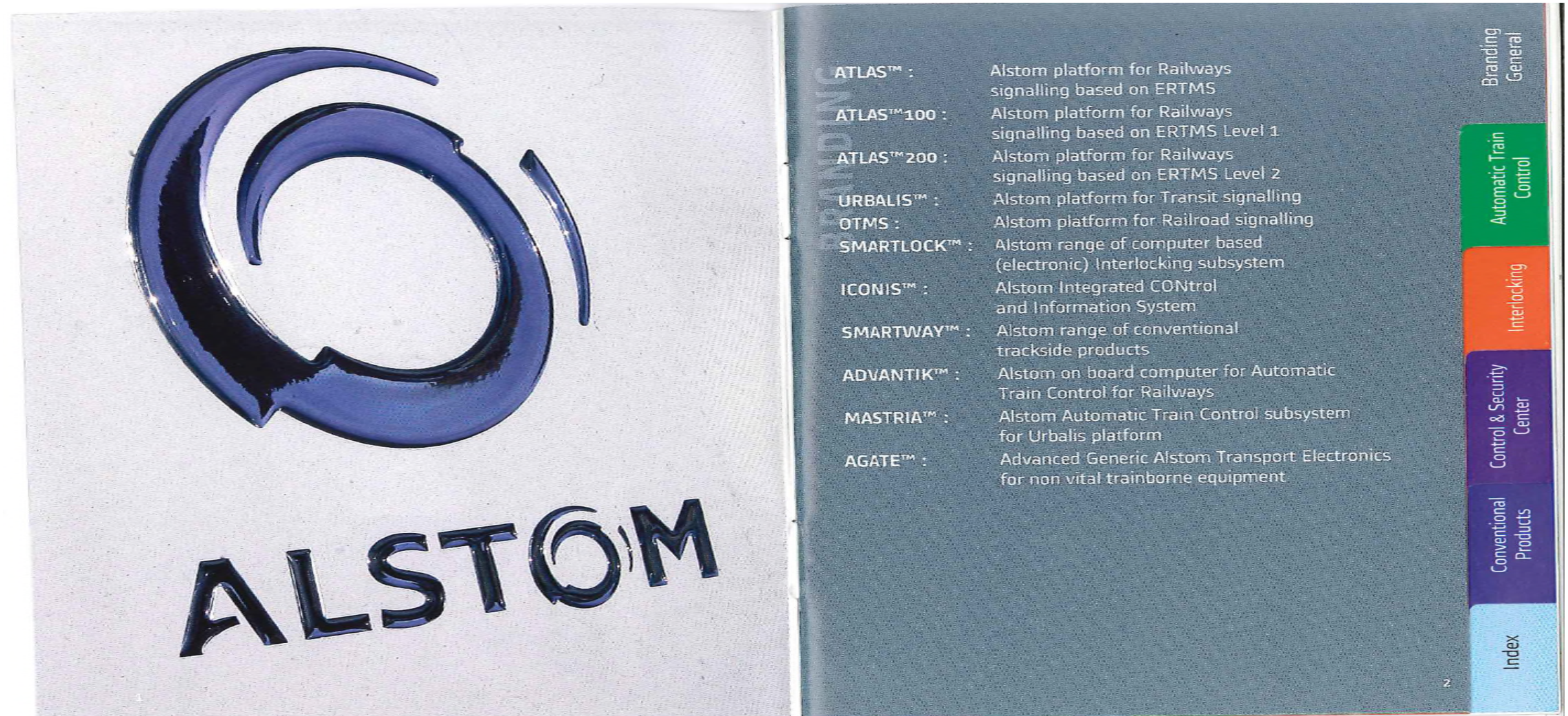
SMARTLOCK – large/medium
ElectroLogIXS - small



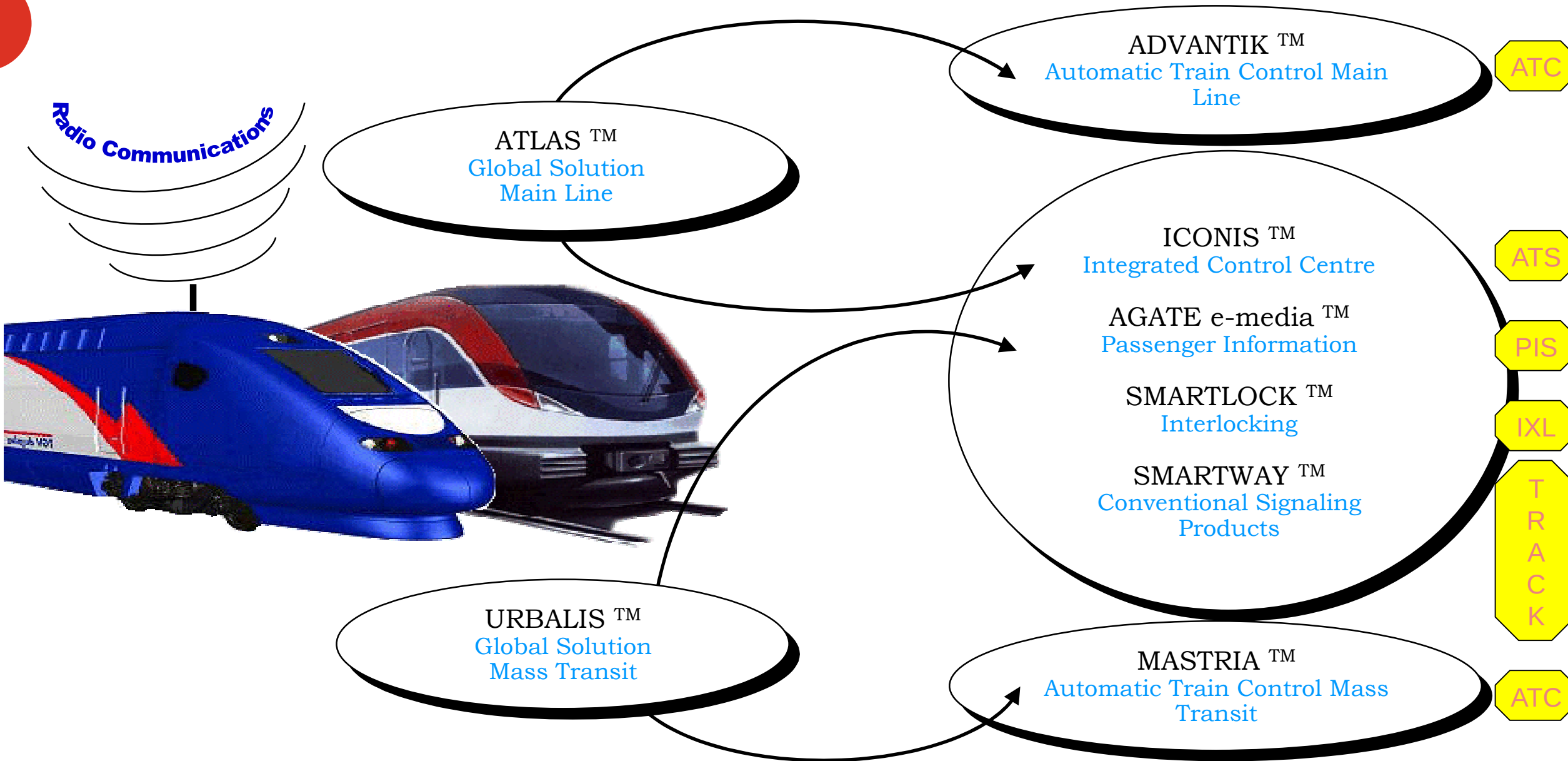
Hierarchy of a signalling system



Branding General



Solutions & products

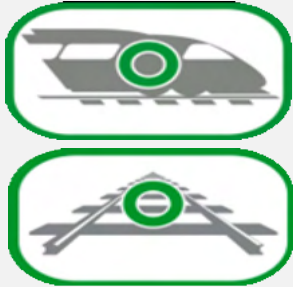


A complete product portfolio

An extensive Product Portfolio serving both **Urban** and **Mainline** Transportation needs

Train Control

Track side
On-Board



Route Control

Interlocking



Track side
Products



Security and Control

Control Centres
(ATS, SCADA)



Security, CCTV
Passenger Information
Public Address
Telecom



Stand alone

ATLAS™ TC

URBALIS™ TC

SMARTLOCK™

SMARTWAY™

ICONIS™

PACIS™

Integrated System

ATLAS™ for Mainline

URBALIS™ for Urban

Functions are allocated to one of 3 main subsystems

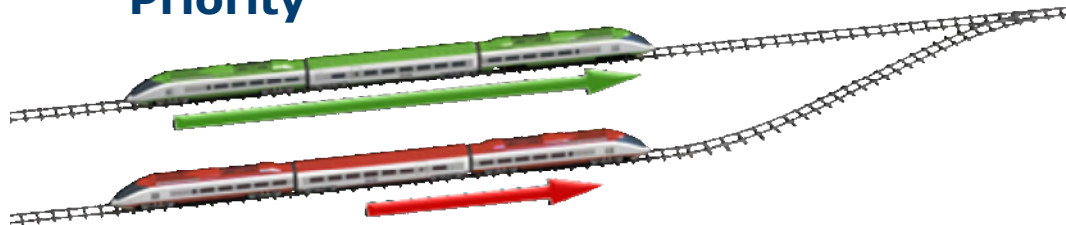
Security and Control ATS

Traffic
Management

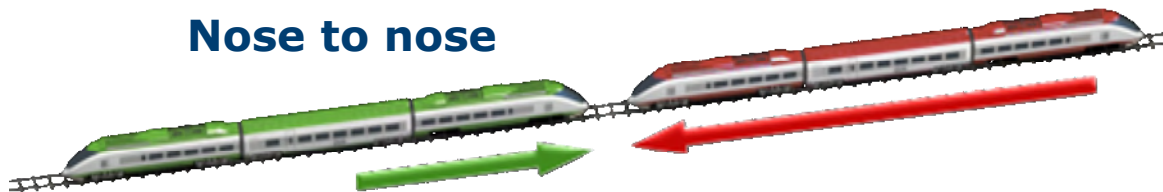
Route Control IXL

Train Control ATC

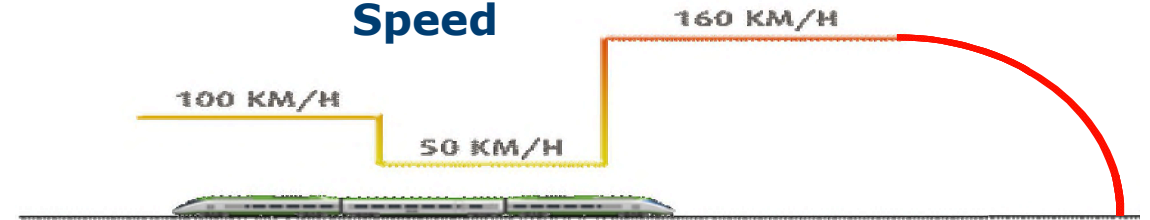
Priority



Nose to nose



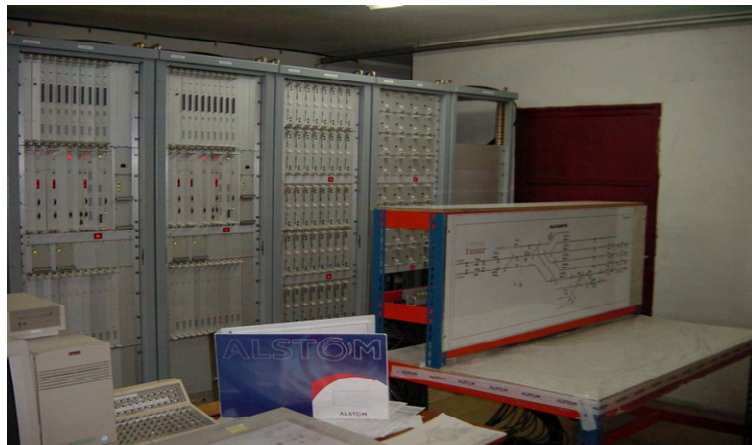
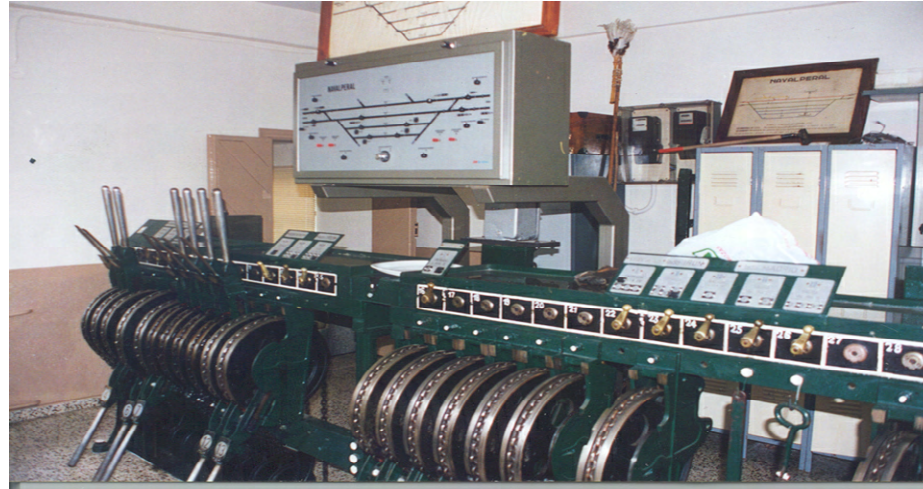
Speed



Catching each other

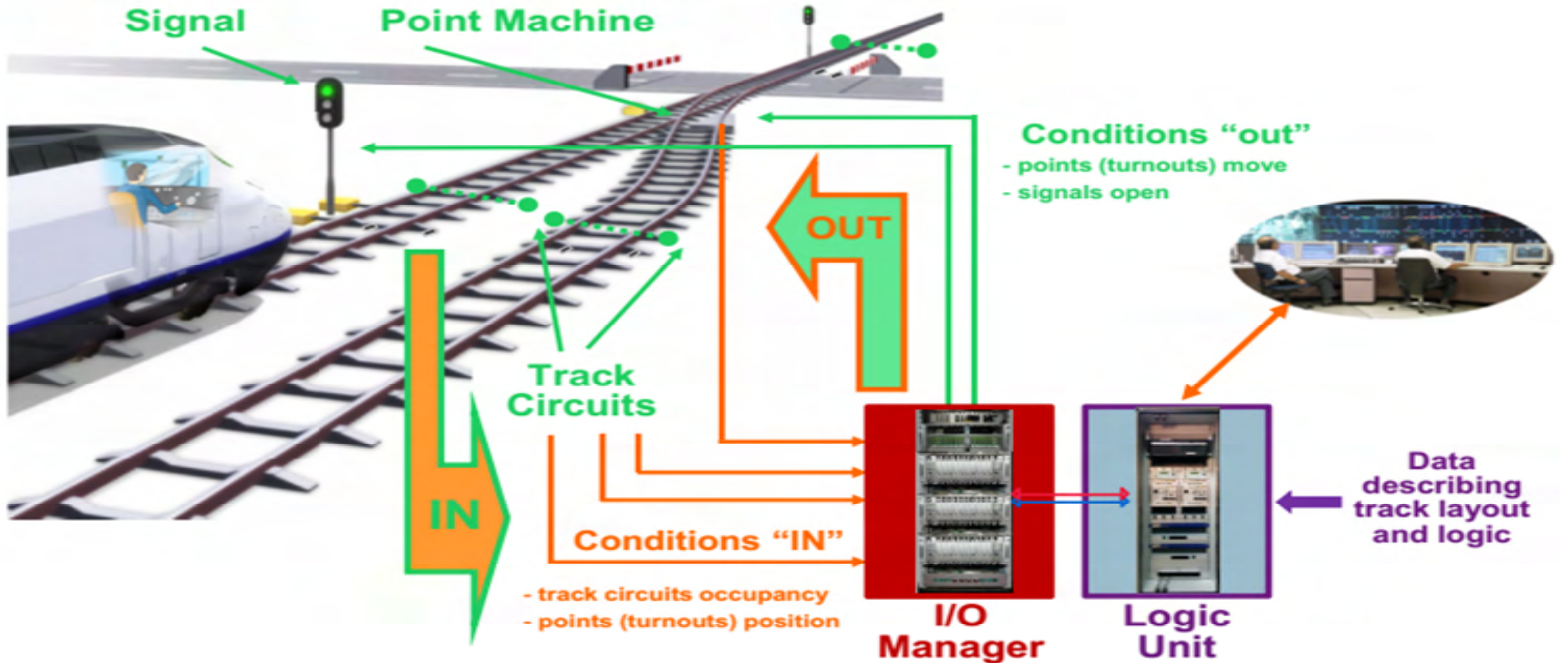


Interlocking: SMARTLOCK™

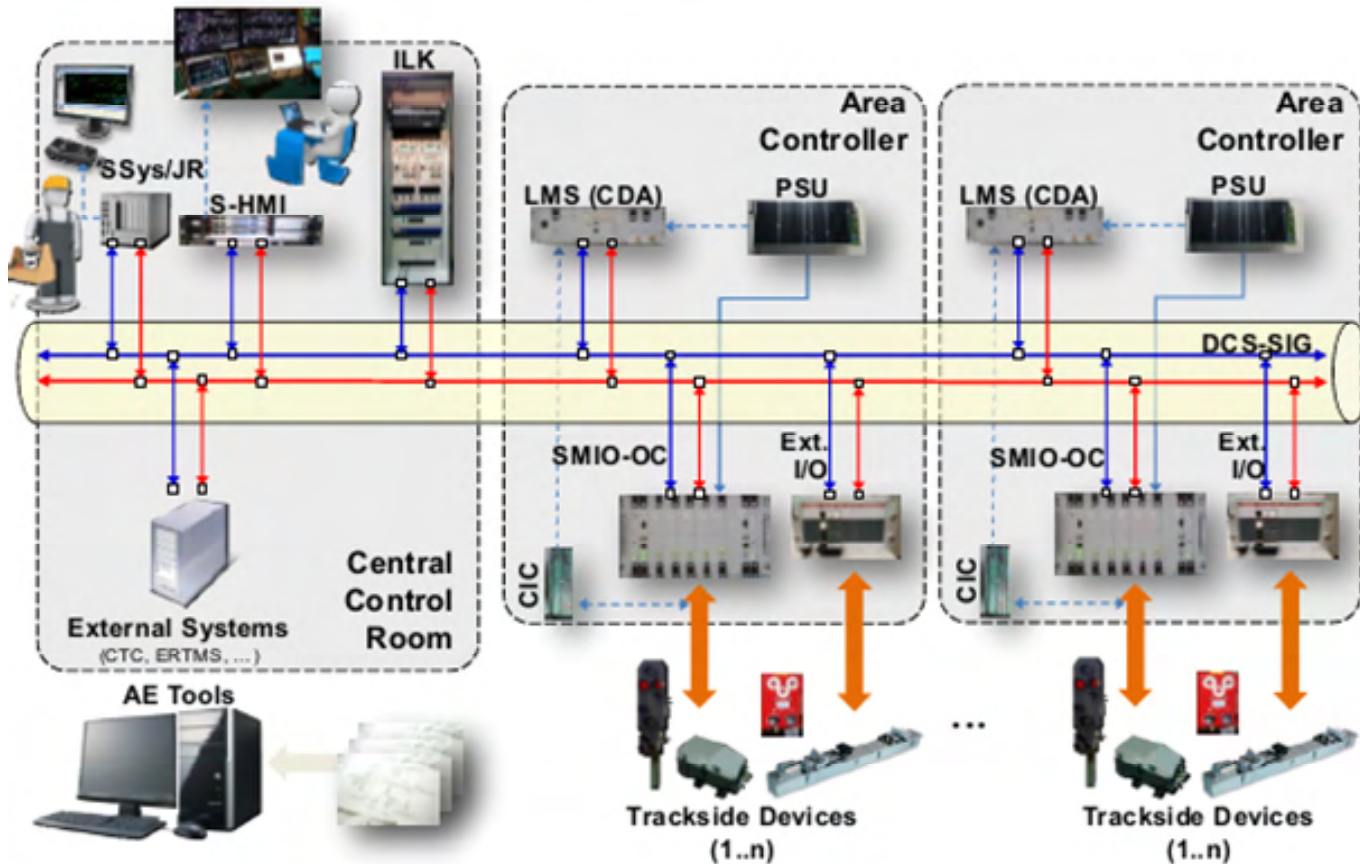


Safe Computing Unit - SCU

Interlocking: SMARTLOCK™ - how does it work ?



SML 400 GP Overview

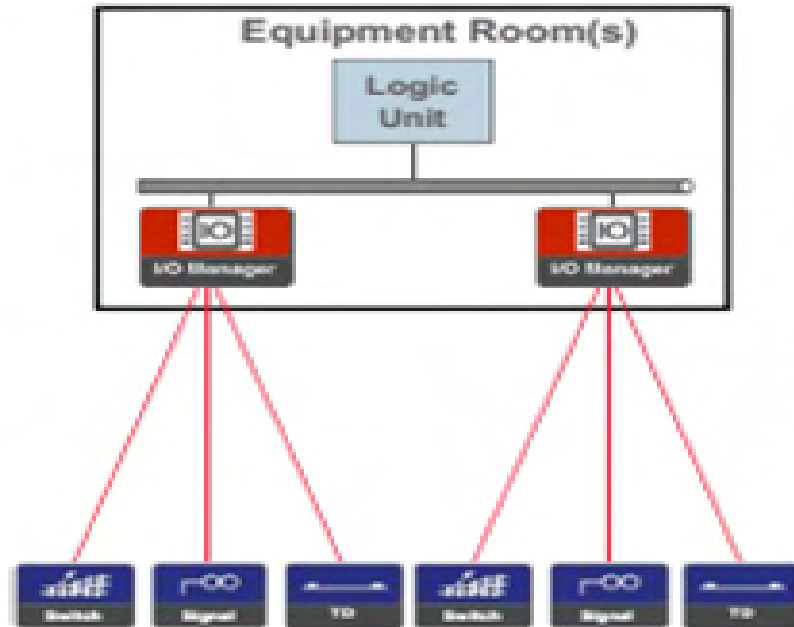


The SML400GP architecture foresees distributed I/O controllers (area controllers) managed by a centralized signaling computer (interlocking kernel). The control electronic consists of two main parts:

- Equipment installed in the **Central Control Room** (ILK, S-HMI, SSys, JR, ...): these devices are hosted in cubicles. A control room needs a fixed number of such cubicles, with a predefined configuration for each type of cubicle;
 - **Area controllers**, that might be distributed in the yard, close to the trackside devices to be managed: these area controllers are generally installed in open frames (IOF), suitable for installation inside technical rooms or shelters. Their cardinality is defined depending on the complexity of the yard to be handled.
- All above devices are connected together by a redundant Ethernet backbone (DCS-SIG network), with standard IP protocols. They are configured using a set of dedicated Application Engineering tools.

Interlocking: SMARTLOCK™ - Centralized or distributed?

Large Station

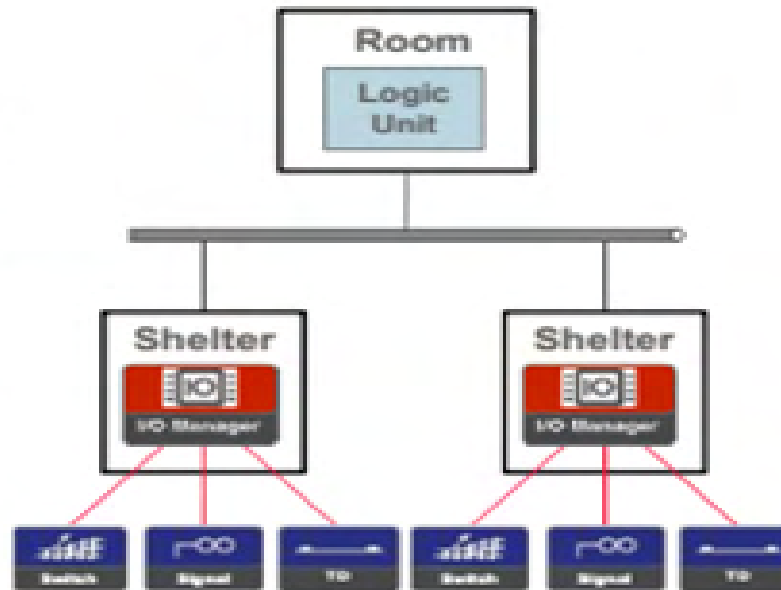


Central Logic Unit

- to meet processing requirement
- To simplify engineering

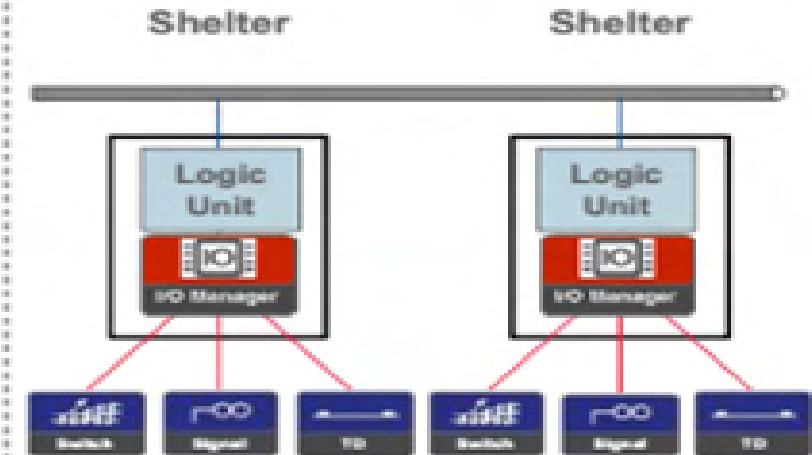
Line or Small stations

Centralised



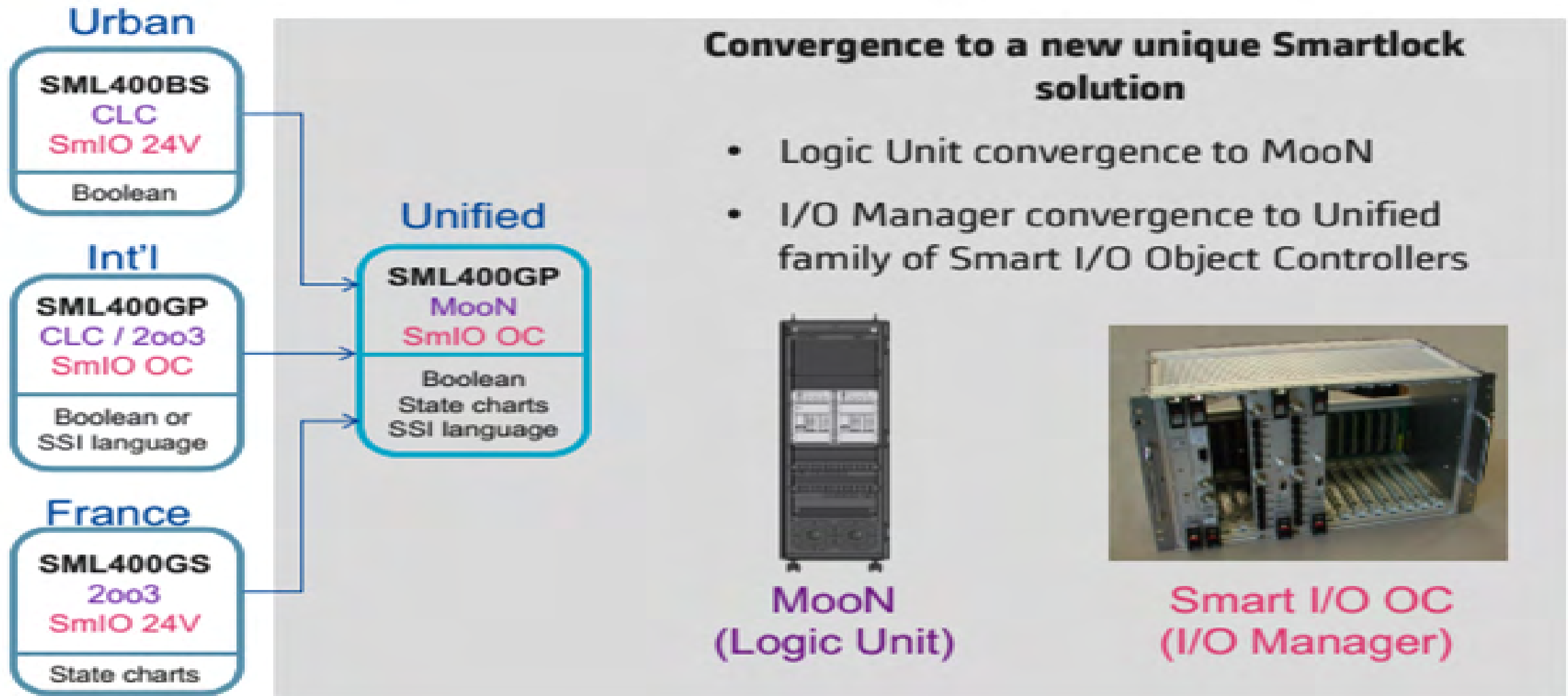
- **Pros:**
 - Only one engineering cycle
 - Well suited for complete lines
- **Cons:**
 - Additional central room

Distributed with small Logic Unit

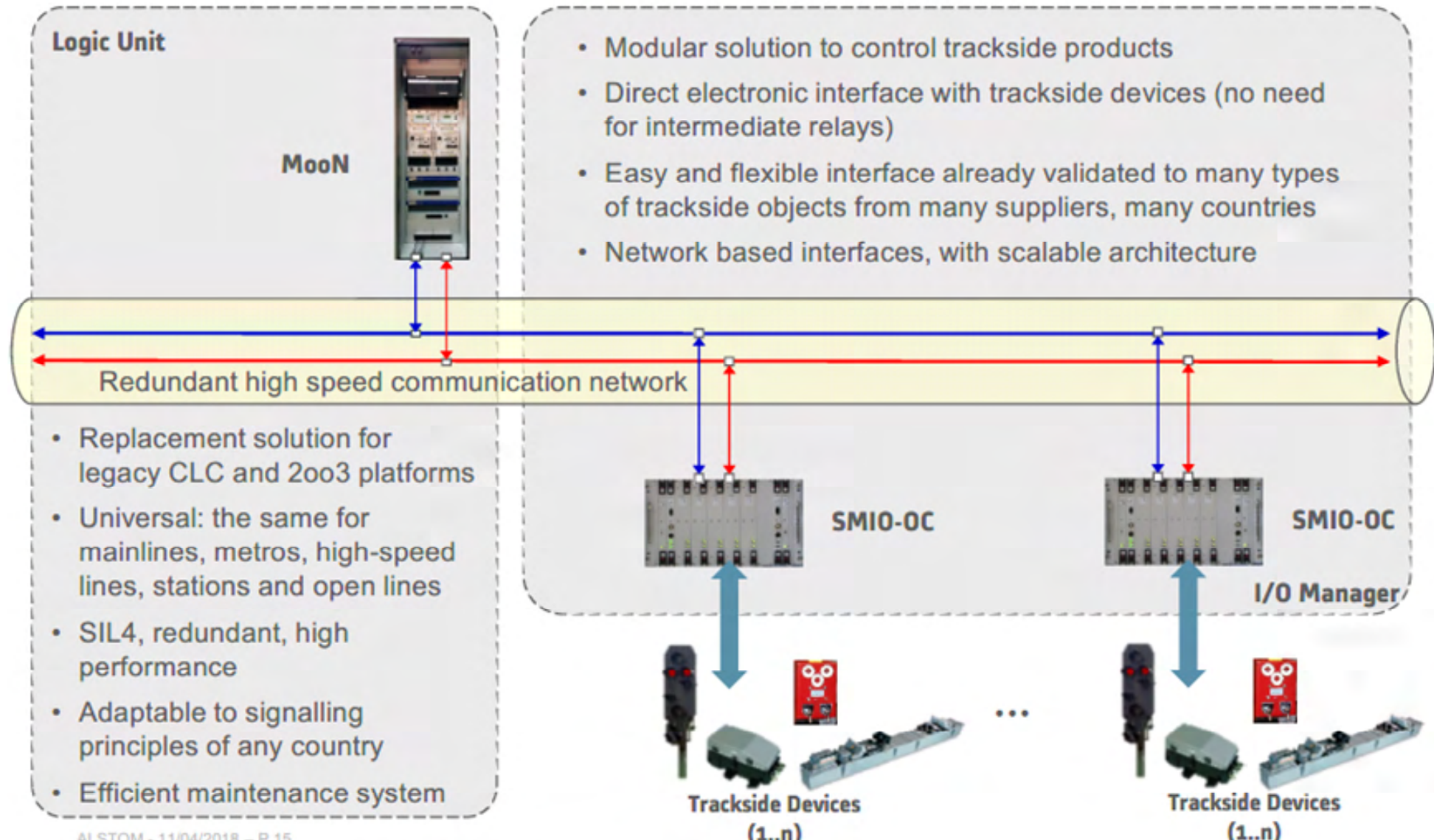


- **Pros:**
 - Well suited for separate station upgrades
 - No central failure point
- **Cons:**
 - Multiple engineering cycles

New Smartlock 400GP with MooN and Smart I/O OC for all Domestic & International Markets



SML 400 – MooN architecture

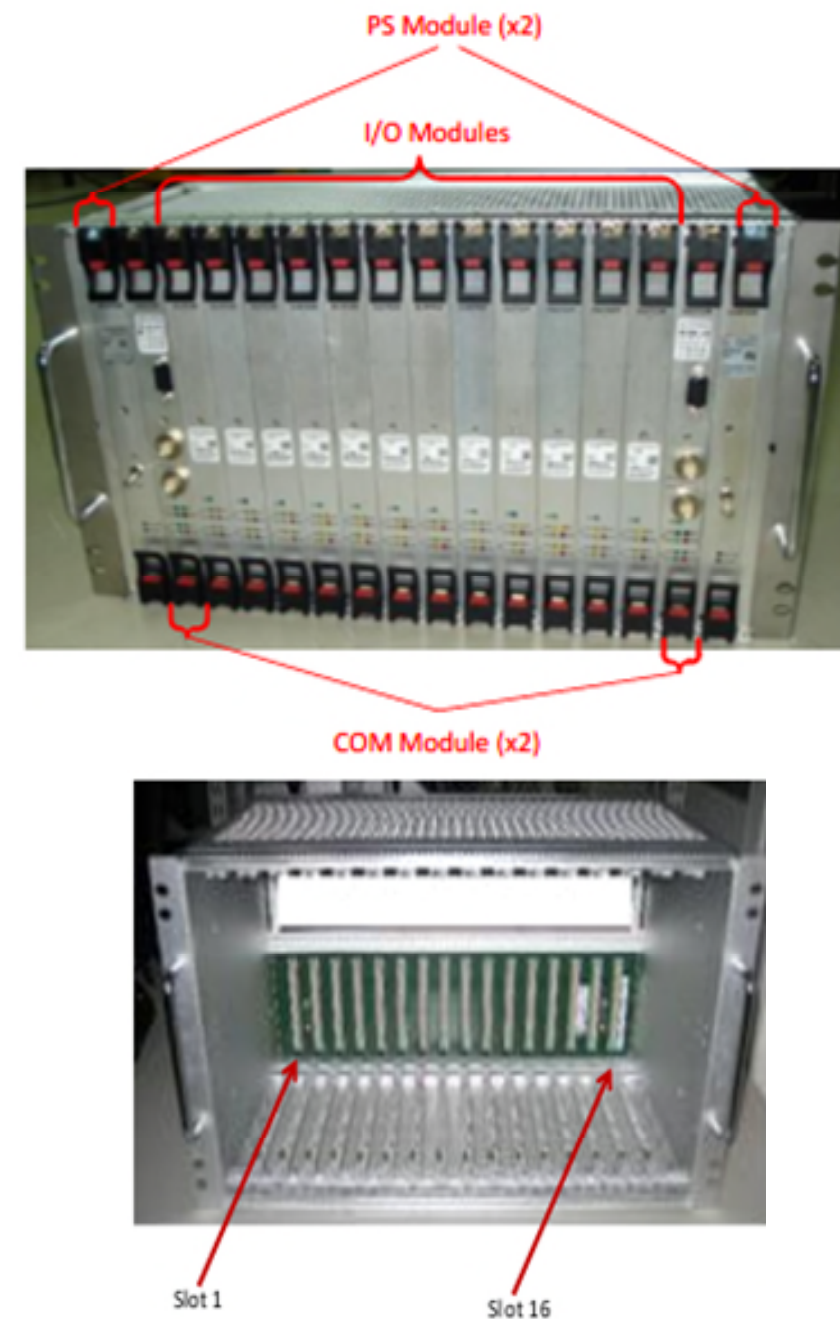


Actuators – SMIO OC

Guest board	Description	Size (slots)
IM module	interfaces up to 12 vital input with relays	1
OM module	interfaces up to 8 vital output with relays	1
SM module	interfaces up to 8 Signal Lamps	1
PM-4W module	interfaces up to 2 AC Point Machines using a 4-wires interface	2
PM-AC module	interfaces up to 2 AC Point Machines using a 7-wires interface (3 wires for motors and 4 wires for point position detection)	2
PM-DC module	<i>same than above for DC Point Machines</i>	2
GI-LR module	Field vital input interfaces without relays	2
GO-LLR module	Field vital output interfaces without relays or direct interfaces with Signal Lamps at 150Vac	2

The SMIO-OC is safety-related (SIL4), it adopts a Composite fail-safe architecture based on 2oo2 modules. Safety is assured by duplication of elaboration functions and diagnostic fault coverage managed by the 2oo2 safety core.

SMIO-OC is realized using the STEP technical platform



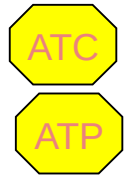
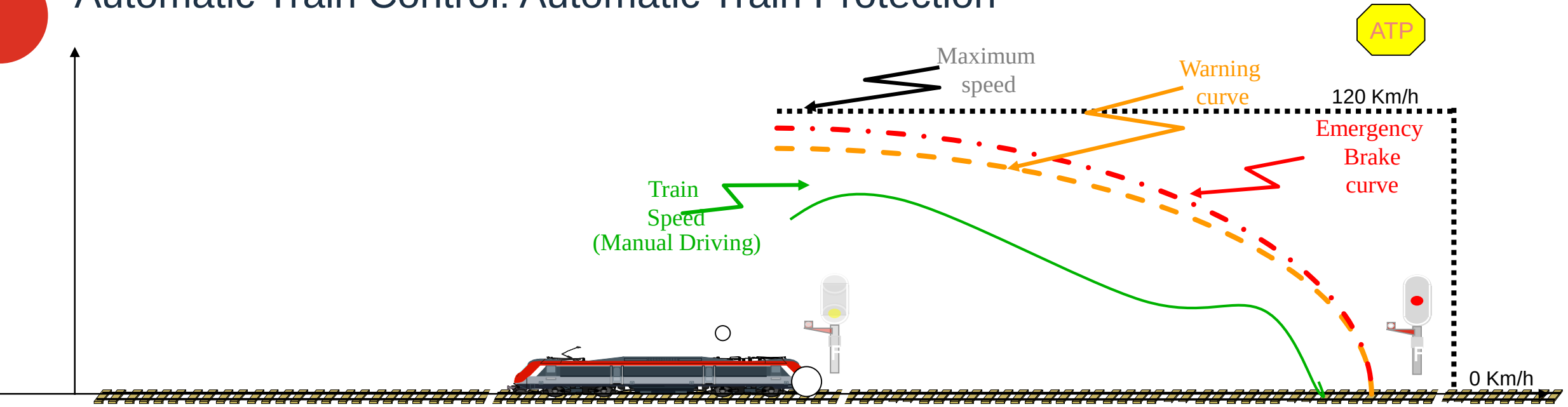
SML 400 GP external interface summary

SML 400 External Interfaces				
SML 400 GP	External	Physical Layer	Protocol Layer	Description
ILK	Non Alstom Electronic IXL	DCS-SIG	Not standardized	In most of the cases the Application data protocol (based on signalling rules) is a gap. Sometime (e.g. PVS protocol in Italy), the Customer defined standard protocols to interface IXL of different suppliers and an existing SML400 GA (i.e. Italy) implement it.
ILK	Non Alstom ATS	DCS-SIG	Not standardized	In most of the cases the Application data protocol (based on customer's way of working) is a gap. Sometime (e.g. Italy), the Customer defined standard protocols to interface IXL and ATS of different suppliers and an existing SML400 GA (i.e. Italy) implement it.
SSys	Non Alstom SCADA	DCS-SIG	SNMP/ OPC	Request to fit with non-standard protocols (e.g. proprietary protocols of other suppliers) are normally gaps. The application level could be linked to the specific tender.

SML 400 External Interfaces				
SML 400 GP	External	Physical Layer	Protocol Layer	Description
ILK	Alstom ATS (ICONIS)	DCS-SIG	NVFB3/ NVFB2	At Application level the following data are exchanged: alerts from Maintenance function to ATS, Safe/Simple Controls from TCS to IXL, Safe/Simple Indications from IXL to TCS operational mode from IXL function to TCS
SSys	Alstom SCADA (ICONIS)	DCS-SIG	SNMP/ OPC	Ref. for Physical Layer: Networks Interface Description - SML400-GP-ID-A407480 with the standard TCP/IP At <u>application level</u> it may exchange data with the SCADA among the following set: Health Status, Identification, Alarms, Logs, Operational Mode, Events, Statistics counters, Measurements.
CIXL-B	TCMS (RBC)	DCS-SIG	Subset-098 EURORadio	At Application level vital controls and vital indications are exchanged (Specific Application).

interfacing external system through not standard protocols is often a source of gaps

Automatic Train Control: Automatic Train Protection



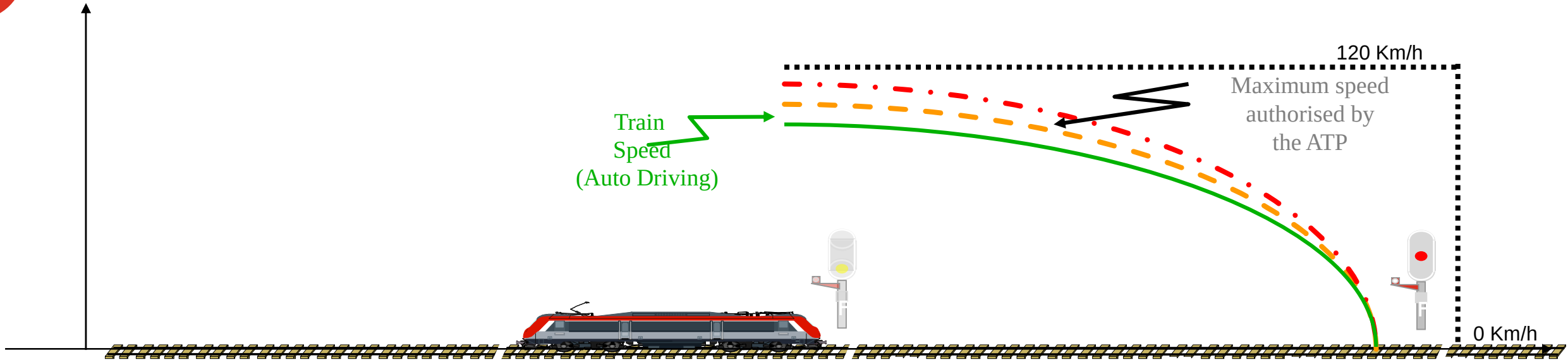
Automatic Train Control
Automatic Train Protection

○ ATP = Supervision of Train Speed

If **TrainSpeed** > **Emergency Brake Speed**
Then **Emergency Brake** = ON
Else If **TrainSpeed** > **Warning Speed**
Then **Warning Sound** = ON
End

Architecture

Automatic Train Control: Automatic Train Operation



Automatic Train Control



Automatic Train Operation

ATO = Drives the Train

Traction and Breaking of the train is no more managed by the driver, but

by the ATO train equipment
(Automatic Pilot)

under the safe supervision of the ATP

Architecture

Automatic Train Supervision

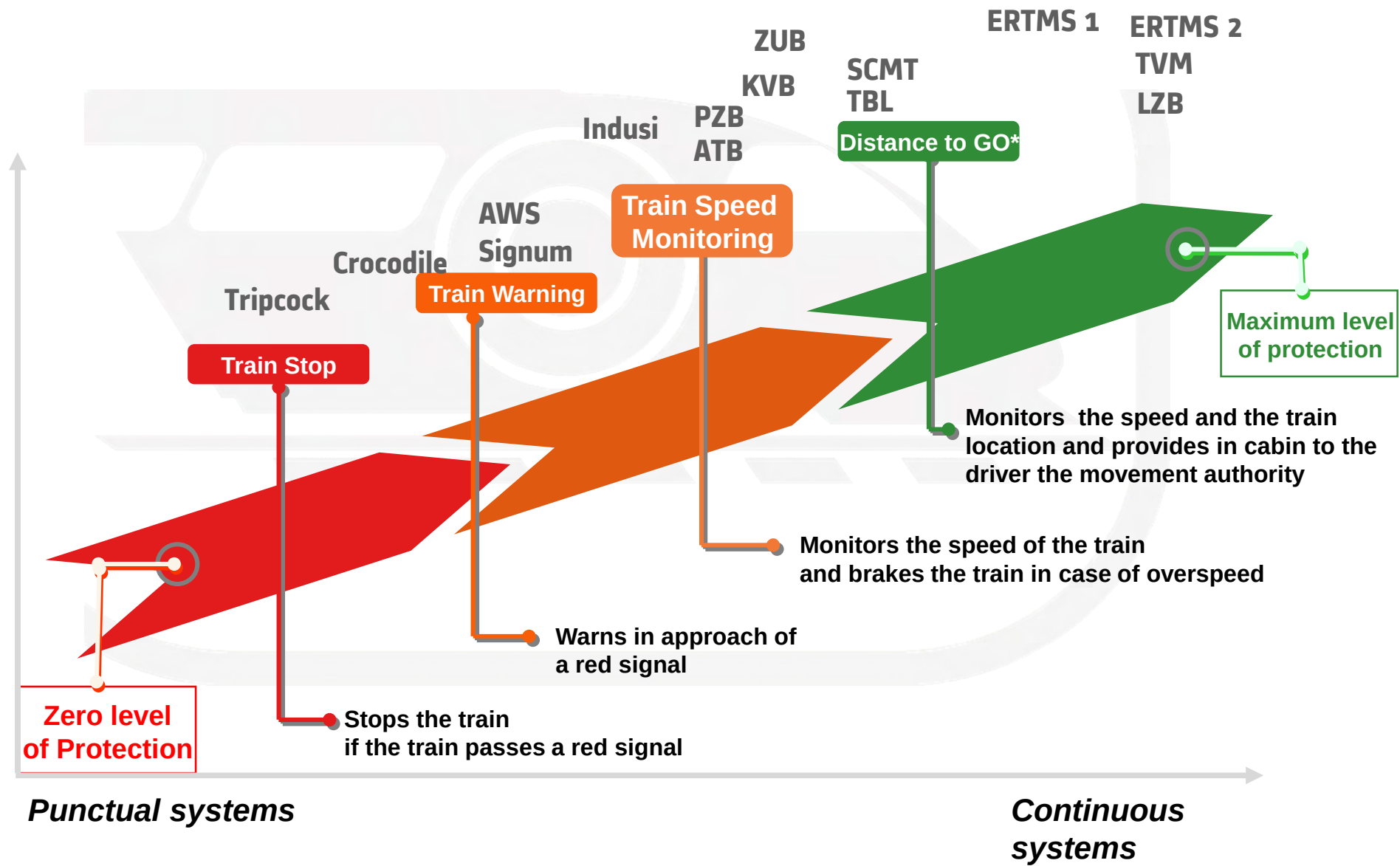
Automatic Train Supervision



OPERATION CONTROL CENTER

ATS = Centralised Supervision & Control of Train Movement

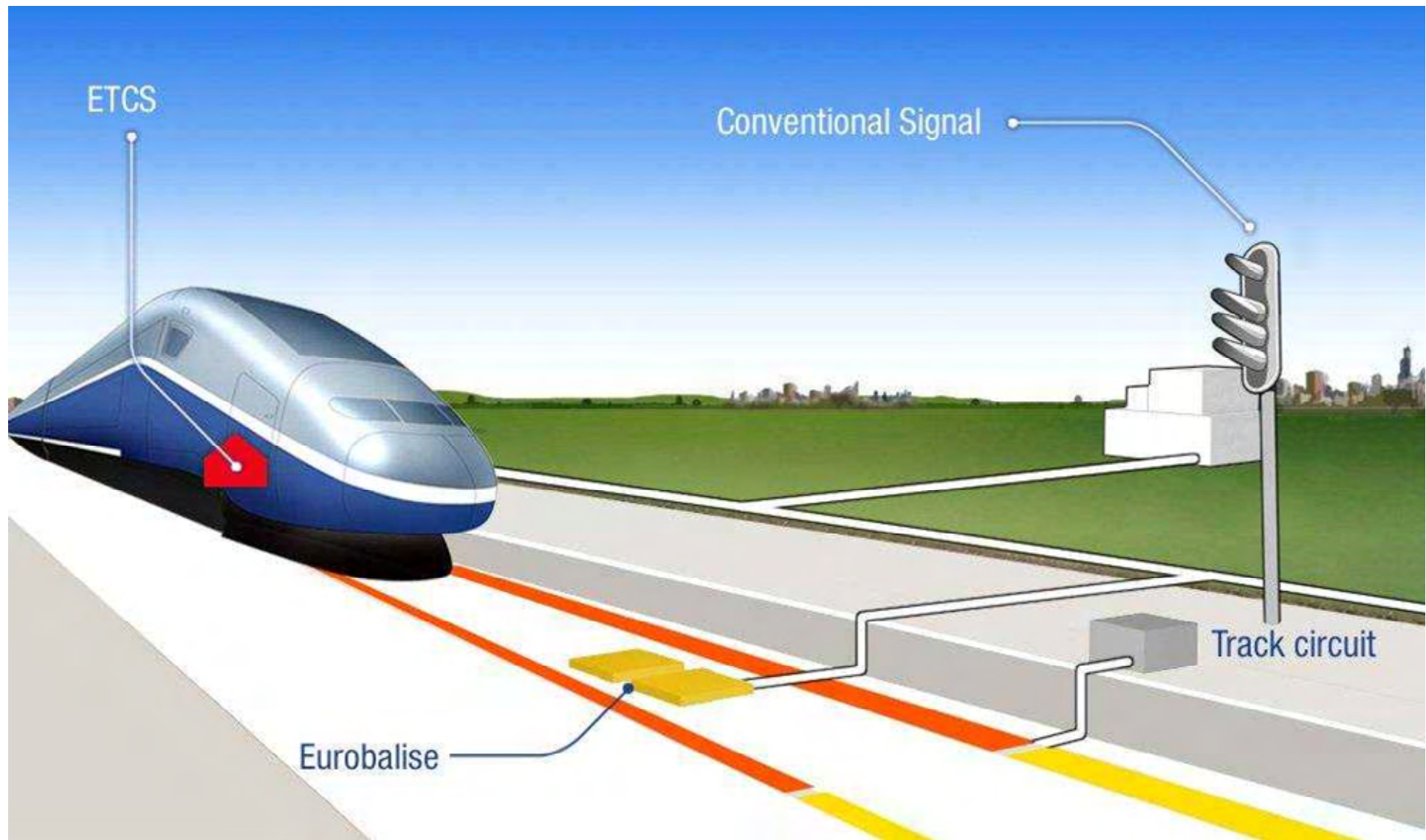
Different Levels of Train Protection (ATP)



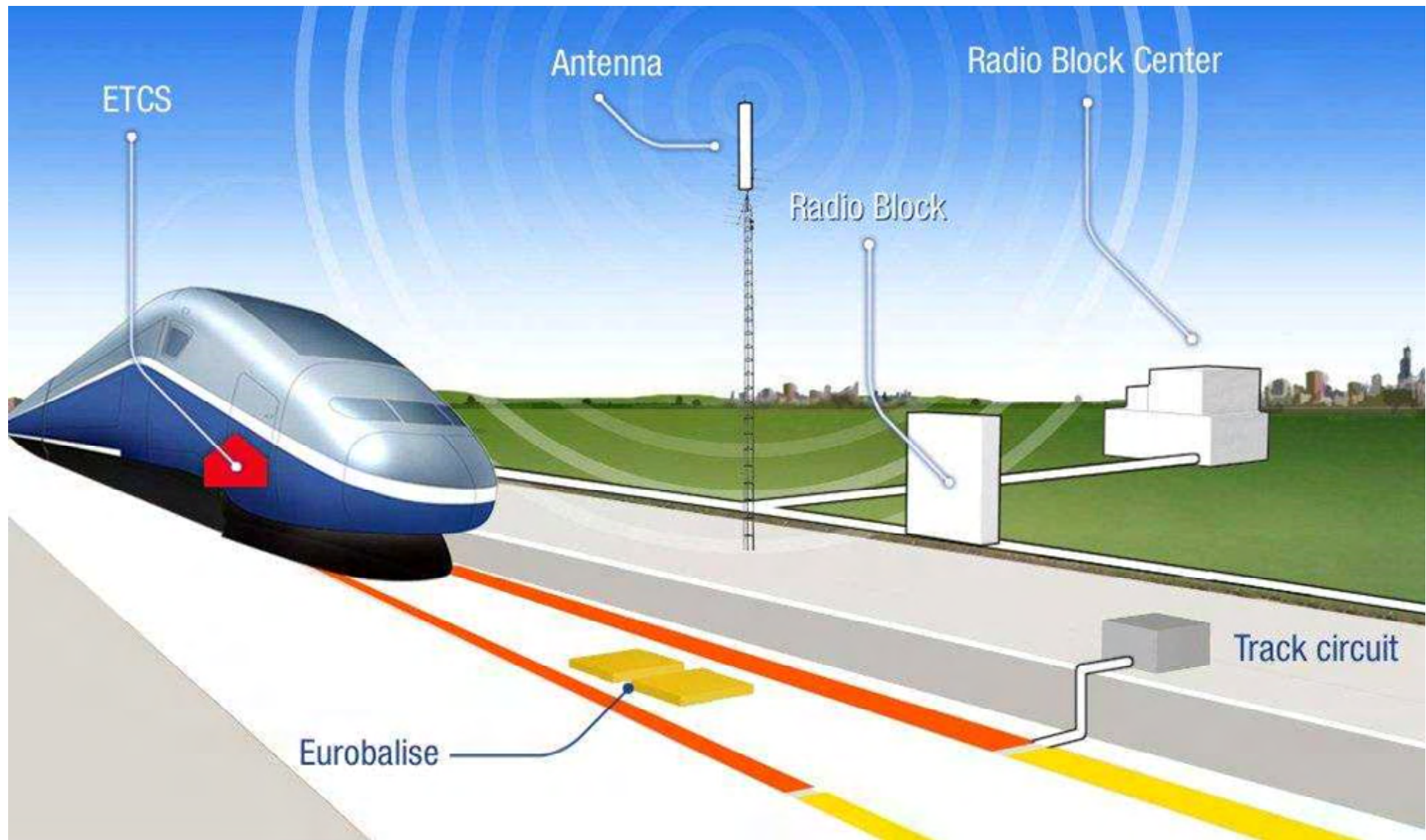
Creating a new legacy for the railway

- Investment in introducing the latest signaling and control technology is part of the European Train Control System (ETCS) and traffic management program.
- Moving to the ETCS will allow trains to run closer together and travel at their best speeds while maintaining safe braking distances.
- The traffic management program, will greatly increase the capacity on the network by maximizing the throughput that existing track can support.
- These future signaling technologies will mean more capacity, greater reliability and a safer network for all – **fundamentally transforming the way that we operate the railway in the future**
- Improved signaling technology means more reliable journeys with fewer delays, so more trains can run and more people will be able to travel on the railway across the country.

ERTMS Level 1 - Equipment

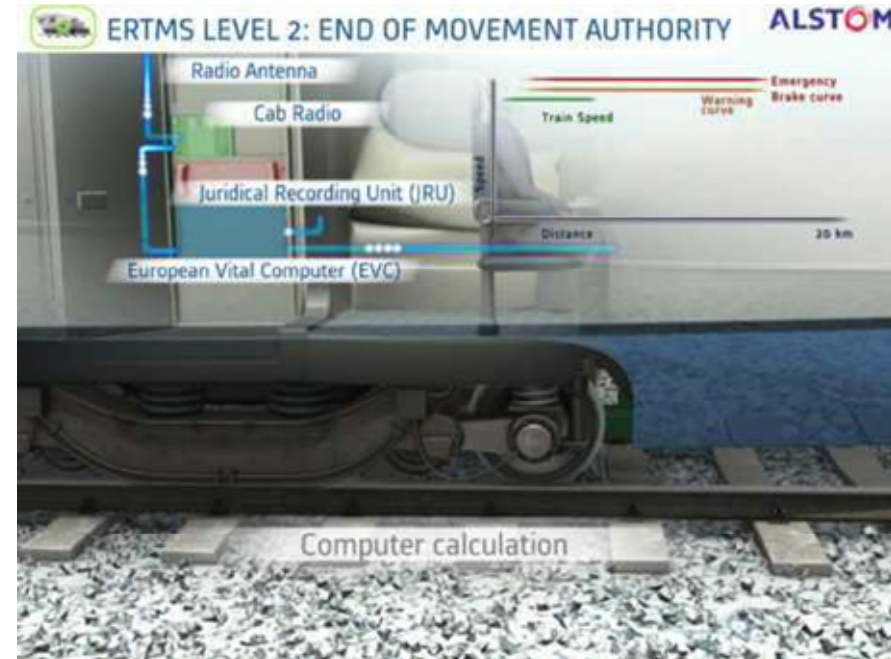
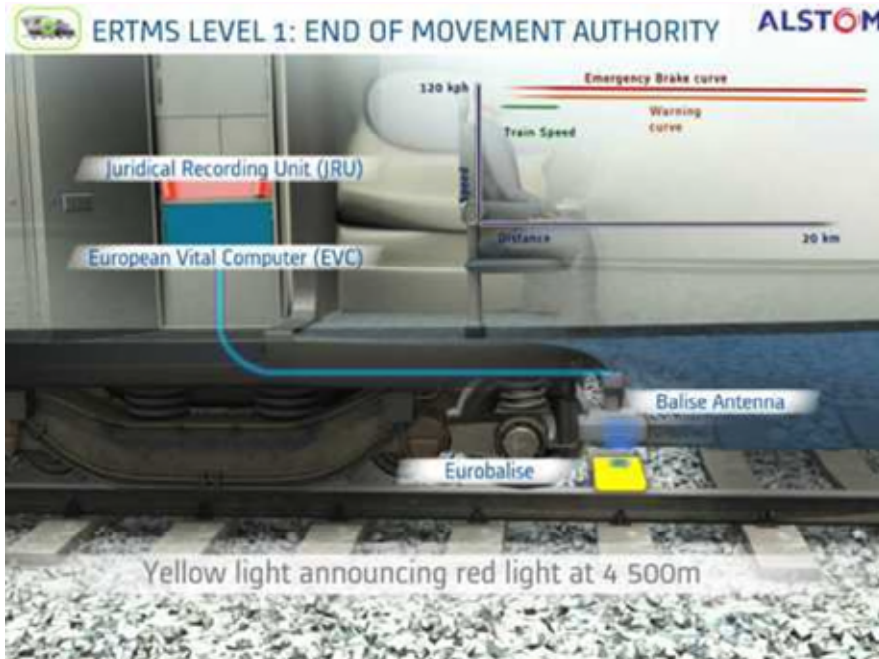


ERTMS Level 2 - Equipment



ERTMS Level 1 & 2

- Communication between the track and train via balise
- GSM-R communication between track and train



What is ICONIS?

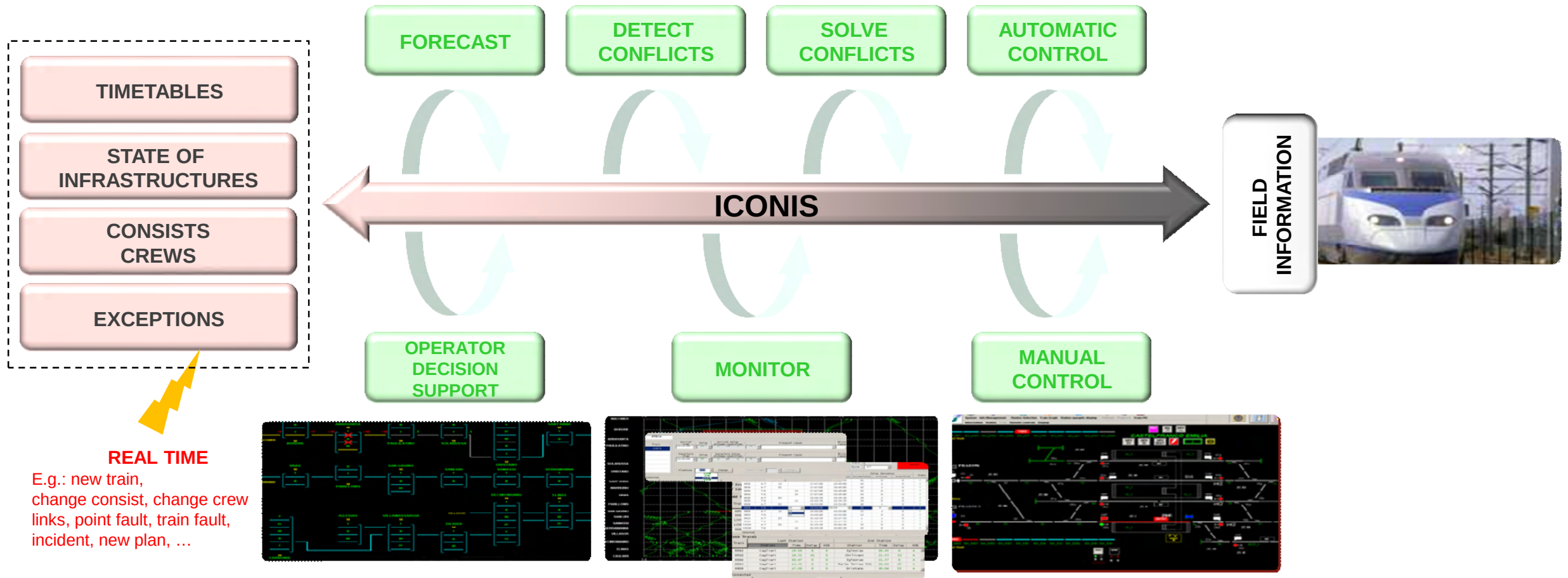
Real-time Traffic Management System

- Diagnostic Functionalities:
 - Health status detection of equipment (interlocking, RBC, field, ...);
 - Predicted and Preventive management;
- Centralized Monitor and Control functionalities:
 - Manual control;
 - Automatic route setting and point pre-movement for early fault detection;
 - Route feasibility analysis with detection of reasons why a route didn't clear;
- Decisional Support System functionalities:
 - Short-medium-long term timetable/rolling-stock/crew/infrastructure planning;
 - Real-time forecast function based on detailed topology (routes-platforms-sections)-rolling stock characteristics-infrastructure;
 - Real-time Conflict Detection;
 - Real-time Conflict Local Solution and Timetable Optimization: train priority, delay reduction, energy saving;
 - Task management for in-office/out-office (tablet/mobile) Operator task coordination;
 - Incident management with predefined default action and remedy plan management;
 - Service Intention as constraints for Optimizations.

} Integrated with DSS;

What does ICONIS?

Monitor & Control → Plan → Automate



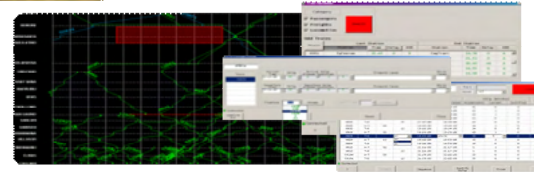
Why does ICONIS help?

Automatic Support, Rapid Decision, Global View, Fast Communication

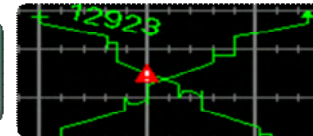
Real Time and Reliable Automatic Infrastructure and Trains Status Acquisition



Precise and faster future trip estimation



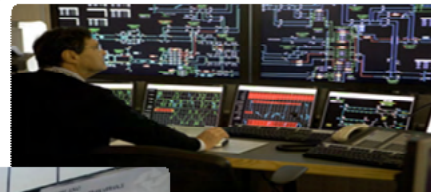
Early problem identification



Problem solved in advance



Efficient traffic management



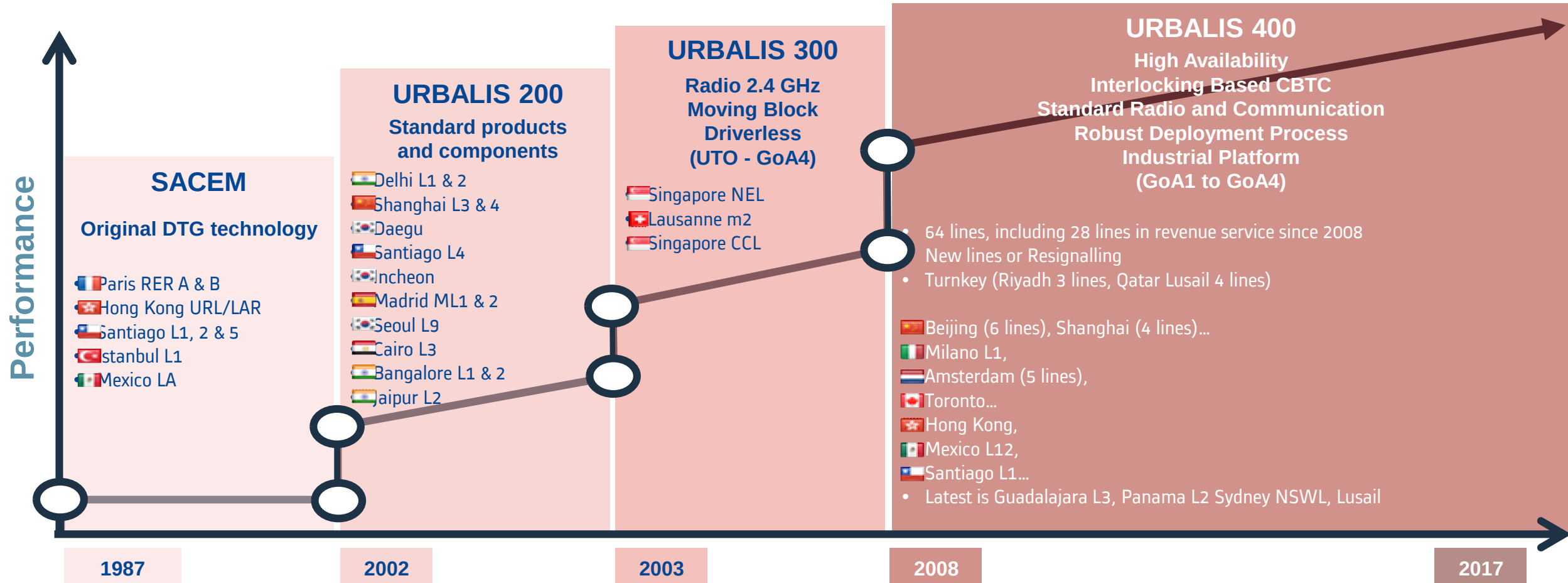
Reliable communication with user



Urban applications

Over 30 years of innovative leadership in train control

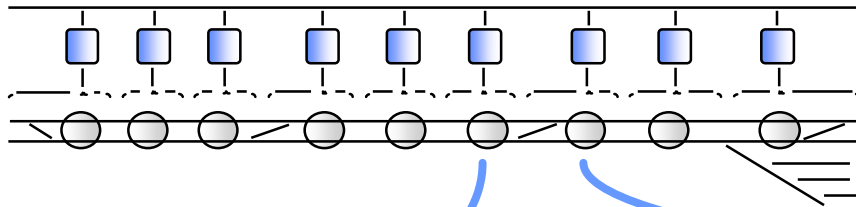
Increasing performance since 30 years



URBALIS 100 : Basic version



- Fixed block
- Track circuits or axle counters
- ATP distance to go
- Semi-continuous Track-to-train transmission



Wayside
signals

Trainborne
Equipment

Station N

Station N + 1

Train detection
devices

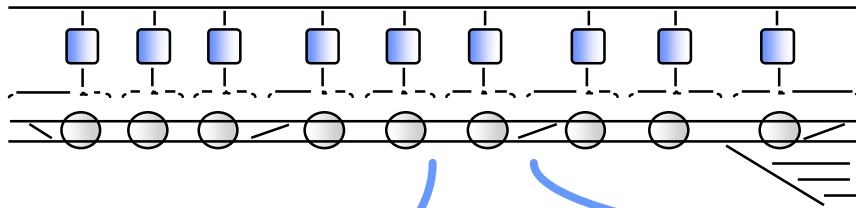
ATP Semi-continuous
transmission area

ATP Semi-continuous
transmission area

URBALIS 200 : Automatic version



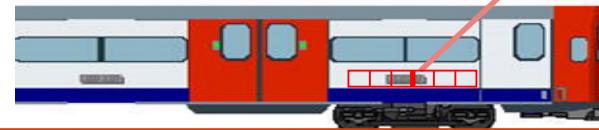
- Fixed or moving block
- Track circuits or axle counters
- ATP / ATO distance to go
- Continuous Track-to-train transmission



Station N



Station N + 1



Trainborne Equipment



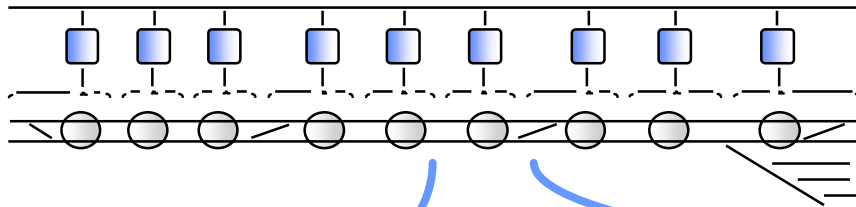
Train detection
devices

ATP Continuous
transmission

URBALIS 300 : Driverless version



- Fixed or moving block
- ATP / ATO distance to go
- Continuous bi-directional transmission



Station N

Moving block
boundary

Station N + 1

Trainborne
Equipment

Continuous bi-directional transmission

06

Rules and regulation

Rules

- For the UK, the operating rulebook is called "GE/RT8000 Rule Book or more commonly known as the "Rule Book" by railway employees. It is controlled by the Rail Safety and Standards Board (RSSB), which is independent from Network Rail or any other train/freight operating company. Most heritage railways operate to a simplified variant of a British Railways rule book
- In Italy, railway signaling is described in a particular instruction called Regolamento Segnali
- In North America and especially the US, operating rules are called method of operation. There are five main sets of operating rules in North America

CENELEC – EN 50129

- This standard is applicable to safety-related electronic systems (including sub-systems and equipment) for railway signaling applications
- the hazard analysis and risk assessment processes defined in EN 50126 and this standard are necessary for all railway signaling systems/sub-systems/equipment
- This standard is not applicable to existing systems/sub-systems/equipment (i.e. those which had already been accepted prior to the creation of this standard). However, as far as reasonably practicable, this standard should be applied to modifications and extensions to existing systems, sub-systems and equipment
- This standard is primarily applicable to systems/sub-systems/equipment which have been specifically designed and manufactured for railway signaling applications. It should also be applied, as far as reasonably practicable, to general-purpose or industrial equipment (e.g.: power supplies, modems, etc.), which is procured for use as part of a safety-related signaling system. As a minimum, evidence shall be provided in such cases to demonstrate either that the equipment is not relied on for safety, or that the equipment can be relied on for those functions which relate to safety

CENELEC Standards

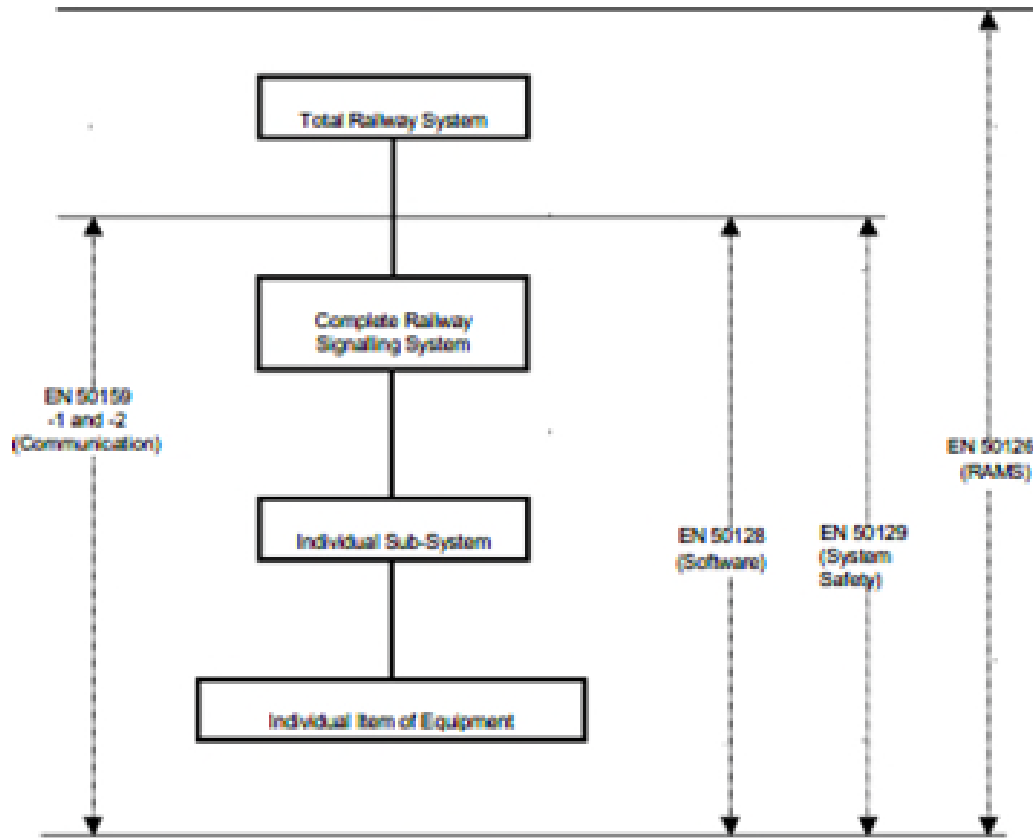


Figure 1 – Scope of the main CENELEC railway application standards

- This document is the first European Standard defining requirements for the acceptance and approval of safety-related electronic systems in the railway signaling field.
- Only some differing national recommendations and general advice of the UIC (International Union of Railways) on this topic were in existence.
- Safety-related electronic systems for signaling include hardware and software aspects. To install complete safety-related systems, both parts within the whole life-cycle of the system have to be taken into account. The requirements for safety-related hardware and for the overall system are defined in this standard. Other requirements are defined in associated CENELEC standards

Signaling Cables

Country	Operator	Standard	Cable type
Germany	DB	PH 416.0113 v1.1	A-2Y2YB2Y n x 1 x 0.9/1.4/1.8 mm S (H115/H145)
			AJ-2Y2YDB2Y n x 1 x 0.9/1.4/1.8 mm S (H115/H145)
		PH 416.0113 v2.1	A-2Y0F(L)2YB2Y n x 1 x 0.9/1.4/1.8 mm S (H115/H145)
			AJ-2Y0F(L)2YDB2Y n x 1 x 0.9/1.4/1.8 mm S (H115/H145)
		PH 416.0114 v2.1	A-2Y0F(L)2YB2Y n x 1 x 1.4/1.8 mm S (H95)
			AJ-2Y0F(L)2YDB2Y n x 1 x 1.4/1.8 mm S (H95)
		PH 416.0115 v1.1	A-2Y(L)2YB2Y n x 4 x 0.9/1.4 mm S (H45)
			AJ-2Y(L)2YDB2Y n x 4 x 0.9/1.4 mm S (H45)
		PH 416.113 v1.1 (based on)	A-HHBH n x 1 x 0.9/1.4/1.8 mm S (H115/H145)
			A-H(L)HBH n x 1 x 0.9/1.4 mm S (H45)
		PH 416.0120	A-2Y(L)2YB2Y 1 x 4 x 1.4 / 1.53 mm Balise
			AJ-2Y(L)2YDB2Y 1 x 4 x 1.4 / 1.53 mm Balise
		PH 416.0118	A-2Y(L)2Y2YV n x 4 x 0.9/1.4 mm + m x 1 x 0.9/1.4/1.8 mm combi cable
			A-2Y(L)2YB2Y n x 4 x 0.9/1.4 mm + m x 1 x 0.9/1.4/1.8 mm combi cable
			AJ-2Y(L)2YDB2Y n x 4 x 0.9/1.4 mm + m x 1 x 0.9/1.4/1.8 mm combi cable
		Arcor TNP 02/05	AJ-02YSTF(L)2YDB2Y n x 4 x 0.9 / 1.4 mm STI LG

Spain	adif/RENFE	E.T. 03.365.051.6	EAPSP n x 1 x 0.9/1.4 mm
			EAPSP n x 4 x 0.9/1.4 mm
			EAPSP-R n x 1 x 0.9/1.4 mm
			EAPSP-R n x 4 x 0.9/1.4 mm
			EATST n x 1 x 0.9/1.4 mm
			EATST n x 4 x 0.9/1.4 mm
			CCPSSP-FR0.3 n x 1 x 0.9/1.4 mm
			CCPSSP-FR0.1 n x 1 x 0.9/1.4 mm
			CCPSSP-FR0.3 n x 4 x 0.9/1.4 mm
			CCPSSP-FR0.1 n x 4 x 0.9/1.4 mm
			CCTSST-FR0.3 n x 1 x 0.9/1.4 mm
			CCTSST-FR0.1 n x 1 x 0.9/1.4 mm
			CCTSST-FR0.3 n x 4 x 0.9/1.4 mm
France	SNCF	NF F 55-698 & CT445	ZPAU n x 2 x 1.0 mm ²
			ZPFU n x 2 x 1.0 mm ²
			ZC03 4 x 4 x 1.0 mm ²

Italy	RFI	STF DTCSTSSTB SF IS 06 200	UG70G7KNR / RG70G7KNR n x 1 / 2.5 / 4 / 6 / 10 / 16 / 25 mm ²
		Digicode	DIGICODE 2 x 2 x 1.4 + 1 x 2 x 0.6 mm

Designation codes for cables & Symbols

VDE designation codes for cables

1. Cable Type

- A- Outdoor cable
- AJ- Outdoor cable with protection against inductive interference

2. Cable Design

- 2Y Polyethylene (PE) insulation material
- 02Y Cellular polyethylene (PE) insulation material
- 02YS Foam-skin polyethylene (PE) insulation material
- DF Loose tube (fibre optic cable)
- F Petroleum jelly filling compound
- OF Low capacitance filling compound
- TF Filling with water swellable yarns and fleeces
- (L)2Y Moisture barrier sheath (laminated AL-foil bonded to PE sheath)
- (St) Screen of copper tape
- D Screen of concentrically positioned copper wires
- Z Screen of concentrically positioned aluminium wires
- (ZN) Non-metallic tensile strength elements
- (SR) Armouring of corrugated steel tape, longitudinally applied
- B Armouring of helically applied steel tape
- Y Polyvinyl chloride (PVC) sheathing material
- 2Y Polyethylene (PE) sheathing material
- 4Y Polyamid (PA) sheathing material
- H Halogen free, flame retardant sheathing material
- V Reinforced sheathing

LSZH-FR

- LS Low smoke
- ZH Zero halogen
- FR Flame/Fire retardant

Explanation of symbols

Ambient temperature



Permissible ambient temperature ranges.
[°C during installation]
[°C during operation and storage]

Short circuit temperature



Maximum permissible short circuit temperature at conductor [°C]

Fire behaviour



EN/IEC 60332-1 flame retardant or
EN/IEC 60332-3 fire retardant or
EN/IEC 50200 fire resistant (only if applicable)

Smoke emission



EN/IEC 61034

Toxicity/Acidity



EN/IEC 60754-1
EN/IEC 60695

Insect resistance



Insect and termite resistant

Bending radius



Minimum bending radius for installed cables in fixed applications.
[for multiple bending]
[for final bending]
D = outer cable diameter

Signaling cabling rules

- The Signalling installation at wayside and trackside are exposed to heavy “electromagnetic environment” made of a complex distribution of electromagnetic waves
- **Potential impacts of interference:**
- The interfering signal will generate an unexpected equipment behaviour, which may generate an unsafe situation.
- **Evolution of technology and impacts:** New technologies become at the same time:
- ➔ Faster (higher clock frequencies ➔ Sensitive (lower level of energy necessary to change signals states)

The evolving technology make our systems more and more exposed to EMI perturbations, therefore cabling and installing care become essential.

EMC solutions:

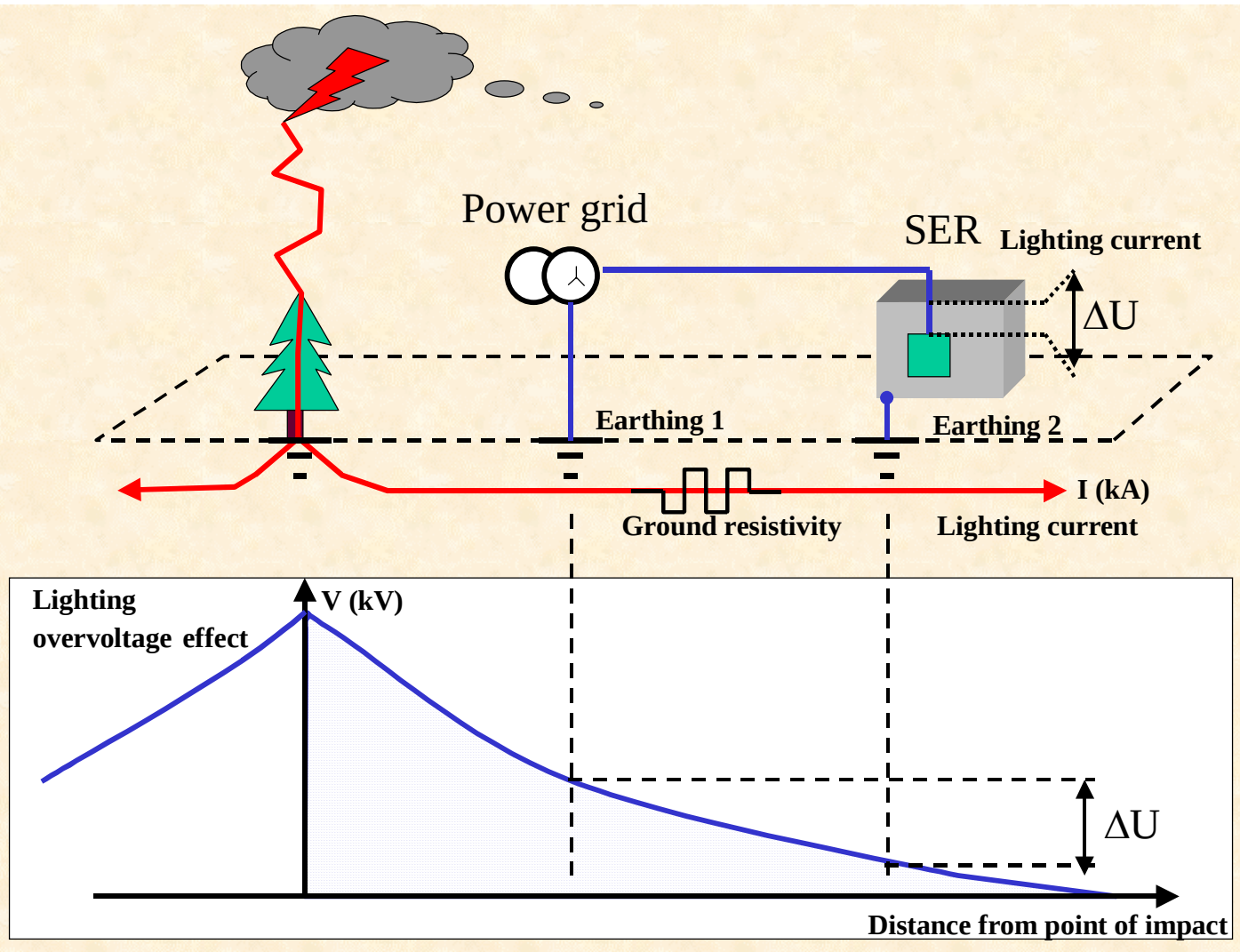
- Compliance with EN50121 Standards (which covers most interference risks).
- Installation as per “state of the art” rules

Equipment fully compliant with the requirements of EN50121 Standard may show unexpected behaviour if installed “wrongly”.

Installation essential rules

- Segregating perturbing from sensitive circuits
- Reducing the loops between conductors – Practically, cables must be arranged in twisted wires, which reduces the surface of the loop to nearly zero
- Reducing the ground impedance (potential reference) : Install internal cables as close as possible to grounding network in order to minimise the inductive loops
- Improving equipotentiality between equipment
- Shielding cables : The attenuating effect is generated by the current flowing in the shield which itself creates a magnetic field in opposite direction. The benefits of shielding only truly appear if both ends of the shield are directly bonded to ground.
 - **Cable shield must be bonded to ground at both ends**
 - **Impedance of shield bonding must be kept low**
- Filtering the sensitive input/outputs

Protection rule : Surge arrester



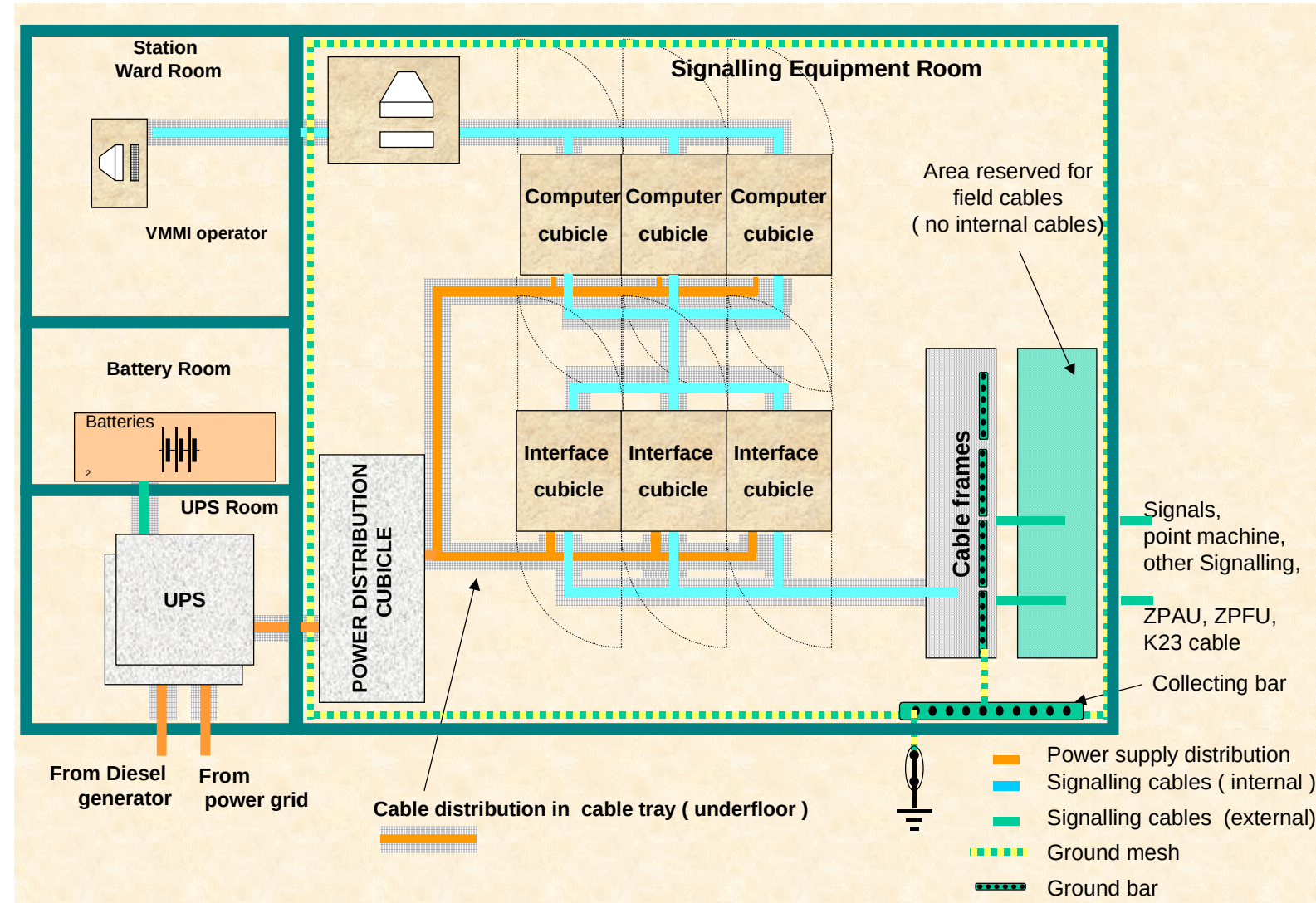
All ports connected to track equipment are concerned with overvoltage lightning effects. In order to overcome the risk of damaging the equipment, the following protections must be implemented :

1. Create a low impedance unique earthing network
2. Fit surge arrester on sensitive ports next to the cable opening of the Signalling Equipment Room.

Implementation of rules in signaling equipment room

Distribution of cubicles must strive to achieve the following:

- The cable frame which performs the filtering function (surge arrestors, etc...) must be located next to the opening for in-coming trackside cables. The Cable frames are used to carry and evacuate the noise picked by external cables, and therefore must remain within a selected noisy area.
- Inside the Cable Frame, the in-coming cables (polluted) must be kept away from the out-coming internal conductors (filtered). Ideally, the trackside cables shall enter the cable frame from the bottom opening, while the internal conductors exit through top opening.
- The power supply cubicles (UPS, power distribution cubicle,...) must be located away from sensitive cubicles. The links between UPS and other power supply units must be kept as short as possible.
- Lay cable tray network on the under floor along the cabling route between cubicles. Provide sufficient spacing on cable tray to achieve the segregation rules between circuits or use distinct cable trays for low class and high class cables.
- Keep the internal earthing network equipotential through the implementation of a complete mesh (all earthed equipment including cable trays are interconnected)

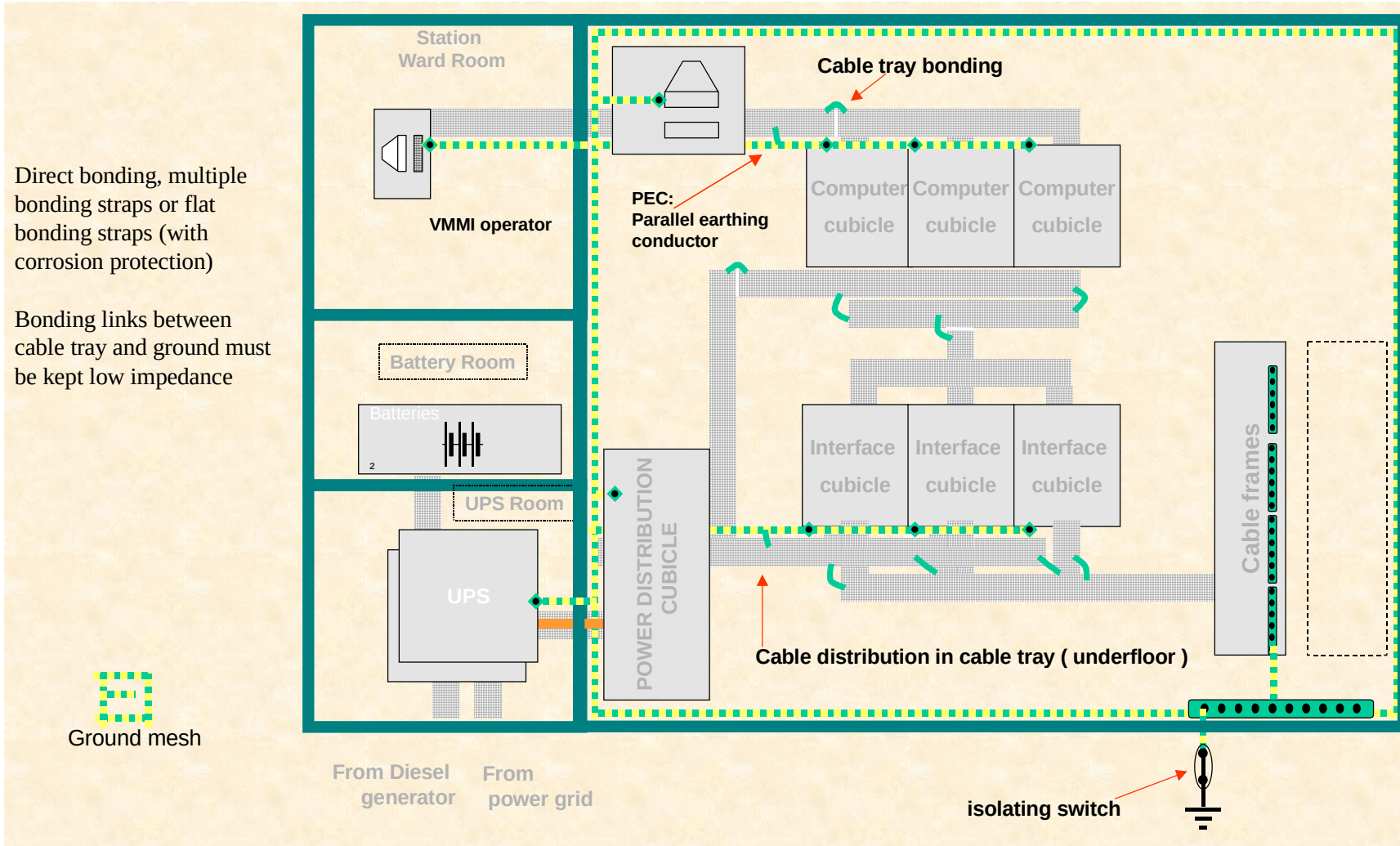


Separate the circuit according to their characteristic

Class	Characteristics of signal	Type of signals	List of signals
Class 1: Sensitive signals	Low level → Sensitive signals (low threshold) → High impedance signals → slowly changing signals	Computers links and analogue messages between track and train	– Serial links – Track to train Digital data transmission – Train detection receiver – Train to wayside coded messages
Class 2 : Medium signals	Medium level signals	All or Nothing signals	– Relays command – 0 / 24V signals – train detection transmitter
Class 3a Polluting signals	High level → High dv/dt or di/dt signals → low impedance signals → high polluting potential → low sensitivity	Primary power cables:	- Power grid to UPS link -UPS to Power distribution cubicle or battery link -UPS to batteries link
Class 3b Noisy signals		Secondary power supply cables:	– secondary power supplies to the trackside equipment : – 380 V to point machines – 220 V to transmitting device – 110V to Signal Lights – etc ...

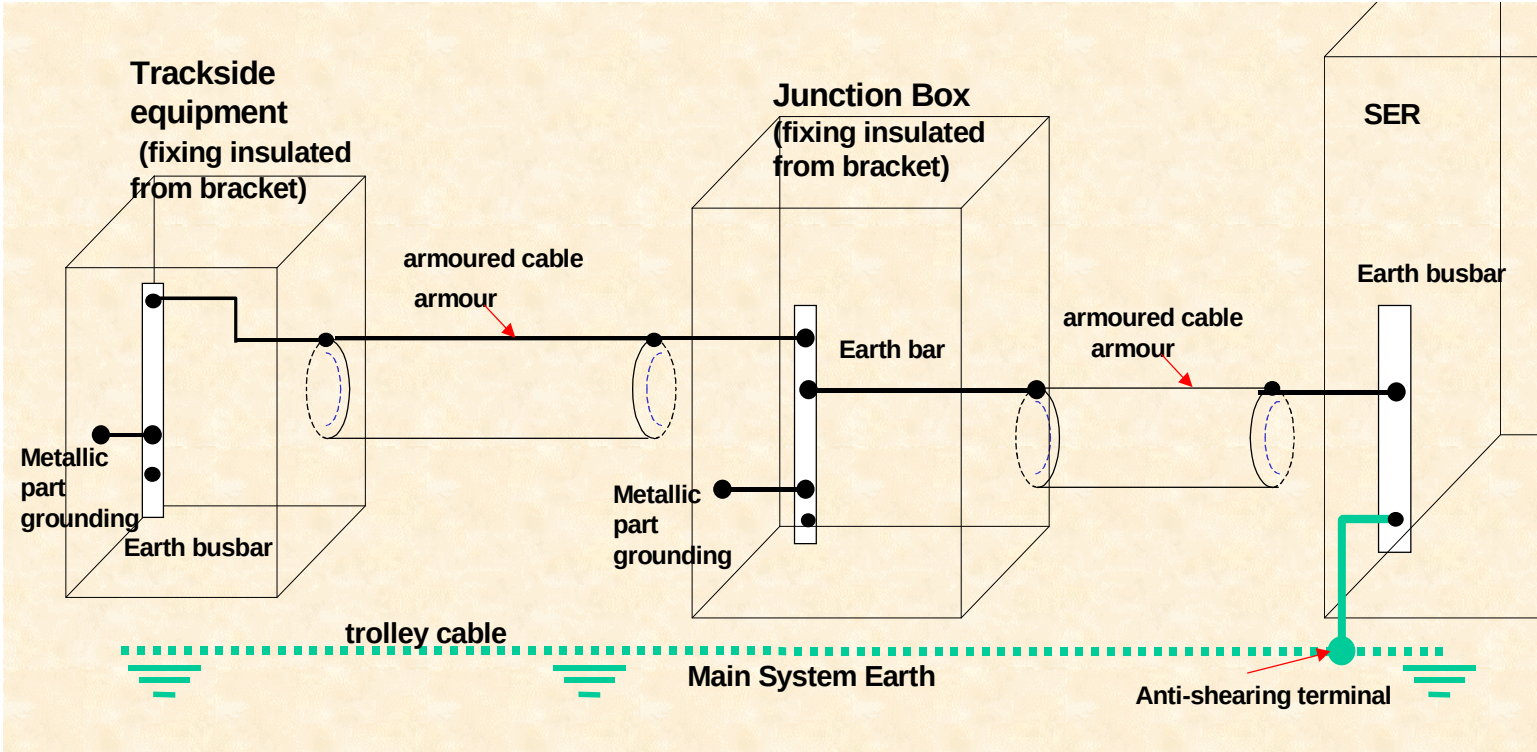
Three classes of circuits are defined by decreasing order of sensitivity and increasing degree of noise generation; the sixth being a special class

Earthing Network



A low impedance conductor runs around the SER to make a ground network. Cubicles are locally connected to this earth reference through short metallic flat braid, which are connected to the metallic sheet of cubicles. This network makes a low impedance mesh. This network is connected to the earth rod via an isolating switch

Earthing of cables armours



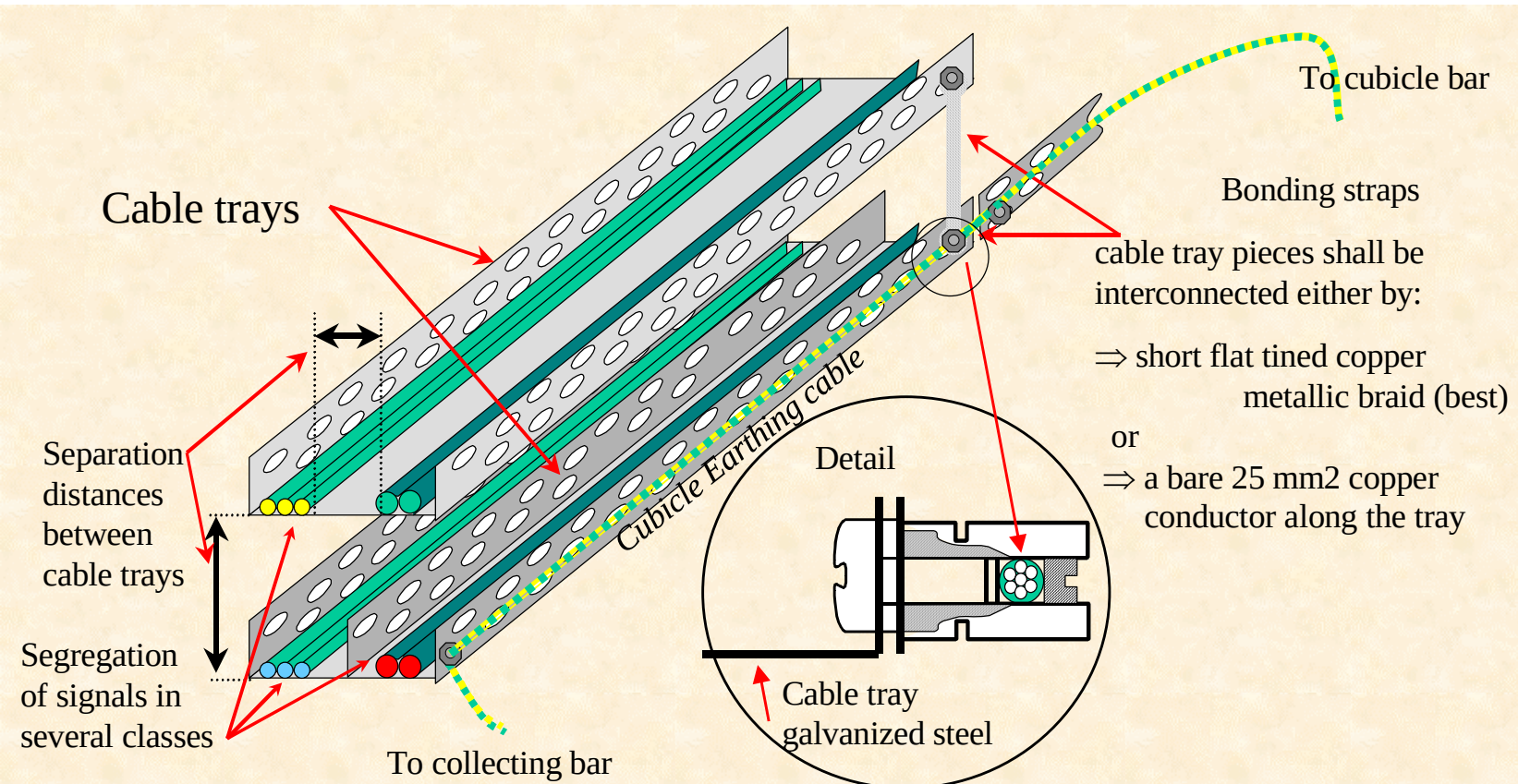
In all cases, links between protective shields or armours must be as short as possible
cables can generally be replaced by non shielded ZPFU cable (weaker induced voltages on cables)

All the metallic signalling equipment (except point machines) at trackside are connected to SER earth through their armours and shields. Non-metallic signalling equipment at trackside are kept floating.

In order to avoid corrosion effects, all metallic part at trackside are connected through armour of successive cable at SER end.

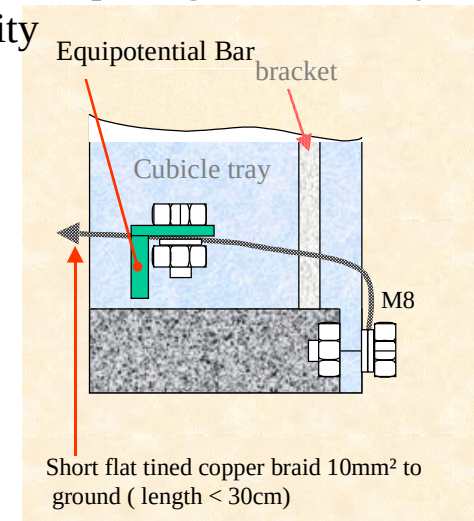
Metallic trackside junctions boxes and equipment are mounted insulated from support

Installation and use of cable trays



1. Lay cables according to their category
2. Keep the cables of distinct categories segregated
3. Flatten and tie up cables on the cable tray
4. Create a ground mesh by bonding the trays to each other with metallic braid
5. Earthing of cubicle is achieved through an earth bus made of a yellow/green 25mm² copper conductor. As it is shown above, the earth bus must be connected to the nearby cable tray to ensure touch voltage protection.
6. If cables of distinct categories must cross this shall be done at right angle
7. Do not fill the cable spacing on cable trays over their capacity

Each cubicle shall be equipped with a local equipotential bar to provide equipotentiality inside the frame



Separate field cables from internal conductors

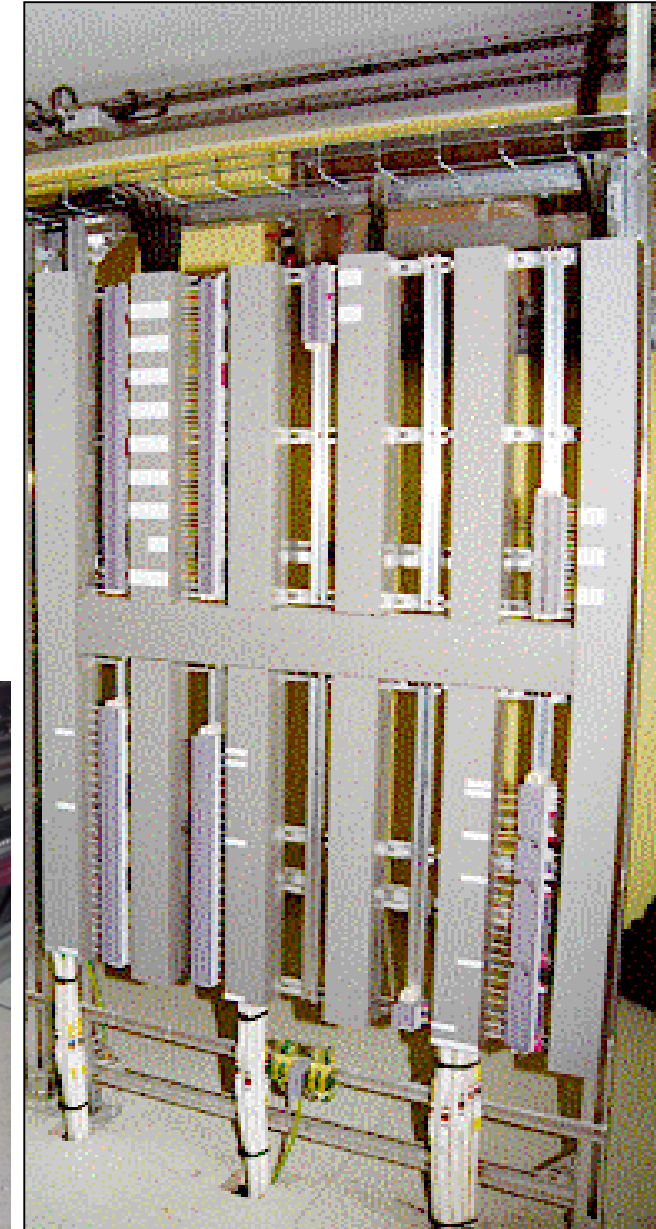
The trackside cables (field ZPAU or K23 cables) enter the cable frame through the bottom opening. The shields are terminated on the equipotential bus bar and the conductors are wired internally.

Filters and surge arrestors are located at the bottom next to the conductors wiring. The upper part of the Cable Frame is dedicated to filtered and un-screened sensitive conductors.

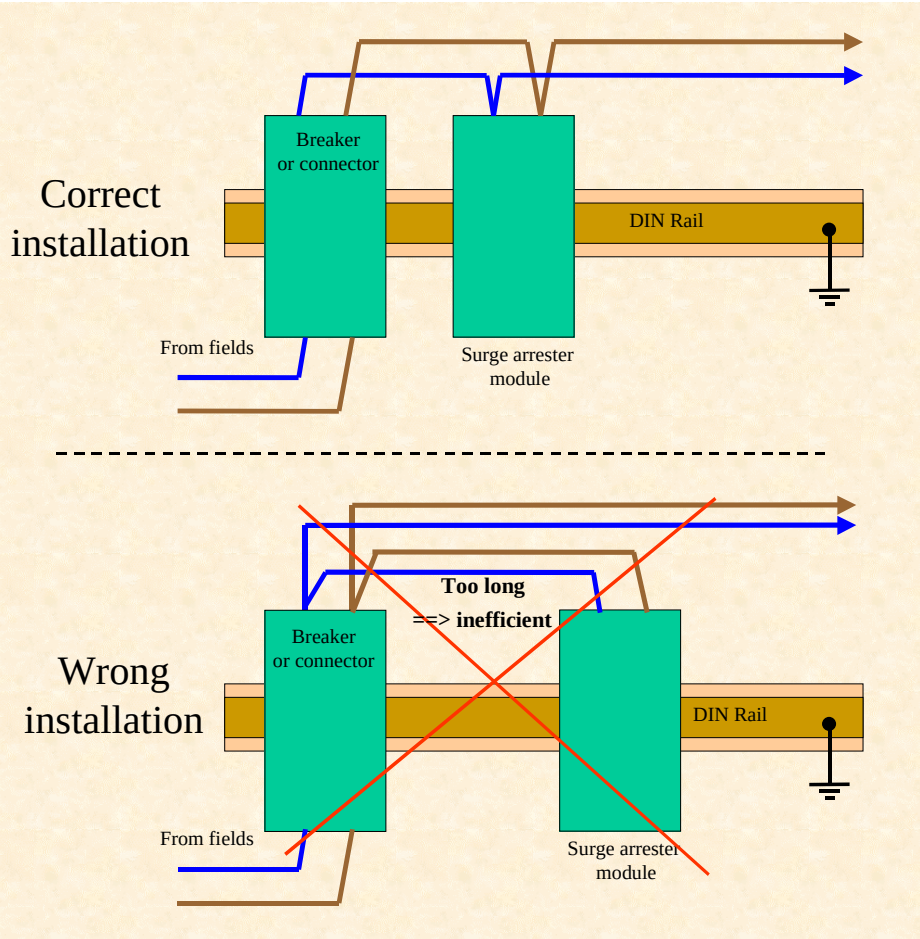
The conductors exit the cable frame through an aperture at the top of the cubicle. They shall run internally to their cubicle on dedicated cable tray network as per segregation rules

The shields and armours of field cables shall be terminated on the equipotential busbar of the Cable Frame.

The shields and armours are bonded with a 6mm² Yellow/green wire which must be kept as short as possible.



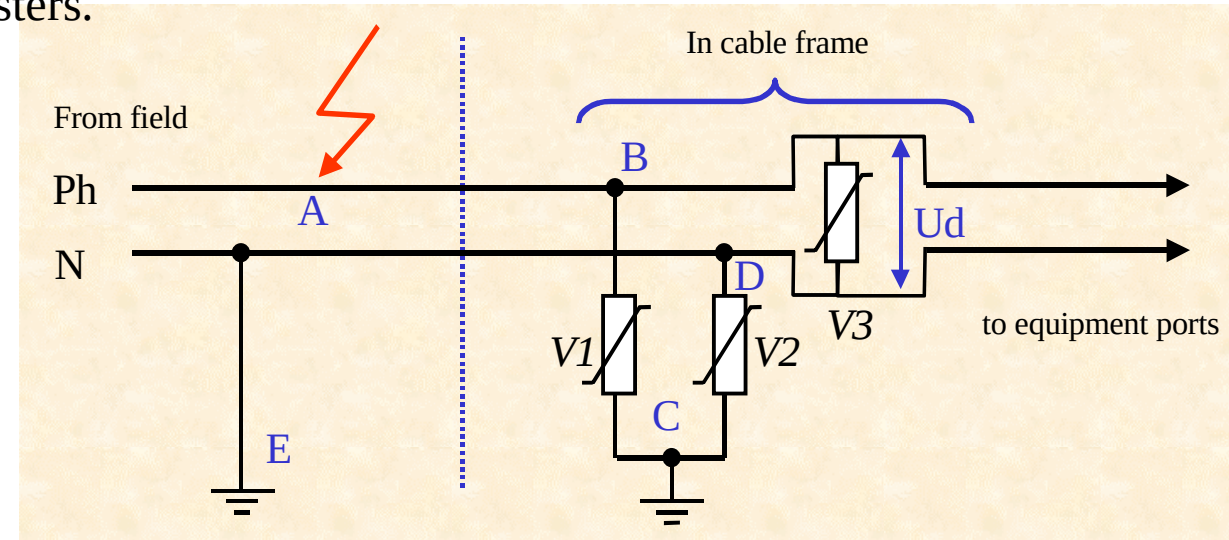
Installation of surge arrestors



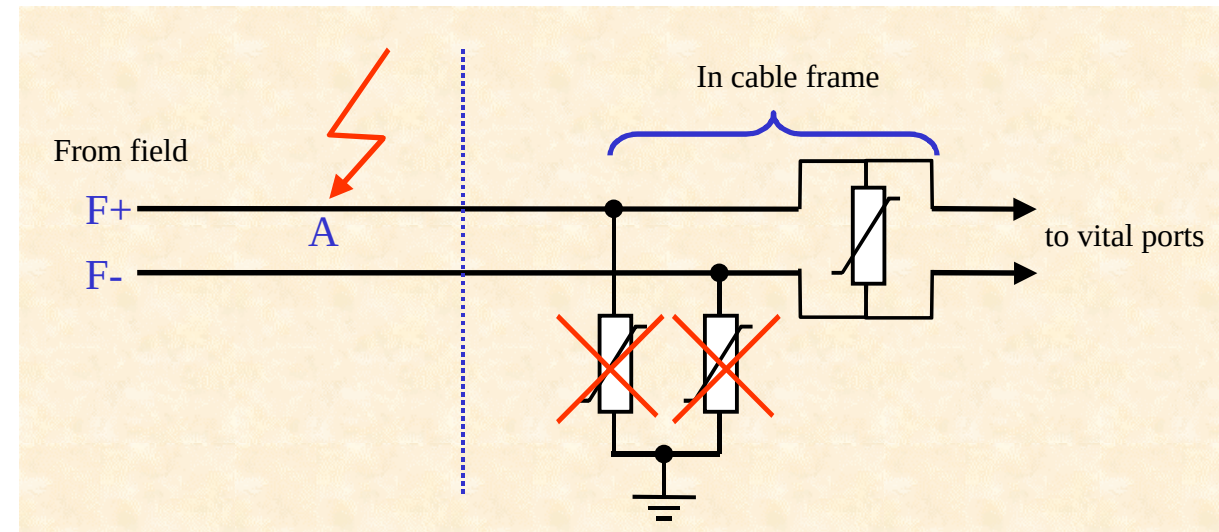
Surge arrester cabling rule:

- ➔ Keep away protected cables from unprotected cables
- ➔ Use short wire to connect surge arresters

Fit non vital ports with both differential and common mode surge arresters.

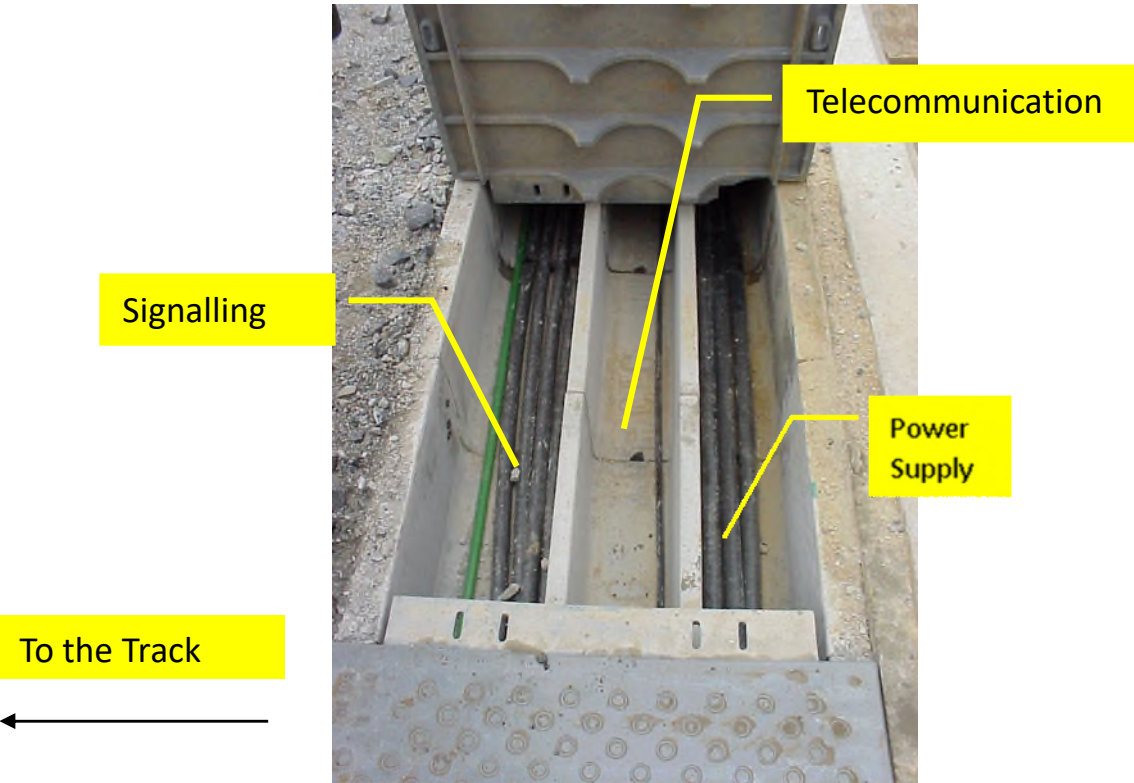
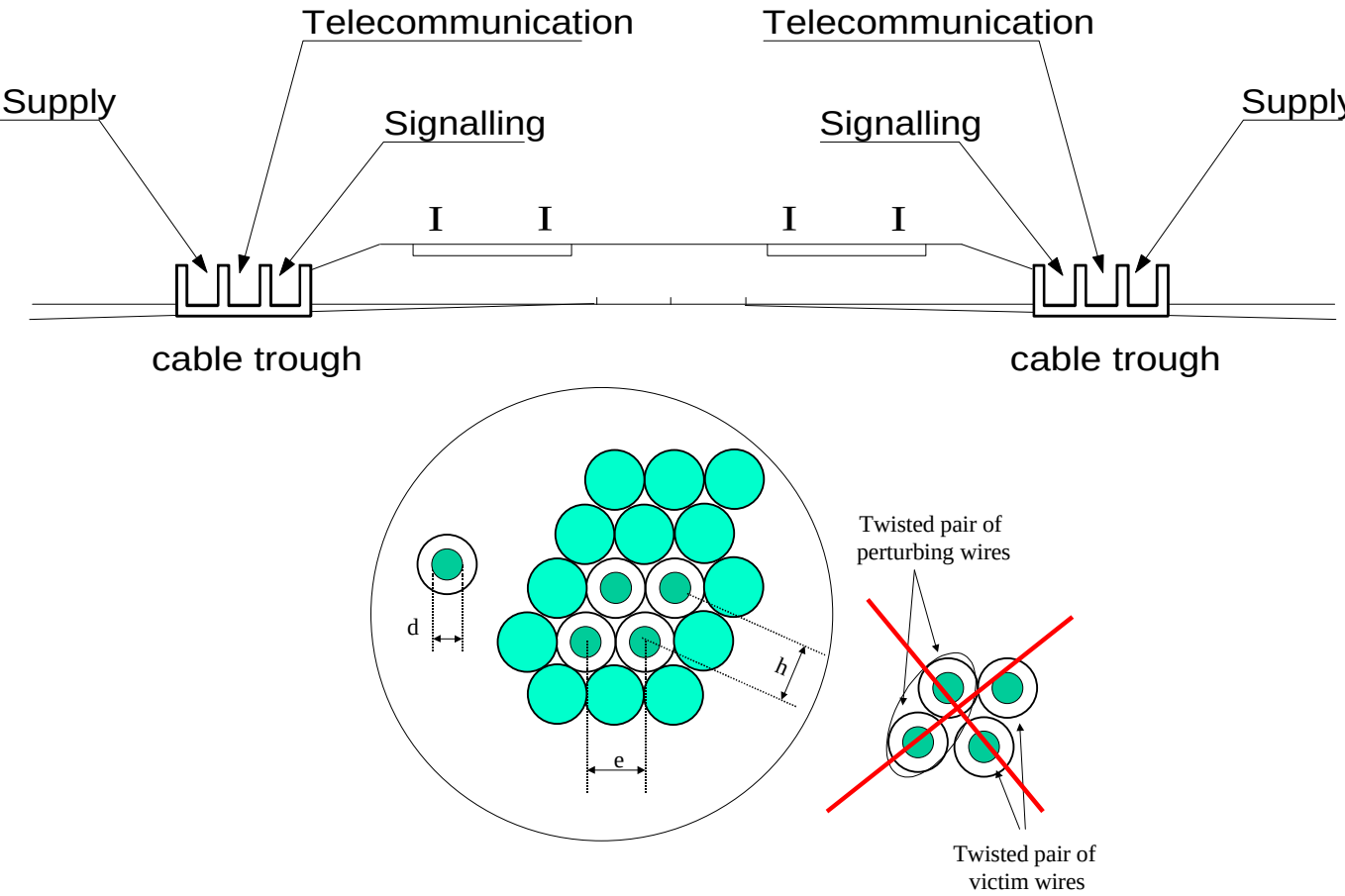


Fit ports with differential surge arresters. (for safety reason, common mode surge arresters cannot be used with vital ports)
Keep protected cables away from unprotected cables



Implementation of rules at trackside

Separation of signals at trackside



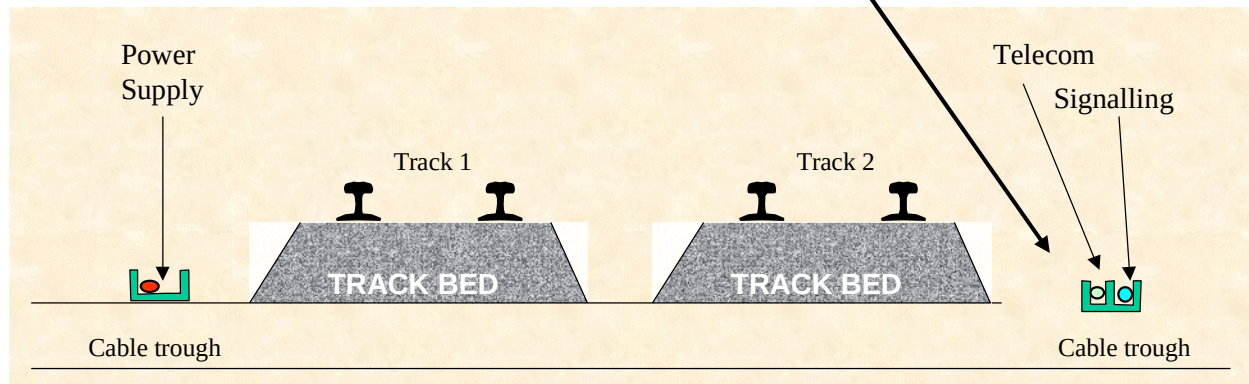
Trackside circuit separation

In order to prevent cross-talk coupling between circuits, avoid coexistence of high class and low class signals within the same trackside cable

Installation of trackside cables

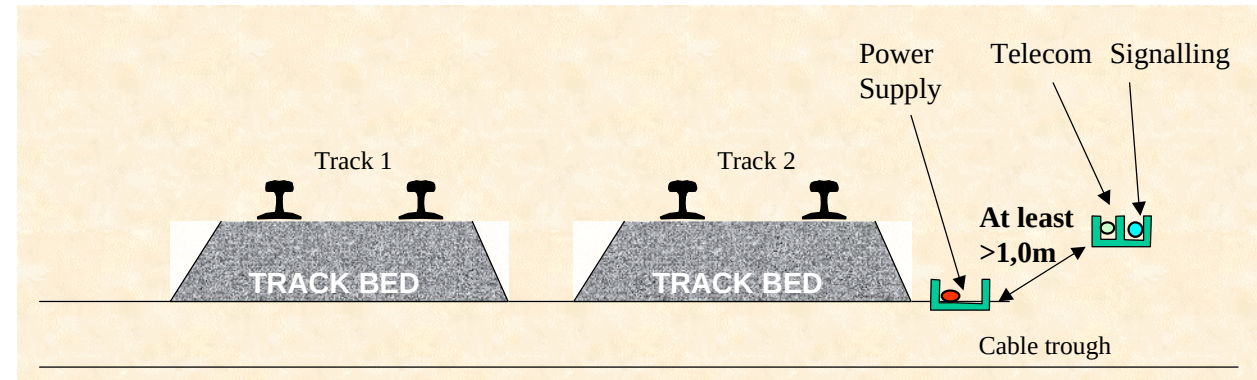


Inside the cable through, a dedicated conduit shall be used by each contractor



Signalling cables arrangement between Stations (best case)

Whenever possible, the high power signals shall run on the opposite side of the track



Signalling cables arrangement between Stations (acceptable case)

When such arrangement is physically impossible due to space constraints, the lineside signalling cables shall be segregated from power cables with at least 1m separation

Coexistence of conductors in multi-conductor cables at trackside

Example of coexistence rules on U200 project

		S0	S1	S2	S3	S4	S5	S6	S7	S8
Point machine relay circuit	S0	NLC	NLC	NLC	FC	LC	FC	LC	LC	FC
Light control relay circuit	S1		NLC	NLC	FC	LC	FC	LC	LC	FC
Track circuit relay circuit	S2			NLC	FC	LC	FC	LC	LC	FC
Point Ma-chines power	S3				NLC	LC	FC	FC	FC	NLC
Power to light indicators	S4					NLC	LC	NLC	NLC	FC
SDTC Trans-mitter/receiver	S5						NLC	LC	LC	LC
Power to route indicator	S6							NLC	NLC	FC
Power to STIB beacon	S7								NLC	FC
Down Link Loop	S8									FC

The K23, ZPFU and ZPAU cables are made of twisted pairs of wires which shall run in close proximity to each other. Care must be taken regarding the choice of conductors installed within the same cable.

Cross-talk coupling can be either from capacitive or from inductive origin.

Inductive coupling depends on:

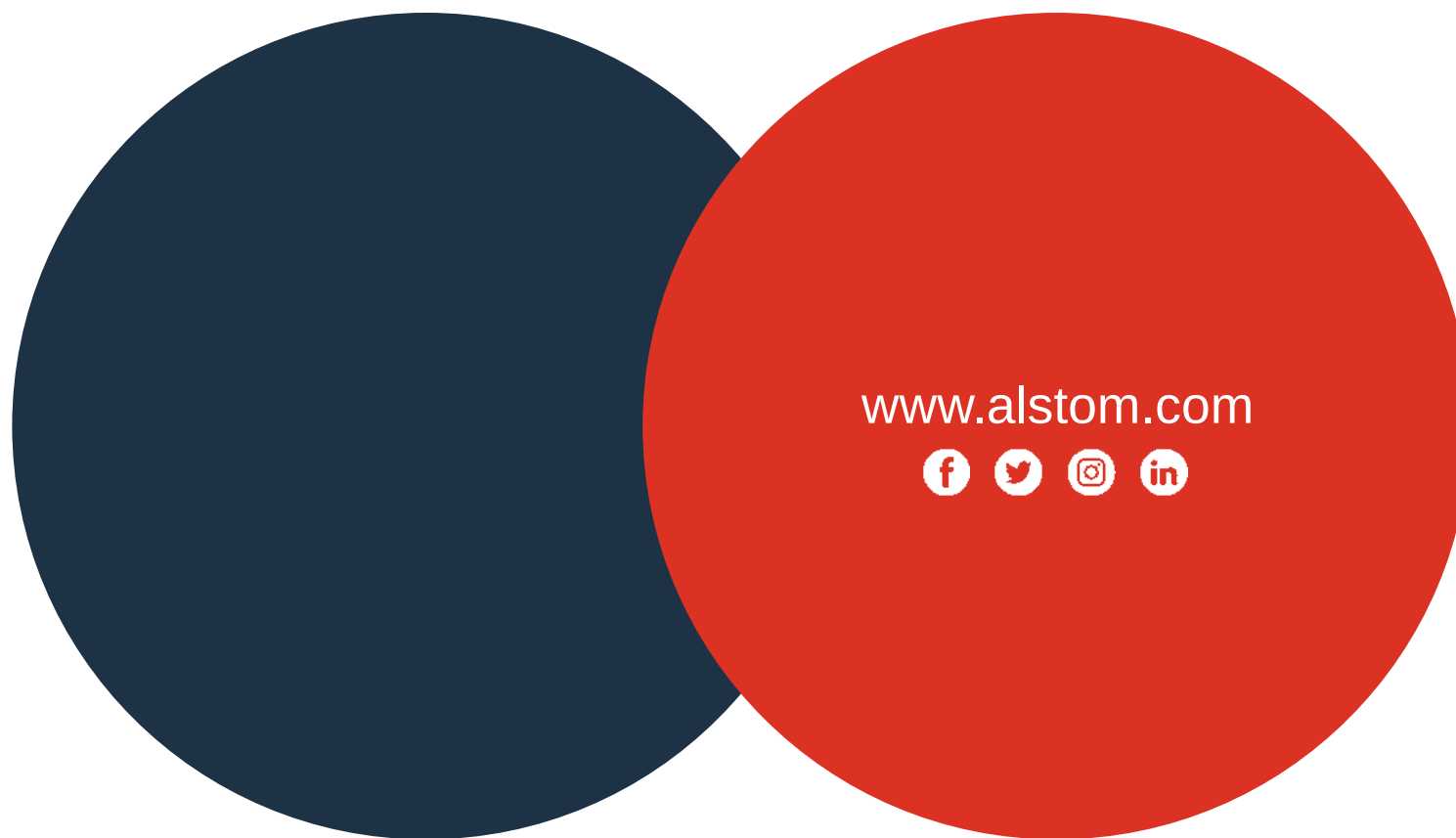
- The current value and frequency in the noisy circuit,
- The mutual inductance between perturbing and victim circuit (function of the loop surfaces of circuits),
- Impedance of the victim circuit

Capacitive coupling depends on :

- Perturbing voltage and frequency in the circuit,
- Parasitic capacity between perturbing and victim circuit,
- Impedance of the victim circuit

Legend

FC	Forbidden Coexistence
LC	Limited Coexistence
NLC	Non Limited Coexistence



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